

Reductive Groups

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REDUCTIVE GROUPS

by ARMAND BOREL and JACQUES TITS¹

This work is mainly devoted to the study of the properties of a connected reductive linear algebraic group G which are relative to a field of definition k of G . In particular, we introduce, using k -subgroups (i.e., closed subgroups defined over k), notions that naturally generalize those of maximal torus, roots, Weyl group, etc., of the classical theory, to prove the existence of a Bruhat decomposition for the group G_k of k -rational points of G , and theorems of conjugacy “over k ”, i.e., by elements of G_k . For this purpose, we essentially use two families of subgroups of G : the maximal tori split or decomposed over k (i.e., isomorphic over k to a product of multiplicative groups $\mathbf{G}_m$) and the minimal parabolic k -subgroups P (i.e., such that G/P is a projective variety). They play, respectively, the role of maximal tori and of maximal connected solvable subgroups over an algebraically closed field.

These results are the subject of §§ 4 to 7 of I. The second part provides various complements or applications in which G is not always reductive or k is subject to certain restrictions.

The §§ 0 to 2 are of a preliminary nature. § 0 fixes conventions and notation; § 1 gathers some results on tori; § 2 recalls the fundamental properties of a reductive group G that we will need: structure of G when G is split over k (which is equivalent to the existence of a maximal torus split over k), density of G_k if k is infinite, existence of a maximal torus defined over k and of a separable extension of k over which G is split (2.14). § 3 is devoted to the study of certain k -subgroups H normalized by a maximal torus of G defined over k . In particular, we establish the existence and conjugacy over k of Levi decompositions (3.13, 3.14), then, in certain cases, the existence of a composition series with successive vector quotients on which a given torus operates by homotheties when H is unipotent (3.17, 3.18), and of local k -sections of the fibration of G by H (3.24).

In § 4, these results are applied to the intersection of two parabolic k -subgroups P, P' . We show moreover in particular that: the fibration of G by P has local k -sections; if P and P' are conjugate over an extension K of k , they

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are conjugate over k ; if P and P' are minimal, they are conjugate over k (4.13). One also establishes the conjugacy over k of maximal k -split tori (4.21) and connects these subgroups to the minimal parabolic k -subgroups (4.16). The proofs give an important role to the notion of opposite parabolic subgroups P, P' ($P \cap R_u(P') = P' \cap R_u(P) = \{e\}$, cf. 4.8, 4.10).

Let S be a maximal k -split torus. By definition, the Weyl group ${}_kW(G)$ of G relative to k is the quotient $\mathcal{N}(S)/\mathcal{Z}(S)$ of the normalizer of S by the centralizer of S , and the k -roots of G with respect to S are the non-trivial characters of S in the adjoint representation of G . It is shown (5.3) that ${}_kW(G)$, viewed as a group of automorphisms of $X^*(S) \otimes_{\mathbf{Z}} \mathbf{R}$ (where $X^*(S)$ is the group of rational characters of S), equipped with a suitable scalar product $(\cdot, \cdot)$, is the group generated by reflections with respect to the hyperplanes orthogonal to the k -roots, and (5.8) that the set ${}_k\Phi(G)$ of the latter is a “root system” (i.e., besides invariance under the Weyl group, satisfies the condition $2(a, b)(b, b)^{-1} \in \mathbf{Z}$ ($a, b \in {}_k\Phi(G)$)). Furthermore, we have $\mathcal{N}(S) = \mathcal{N}(S)_k \cdot \mathcal{Z}(S)$ and the double cosets of G_k modulo P_k (P a minimal parabolic k -subgroup) correspond bijectively to the elements of ${}_kW(G)$ (5.15).

§6 relates the roots and Weyl groups relative to k and to an extension K of k , the parabolic subgroups over K and over k , and the Bruhat decompositions of G_K and G_k . It is also shown that $G_K = R_u(P)_K \cdot U_{\bar{K}} \cdot P_K$ where P is a minimal parabolic k -subgroup and U^- is the unipotent radical of a K -subgroup opposite to P (6.25).

The set Φ_0 of elements of ${}_k\Phi(G)$ whose double is not a root is also a root system. In §7, one constructs a k -split subgroup of G of which S and Φ_0 are respectively a maximal torus and the root system.

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In §10, it is shown (10.2) that in any algebraic group G , the conjugacy class of a semisimple element is closed (and conversely if G is semisimple) and that the centralizer of a k -torus, k being a field of definition for G , is also defined over k (10.5). In §11, we recover results of Grothendieck and Rosenlicht on the existence and conjugacy over k of Cartan k -subgroups of G when G is solvable.

In the remaining paragraphs, we make restrictive assumptions on the base field k . Section 8 assumes it to be perfect and proves, by a fixed-point argument, the conjugacy over k of maximal connected subgroups triangulable over k (i.e., isomorphic over k to a group of triangular matrices) (8.2).

If k is locally compact of characteristic zero, G_k is naturally equipped with a topological group structure. Sections 9, 13, and 14 are devoted to various questions involving this topology; in particular: a necessary and sufficient condition for G_k/H_k or $(G/H)_k$ to be compact (9.3) or for G_k to be generated by a compact neighborhood of the identity element (13.4), structure of the group of connected components of G_k when $k = \mathbf{R}$ and G is irreducible (14.5). In §14, we also make the connection between the point of view developed here and the transcendental theory (Iwasawa decomposition, Weyl group and roots of a “symmetric pair”).

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Finally, § 12 assumes G to be connected semi-simple, k of characteristic zero, and discusses questions of rationality of irreducible linear or projective representations of G and the properties of their weights restricted to S . In particular, we characterize the dominant weights of projective representations defined over k (12.6) and of strongly rational linear representations over k , that is to say, those which are defined over k and whose representation space contains a line defined over k stable under a parabolic subgroup of G (12.10).

The results of this work were partly announced previously [2, 34, 35], often assuming k to be perfect, a restriction which could be removed thanks to the results of Grothendieck [11] recalled in 2.14. Let us note in this regard that a certain number of technical complications or detours are due to the necessity of showing that certain k -closed subgroups are defined over k , or that certain classes of subgroups contain a representative defined over k , results which are obvious or well known if the base field is assumed to be perfect. Under this latter hypothesis, a part of the results of §§ 4, 5, 8 is also proved by Satake [28].

§ 0. NOTATIONS AND PRELIMINARIES

0.1. Fields are always commutative. If k is a field, k^* , k_s , and $\bar{k}$ denote respectively the multiplicative group of non-zero elements of k , a separable closure of k , and an algebraic closure of k ; if k' is a separable normal extension of k , $\text{Gal}(k'/k)$ denotes the Galois group of k' over k .

0.2. Let G be a group, H a subset of G . We will often denote by gH the transform of H by the inner automorphism $\text{Int } g: x \mapsto g \cdot x \cdot g^{-1}$ ($g, x \in G$). The image of a subgroup L of G in the group $\text{Int } G$ of inner automorphisms of G will be denoted by $\text{Int}_G L$.

$\mathcal{N}_G(H)$ or $\mathcal{N}(H)$ is the normalizer of H in G , and $\mathcal{Z}_G(H)$ or $\mathcal{Z}(H)$ the centralizer of H in G . The i -th derived group (resp. the i -th term of the lower central series) of G is denoted by $\mathcal{D}^i G$ (resp. $\mathcal{C}^i G$), and we set $\mathcal{D}^\infty G = \bigcap_i \mathcal{D}^i G$, $\mathcal{C}^\infty G = \bigcap_i \mathcal{C}^i G$. Thus we have

$$\mathcal{D}^i G = (\mathcal{D}^{i-1} G, \mathcal{D}^{i-1} G), \quad \mathcal{C}^i G = (G, \mathcal{C}^{i-1} G) \quad (i \geq 1)$$

with the convention that $G = \mathcal{D}^0 G = \mathcal{C}^0 G$, where (A, B) denotes the subgroup generated by the commutators $(a, b) = a \cdot b \cdot a^{-1} \cdot b^{-1}$ ($a \in A, b \in B$), A and B being two subsets of G . The identity element of G will be denoted by e or 1 . Let A_i ($1 \leq i \leq m$) be subsets of G . The map $(a_i) \mapsto a_1 \dots a_m$ from $A_1 \times \dots \times A_m$ into G will be called the *product map*.

0.3. By algebraic group over k , or defined over k , or k -algebraic group², we mean in this work a linear algebraic group defined over k , in the sense of [1], or, if one wishes to avoid recourse to the universal field, an algebraic subgroup over k .

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of the group $\mathbf{GL}_n$ of invertible matrices of order n , in the sense of [3]. We leave to the reader the choice of the framework in which they wish to work, or to consider the k -algebraic groups as absolutely reduced affine group schemes over k [11] if they wish to be more intrinsic.

Let G be a k -algebraic group. G^0 denotes the connected component of e in G . It is also defined over k . If H is a closed subset of G , defined over k , then $\mathcal{N}_G(H)$ and $\mathcal{Z}_G(H)$ are algebraic, k -closed (i.e., defined over a purely inseparable extension of k). We will denote by $\mathcal{N}_G(H)^0$ (resp. $\mathcal{Z}_G(H)^0$) the connected normalizer (resp. centralizer) of H in G .

For any extension K of k , G_K denotes the set of points of G which are rational over K (and similarly for an algebraic variety defined over k). In general, we will identify G and $G_{\bar{k}}$. Two subsets M, M' of G are *conjugate* (resp. *conjugate over k*) if there exists $g \in G$ (resp. $g \in G_k$) such that $M' = {}^g M$.

We will often write $\mathbf{G}_m$ for $\mathbf{GL}_1$, and $\mathbf{G}_a$ will denote the one-dimensional additive group. $\mathbf{GL}(\mathfrak{v})$ is the group of automorphisms of a vector space $\mathfrak{v}$.

The Lie algebra of an algebraic group $G, H, L, \dots$ will be denoted by the corresponding gothic letter $\mathfrak{g}, \mathfrak{h}, \mathfrak{l}, \dots$.

0.4. Let G, G' be k -algebraic groups. A *morphism* $f: G \rightarrow G'$ of *algebraic groups* is a morphism of the underlying varieties which is a group homomorphism. We will often omit “of algebraic groups” when this does not lead to confusion, and we will say *k -morphism* if f is defined over k .

We say that G *acts morphically* on the algebraic k -variety V if it acts on V via a morphism $\varphi: G \times V \rightarrow V$, and that it acts *k -morphically* if φ is defined over k . If V is a k -group, it will be understood that G acts as a group of automorphisms.

We denote by $\text{Ad } g$ the image of $g \in G$ under the adjoint representation $G \rightarrow \mathbf{GL}(\mathfrak{g})$. It is therefore the differential at e of $\text{Int } g$.

0.5. Let G be an algebraic group. A *character*, or *rational character*, of G is a morphism from G to $\mathbf{G}_m$. The set of characters, equipped with the law of composition defined by the product of values, is a finitely generated commutative group [23, §1], free if G is connected, denoted by $X^*(G)$, and whose subset of elements defined over k is denoted by $X^*(G)_k$. For characters, we will frequently use exponential notation, denoting by g^a the value at g of a character a ; the law of composition in $X^*(G)$ will then be written additively.

A *one-parameter multiplicative group* of G is a morphism $\mathbf{G}_m \rightarrow G$. If G is commutative, the set of one-parameter multiplicative groups, equipped with the law of composition defined by the product

²On this point, we therefore depart from the terminology of [1].

of values, is a free finitely generated commutative group, denoted by $X_*(G)$, whose subset of elements defined over k is denoted by $X_*(G)_k$.

A morphism $f : G \rightarrow G'$ of algebraic groups canonically induces morphisms $f^* : X^*(G') \rightarrow X^*(G)$ and $f_* : X_*(G) \rightarrow X_*(G')$.

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0.6. An algebraic group G is the *direct product* of closed normal subgroups $G_1, \dots, G_m$ if the product map $G_1 \times \dots \times G_m \rightarrow G$ is an isomorphism of algebraic groups; G is the *almost direct product* of the G_i if this map is an isogeny (i.e., surjective and with finite kernel).

G is the *semi-direct product* of the closed subgroup H and the closed normal subgroup N if the product map induces an isomorphism of algebraic varieties from $H \times N$ onto G . For this, it is necessary and sufficient that: (i) G is the semi-direct product of H and N as abstract groups; (ii) the Lie algebras of H and N are linearly independent. Recall that, in non-zero characteristic, (ii) does not follow from (i), even if G is commutative.

0.7. An algebraic k -group G is *semisimple* (resp. *reductive*) if the radical $R(G^0)$ (resp. unipotent radical $R_u(G^0)$) of G^0 is reduced to the identity element. Recall that the radical $R(H)$ (resp. unipotent radical $R_u(H)$) of a connected algebraic k -group H is the largest connected closed solvable (resp. unipotent) normal subgroup of H ; it is k -closed.

A connected algebraic k -group is *simple* (resp. *almost simple*) if every proper closed normal subgroup is reduced to $\{e\}$ (resp. finite), *k -simple* (resp. *almost k -simple*) if this condition is satisfied by the closed normal k -subgroups. A group $\neq \{e\}$ is semisimple if and only if it is the almost direct product of non-commutative almost simple groups [10, exp. 17, prop. 1, 2]. In particular, if G is semisimple, it is equal to its commutator subgroup.

0.8. Let G be an algebraic k -group. We will call a *Levi subgroup* of G any closed reductive subgroup H such that G is the semi-direct product of H and its unipotent radical $U = R_u(G)$, and a *Levi decomposition* such a presentation of G^3 . If k is of characteristic zero, G always possesses a k -Levi subgroup H , and every reductive k -subgroup of G is conjugate to a subgroup of H by an element of U_k (see [20] and also [5, prop. 5.1]). On the other hand, if k is of non-zero characteristic, G does not necessarily possess a Levi subgroup, according to Chevalley (unpublished), two Levi subgroups are not necessarily conjugate (3.15), and if L is a reductive subgroup such that G is set-theoretically the semi-direct product of L and U , the Lie algebras of L and U are not necessarily transverse (3.15).

0.9. As is customary, $\mathbf{Z}, \mathbf{N}, \mathbf{N}^+, \mathbf{Q}, \mathbf{R}, \mathbf{C}$ denote respectively the integers, the integers ≥ 0 , the integers > 0 , the rational numbers, the real numbers, and the complex numbers.

In all that follows in this work, G denotes an algebraic group, k a field of definition of G , and p the characteristic of k .

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I. REDUCTIVE GROUPS

§ 1. TORI

1.1. An algebraic group T is a *torus* if it is connected and satisfies the following equivalent conditions: (i) T consists of semisimple elements; (ii) T is diagonalizable; (iii) T is isomorphic to the product of n copies of $\mathbf{G}_m$ ($n = \dim T$).

³This is for brevity, and although this terminology is not in complete agreement with the usual one in the theory of Lie algebras, where a Levi subalgebra is a semisimple subalgebra, complementary to the radical.

A torus is *anisotropic over the field K* , or *K -anisotropic*, if it is defined over K and if $X^*(T)_K = 1$. It is *split* over K if it is defined over K and K -isomorphic to a product of groups $\mathbf{G}_m$. A K -torus is always split over $\bar{K}$.

1.2. Let T be a torus. The groups $X^*(T)$ and $X_*(T)$ are then free abelian, of rank equal to $\dim T$. If $a \in X_*(T)$, $b \in X^*(T)$, then $b \circ a \in X^*(\mathbf{G}_m)$, so $b \circ a$ is of the form $x \mapsto x^q$ ($x \in \mathbf{G}_m$; $q \in \mathbf{Z}$). By associating q to the pair (a, b) , we define on $X_*(T) \times X^*(T)$ an integer-valued bilinear form, denoted $\langle \cdot, \cdot \rangle$, which puts these two groups in duality [10, Exp. 9, § 5]. If f is a morphism from T into a torus T' , then

$$\langle f_*a, b \rangle = \langle a, f^*b \rangle \quad (a \in X_*(T), b \in X^*(T')).$$

If f is surjective, then f^* is injective. Since a subtorus of a torus is always a direct factor [1, 7.4, 7.5], an exact sequence of morphisms

$$1 \rightarrow T'' \xrightarrow{i} T \xrightarrow{p} T' \rightarrow 1$$

gives rise to exact sequences

$$\begin{aligned} 1 \rightarrow X_*(T'') \xrightarrow{i_*} X_*(T) \xrightarrow{p_*} X_*(T') \rightarrow 1 \\ 1 \rightarrow X^*(T') \xrightarrow{p^*} X^*(T) \xrightarrow{i^*} X^*(T'') \rightarrow 1. \end{aligned}$$

If ψ is a subset of $X^*(T)$, then T_ψ will denote the identity component of the intersection of the kernels of the characters $a \in \psi$.

1.3. Proposition. — *Let T be a k -torus. Then the following conditions are equivalent:*

- a) *The image of T under any k -morphism $T \rightarrow \mathbf{GL}_m$ ($m = 1, 2, \dots$) is diagonalizable over k .*
- b) *T is diagonalizable over k .*
- c) *T is split over k .*
- d) *We have $X^*(T) = X^*(T)_k$.*
- e) *We have $X_*(T) = X_*(T)_k$.*

It is clear that a) $\Rightarrow$ b). For b) $\Rightarrow$ c), see [1, Prop. 7.4]. Since the characters of $\mathbf{G}_m$ are of the form $x \mapsto x^q$ ($q \in \mathbf{Z}$), it is clear that c) $\Rightarrow$ d), e). If f is a morphism of T into $\mathbf{GL}_m$, the eigenvalues of $f(t)$ ($t \in T$) define characters of T , so d) $\Rightarrow$ a) follows from Prop. 5 of [23]. Finally, if e) is true, T is the image of $(\mathbf{G}_m)^n$ ($n = \dim T$) under a k -morphism, hence e) $\Rightarrow$ a) by virtue of the equivalence, already established, of the first four conditions, applied to $(\mathbf{G}_m)^n$.

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1.4. Proposition 1.3 implies that the image of a split torus over k under a k -morphism is a split torus over k , and that a k -torus T possesses a unique largest split subtorus over k , the group generated by the images of the k -one-parameter subgroups, which will be denoted by T_d . The intersection of the kernels of the elements of $X^*(T)_k$ will be denoted by T_a . We shall see below that T_a is the largest k -anisotropic subtorus of T and that the latter is the almost direct product of T_d and T_a .

Proposition 0.1 (1.5). *Let T be a k -torus. Then there exists a finite Galois extension of k over which T splits.*

This proposition is known [21, Prop. 1.2.1]. For the convenience of the reader, we indicate a proof of it, communicated by J. Tate.

Since $X^*(T)$ is of finite type, it suffices, in view of 1.3, to show that every element $a \in X^*(T)$ is defined over k_s . Now, T is diagonalizable over $\bar{k}$, so a is in any case defined over $\bar{k}$. Replacing k by k_s , we see that it suffices to prove that if a is defined over a purely inseparable extension of k , then it is defined over k .

There is nothing to prove if $p = 0$. Otherwise, let $q = p^s$ ($s \in \mathbf{Z}, s > 0$) be a power of p large enough so that a is defined over $k^{1/q}$. We have $a(t^q) = a^q(t) \in k(t)$, hence

$$a(t^q) \subset k(t) \cap k^{1/q}(t^q) \quad (t \in T).$$

But if t is generic over k , the field $k(t)$ is linearly disjoint from $\bar{k}$, hence $k(t) \cap k^{1/q}(t^q) = k(t^q)$, and $a(t^q) \subset k(t^q)$. The element t^q is also generic over k , since $x \mapsto x^q$ is a bijective morphism from T to itself; the inclusion $a(t^q) \subset k(t^q)$ then shows that a is defined over k .

Corollary 0.2 (1.6). *Every closed subgroup of T is defined over k_s , and is defined over k if T is split over k . Every morphism $f : T \rightarrow T'$ from T into a k -torus is a k_s -morphism, and a k -morphism if T and T' are split over k .*

The first assertion follows from the fact that T is diagonalizable over k_s (resp. over k) and that every closed subgroup of a diagonal torus is defined over the prime field [1, Prop. 7.4]. The second is a consequence of the first, applied to the graph of f in $T \times T'$.

1.7. The Galois group $\Gamma = \text{Gal}(k_s/k)$ operates in a natural way on $X^*(T)$ and $X_*(T)$. We shall often write γu for the transform by $\gamma \in \Gamma$ of an element u of $X^*(T)$ or $X_*(T)$. In particular, we have

$$\begin{aligned} \gamma u(t) &= \gamma(u(\gamma^{-1}(t))), & (u \in X^*(T), t \in T_{k_s}), \\ \langle \gamma u, v \rangle &= \langle u, \gamma^{-1} v \rangle, & (u \in X_*(T), v \in X^*(T)). \end{aligned}$$

$X^*(T)_k$ (resp. $X_*(T)_k$) is the set of fixed points of Γ in $X^*(T)$ (resp. $X_*(T)$). It is therefore a direct factor, which shows that the group T_a introduced in 1.4 is indeed connected. A subset of $X^*(T)$ will be said to be *defined over k* if it is invariant under Γ .

The operations of Γ on $X^*(T)$ and $X_*(T)$ are continuous (which here is equivalent to the fact that the orbits of Γ are finite). Since $X^*(T)$ and $X_*(T)$ are of finite type, there exists

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in Γ an open normal subgroup Γ' of finite index acting trivially on these groups. Then $K = (k_s)^{\Gamma'}$ is a finite Galois extension of k over which T splits. The action considered is therefore in fact that of a finite group $\Gamma/\Gamma' \cong \text{Gal}(K/k)$.

1.8. Proposition. — *Let T be a k -torus and S a k -subtorus. Then $X^*(T)_k$ and $X_*(T)_k$ have the same rank. There exists a k -subtorus S' of T such that T is the almost direct product of S and S' . If $S = T_d$, then S' is unique, equal to T_a , and T_a is k -anisotropic. If f is a k -morphism from T into a k -torus T' , then $f(T_d) \subset T'_d$, $f(T_a) \subset T'_a$. For the proof, see [4, §§ 8, 13].*

1.9. Corollary. — *a) Let T, T' be two k -tori isogenous over k . Then T is split over k if and only if T' is.*

b) Let $1 \rightarrow T' \rightarrow T \rightarrow T'' \rightarrow 1$ be an exact sequence of k -tori and k -morphisms. Then T is split over k if and only if T' and T'' are.

a) The hypothesis means that there exists a k -torus T'' and k -isogenies from T'' to T and T' . It therefore suffices to prove a) when there exists a k -isogeny $f : T \rightarrow T'$. In this case, f^* maps $X^*(T')$ (resp. $X^*(T')_k$) injectively onto a subgroup of finite index of $X^*(T)$ (resp. $X^*(T)_k$). Since $X^*(T)_k$ and $X^*(T')_k$ are direct factors in $X^*(T)$ and $X^*(T')$ by 1.7, it is then clear that the equalities $X^*(T)_k = X^*(T)$ and $X^*(T')_k = X^*(T')$ are equivalent, and a) follows from 1.3.

b) If T is split over k , the subtori and quotient tori of T are also split (1.4, 1.6), so T' and T'' are split over k . Conversely, if T'' is split over k , then $T_a \subset T'$ by 1.8; if moreover T' is split over k , then $T_a = \{e\}$ and $T = T_d$ by 1.8.

1.10. Proposition. — *Suppose k is infinite, and G is connected, and let T be a k -torus acting k -morphically on G . Then there exists an element $t \in T_k$ whose set of fixed points in G is equal to that of T . In particular, if T is contained in G and acts on it by inner automorphisms, $\mathcal{Z}_G(t) = \mathcal{Z}_G(T)$.*

Up to replacing G by its semi-direct product with T (relative to the given action of T on G), we can assume that $T \subset G$, and that T acts on G by inner automorphisms. Then, the set V of $t \in T$ such that $\mathcal{Z}_G(t) \neq \mathcal{Z}_G(T)$ is contained in the union of the kernels of a finite number of non-trivial characters of T [1, Prop. 7.10]; it is therefore a proper algebraic subset, and it suffices to know that T_k is Zariski-dense in T , which follows from the fact that T is unirational over k [23, Prop. 10].

§ 2. SPLIT GROUPS

2.1. Root systems. — Let M be a free module of finite type, N a direct factor submodule, Φ a finite subset of M , and P the submodule generated by Φ . We shall say that Φ is a root system in (M, N) if Φ is a root system in the usual sense [6, § 7] in $P \otimes \mathbf{Q}$, if $M \otimes \mathbf{Q}$ is the direct sum of $P \otimes \mathbf{Q}$ and $N \otimes \mathbf{Q}$, and if the Weyl group $W(\Phi)$ of Φ is the restriction of a group of automorphisms of $M \otimes \mathbf{Q}$.

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leaving M stable and acting trivially on N^4 . Let us equip $M \otimes \mathbf{R}$ with a positive non-degenerate and *admissible* scalar product $(\cdot, \cdot)$, i.e., invariant under $W(\Phi)$. If Φ is a root system in (M, N) , then: $0 \notin \Phi$, $\Phi = -\Phi$, Φ is invariant under $W(\Phi)$, the numbers $2(a, b) \cdot (b, b)^{-1} = n_{ab}$ are integers ($a, b \in \Phi$), and $W(\Phi)$ is generated by the reflections r_a with respect to the hyperplanes orthogonal to the roots, which are hyperplanes containing $N \otimes \mathbf{R}$. In particular, $N \otimes \mathbf{R}$ and $P \otimes \mathbf{R}$ are necessarily orthogonal.

A subset of Φ is *connected* if it is not the disjoint union of two non-empty subsets orthogonal to each other.

Let us recall that if $a = \lambda \cdot b$ ($a, b \in \Phi$, $\lambda \in \mathbf{R}$), then $\lambda = \pm 1, \pm(1/2), \pm 2$ [6, §7, n° 3, prop. 8]. The system Φ is said to be *reduced* if no root is double another root. A root a is *indivisible* (resp. *non-multipliable*) if $(1/2)a \notin \Phi$ (resp. $2a \notin \Phi$).

The *Weyl chambers* (connected components of the complement of the union of the hyperplanes orthogonal to the roots) are here the products by $N \otimes \mathbf{R}$ of the Weyl chambers in $P \otimes \mathbf{R}$. They are permuted in a simply transitive manner by $W(\Phi)$. Each determines a set Φ^+ of positive roots. We denote by Φ^- (resp. Δ) the set of negative (resp. simple) roots. Let us recall that a root is simple if it is positive and not the sum of two positive roots, and that every root is uniquely a linear combination of simple roots with integer coefficients all of the same sign. We have $r_a(b) = b - n_{ba} \cdot a$ ($a, b \in \Phi$). The group $W(\Phi)$ is generated by the reflections r_a ($a \in \Delta$), called *fundamental reflections*.

The *diagram* of Φ consists of the datum for every $a \in \Delta$ of the largest integer n_a such that $n_a \cdot a \in \Phi$, and for every pair a, b of simple roots, of the integers n_{ab}, n_{ba} . There exists a unique element $w \in W(\Phi)$ which maps Φ^+ onto Φ^- , hence also a unique automorphism i of Φ satisfying $w(i(a)) = -a$ ($a \in \Phi$). The automorphism i is involutive, leaves Φ^+ and Δ stable, and is called *the opposition involution*. Two subsets of Φ^+ mapped to one another by i are said to be *opposite*.

The following proposition is well known. Being unable to provide a reference, we give a proof of it for the convenience of the reader.

Proposition 0.3 (2.2). *Suppose G is connected. Then the following conditions are equivalent:*

- (i) G is reductive (i.e., $R(G)$ is a torus).
- (ii) G is an almost direct product of a torus and its derived group $\mathcal{D}G$, which is semi-simple.
- (iii) G possesses a completely reducible linear representation with finite kernel.

(ii) $\Rightarrow$ (iii). The group G is an almost direct product of almost simple groups $G_1, \dots, G_n$. It is therefore sufficient to take a direct sum of non-trivial irreducible representations of the quotients $G/(G_1 \dots G_{i-1} \cdot G_{i+1} \dots G_n)$.

(iii) $\Rightarrow$ (i). Let $\rho : G \rightarrow \mathrm{GL}(V)$ be a completely reducible representation and let $U = R_u(G)$. The space V_0 of fixed points under U is $\neq 0$ according to Lie's theorem—

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Kolchin, and is obviously stable under G . There exists therefore a complement V_1 of V_0 invariant under G , in which the representation of G is still completely reducible. Reasoning by induction on $\dim V$, we see that U acts trivially on V , hence the announced implication.

(i) $\Rightarrow$ (ii). Let $S = R(G)$. This is an invariant torus in G , hence central. The group G/S is semisimple, hence equal to its commutator subgroup, which implies that $G = S \cdot \mathcal{D}G$. It remains to show that $S \cap \mathcal{D}G$ is finite, which follows from the following lemma:

Lemma 0.4. *Let H be a connected algebraic group, and S a central torus of H . Then $S \cap \mathcal{D}H$ is finite.*

⁴This variant of the definition in [6] is adopted here to be able to speak conveniently of the root system of a reductive group, not necessarily semi-simple. Of course, the general properties of root systems proved in [6] remain valid, and we shall refer to them without further comment.

Proof. Identify H with a group of automorphisms of a vector space V . The latter is a direct sum of subspaces V_i stable under G and on which S acts by homotheties. The lemma then follows from the fact that any morphism $H \rightarrow \mathbf{GL}_m$ maps $\mathcal{D}H$ into $\mathbf{SL}_m$. $\square$

2.3. Reductive groups

We recall here some fundamental properties of reductive groups. The general reference for these results is [10].

Let G be a connected reductive group, T a maximal torus of G , and Z the connected center of G . The group G is therefore an almost direct product of Z and of $\mathcal{D}G$; consequently, $\mathcal{D}G$ is the largest semisimple subgroup of G and contains any unipotent subgroup of G . The set of roots of G with respect to T , i.e., of non-trivial characters of T in the adjoint representation of G , is denoted by $\Phi(T, G)$ or simply $\Phi(G)$, or Φ if this does not lead to confusion. We denote by U_b the one-parameter root subgroup associated with $b \in \Phi$. It is characterized by the existence of an isomorphism $\theta_b : \mathbf{G}_a \rightarrow U_b$ such that:

$$t \cdot \theta_b(x) \cdot t^{-1} = \theta_b(t^b \cdot x). \quad (t \in T, x \in \mathbf{G}_a)$$

Let T_b be the identity component of $\ker b$, $Z_b = \mathcal{Z}_G(T_b)$, and $Z'_b = \mathcal{D}Z_b$. It is known that Z_b is connected, reductive, with connected center T_b , that Z'_b is almost simple, of dimension three, generated by U_b and U_{-b} , and that $H_b = Z'_b \cap T$ is a maximal torus of Z'_b . We will also denote by h_b the one-parameter multiplicative subgroup of T whose image is $Z'_b \cap T$, and such that $\langle h_b, b \rangle = 2$.

Let $W = W(G) = \mathcal{N}(T)/\mathcal{Z}(T)$. This is a finite group, which acts faithfully on $X^*(T)$ and $X_*(T)$. The set of H_b ($b \in \Phi$) (resp. the set Φ) is a root system in $(X_*(T), X_*(Z))$ (resp. $(X^*(T), N)$), where N is the set of characters trivial on $(T \cap \mathcal{D}G)^0$, with Weyl group W . The reflection r_b comes from an element of Z'_b . It maps U_b onto U_{-b} , H_b onto H_{-b} , and acts trivially on T_b .

Fix an order on Φ ; let $U = U^+$ (resp. U^-) be the subgroup generated by the U_a ($a \in \Phi^+$) (resp. $a \in \Phi^-$), and $B = B^+ = T \cdot U$, $B^- = T \cdot U^-$. The group U is unipotent. For any order on Φ^+ , the product map from $\prod U_a$ ($a \in \Phi^+$) into U is an isomorphism of algebraic varieties. The subgroup B is a closed connected solvable maximal subgroup,

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or else, is a Borel subgroup of G^5 ; it is equal to its normalizer and to the normalizer of U , its derived subgroup is U and every Borel subgroup is conjugate to it. Similarly for B^- and U^- . The Borel subgroups of G containing T correspond bijectively (one-to-one) to the sets of positive elements for the different possible orders on Φ , and are permuted in a simply transitive manner by $W(G)$. The product map induces an isomorphism of varieties from $U^- \times B$ onto an open subset of G . The group G is generated by $(\mathcal{Z}(G))^0$ and the groups U_a, U_{-a} ($a \in \Delta$). Every subgroup of U normalized by T is closed, connected, and a multiple semi-direct product of the groups U_a that it contains [10, Exp. 13, th. 1 d].

Remark. — Let H be a connected algebraic group and S a maximal torus of H . Then there exist two closed connected unipotent subgroups U, U' normalized by S such that H is generated by S and U, U' , and such that every connected unipotent subgroup of H is contained in the subgroup generated by U and U' . Indeed, this is true if H is reductive by what precedes, and we reduce immediately to this case by considering the quotient $H/R_u(H)$, which is reductive, and by noting that the image of S in $H/R_u(H)$ is a maximal torus of it.

It follows in particular that every closed connected subgroup H of G normalized by T is generated by $(T \cap H)^0$ and the subgroups U_a that it contains, and is reductive if and only if $U_a \subset H$ implies $U_{-a} \subset H$ for any $a \in \Phi$. For example, the centralizer of a subset of a torus of G is reductive.

2.4. Bruhat Decomposition. — For every element $w \in W(G)$, let us choose a representative n_w in $\mathcal{N}(T)$. On the other hand, let U'_w (resp. U''_w) be the subgroup of U generated by the U_a ($a \in \Phi^+$, $w^{-1}(a) \in \Phi^-$) (resp. $a \in \Phi^+$, $w^{-1}(a) \in \Phi^+$). Then the product map defines an isomorphism of varieties from $U'_w \times U''_w$ onto U and from $U'_w \times n_w \times B$ onto $B.n_w.B = U.n_w.B = U'_w.n_w.B$. The group G is the disjoint union of the double cosets $B.n_w.B$ ($w \in W(G)$). Two double cosets $U.n.U$ and $U.n'.U$ ($n, n' \in \mathcal{N}(T)$) are equal only if $n = n'^6$.

This implies in particular that if B' is a Borel subgroup of G , then $B' \cap B$ contains a maximal torus of G . Indeed, since $G = B.\mathcal{N}(T).B$, one can find $x \in B$ and $n \in \mathcal{N}(T)$ such that ${}^x B' = {}^n B$, whence

$${}^x (B' \cap B) = {}^x B' \cap B = {}^n B \cap B \supset T,$$

and $B' \cap B \supset x^{-1}.T.x$.

The coset $n_w.T$ of T in $\mathcal{N}(T)$ obviously depends only on w . In the following, we will often agree to denote it by $w.T$. More generally, if A is a subset of $\mathcal{N}(T)$, the set $A.T$ depends only on the image C of A in W and 669

⁵One of the authors insisting that we adopt this terminology, now generally accepted, the other author resigns himself to it.

⁶In fact, this last property does not seem to appear explicitly in [10]. It is proved from the preceding ones exactly like the corresponding assertion in 5.15, of which it is, moreover, a special case.

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will often be denoted $C.T$. Similarly, if F is a subset of G normalized by T , the set ${}^n_w F$ will also be denoted ${}^w F$.

Proposition 0.5 (2.5). *We preserve the preceding notation. Let $a, b \in \Phi(G)$ be non-proportional. Let $U_{(a,b)}$ be the subgroup of G generated by the groups U_c , where c runs over the set (a, b) of roots of the form $i.a + j.b$ ($i, j > 0$). Then $U_{(a,b)}$ is unipotent, contains no other one-parameter root group, and we have $(U_a, U_b) \subset U_{(a,b)}$.*

This proposition results from facts established in [9, 10], and we limit ourselves to sketching its proof. One can first fix an order on Φ such that $a, b \in \Phi^+$ [9, Lemma 1]. Suppose $a < b$. It follows from [10, Exp. 17, Cor. to Th. 1] that U_a, U_b normalize $U_{(a,b)}$ and that if we arrange the U_c ($c \in (a, b)$) in increasing order, the product map from $U_a \times U_b \times \prod_{c \in (a,b)} U_c$ into U is an isomorphism of varieties from this product onto the group generated by U_a, U_b and $U_{(a,b)}$. Furthermore, Lemma 1 of [10, Exp. 17] shows that if $x \in U_b$ and $y \in U_a$,

$$x.y \in U_a.U_b.U_{(a,b)},$$

and that the images of the components of $x.y$ in U_a and U_b under θ_a^{-1} and θ_b^{-1} are of the form $c_a.\theta_a^{-1}(x)$ and $c_b.\theta_b^{-1}(y)$ where c_a, c_b are constants. Setting $y = 1$ or $x = 1$, we see that $c_a = c_b = 1$, whence

$$x.y \in y.x.U_{(a,b)},$$

which completes the proof.

Remark. — One will find in [9, p. 27] a much more precise formula, established in characteristic zero, but valid in general in view of the classification theorems of [10]. If G is almost simple, this formula shows that (U_a, U_b) can be a proper subgroup of $U_{(a,b)}$ only in the following cases:

$$p = 2, G = \mathbf{B}_n, \mathbf{C}_n, \mathbf{G}_2, \mathbf{F}_4; \quad p = 3, G = \mathbf{G}_2.$$

For convenience of reference, we insert here a known elementary lemma.

Lemma 0.6 (2.6). *Suppose G is connected and let B be a closed k -subgroup of G , such that the fibration of G by B admits a local section defined over k . If k is finite or G_k is Zariski dense in G , then the canonical projection $G_k \rightarrow (G/B)_k$ is surjective.*

(Recall that a local section over k of the fibration of an algebraic k -group H by a k -subgroup B is a k -morphism σ from an open set V defined over k of H/B into H such that $\pi \circ \sigma = 1$, where $\pi : H \rightarrow H/B$ denotes the canonical projection. The product map then defines a k -isomorphism of $\sigma(V) \times B$ onto an open set of H . In particular, B and V are irreducible if H is.)

If G_k is Zariski dense in G , the translates of the domain of definition V of a local section over k by the elements of G_k form a covering of G/B , whence the

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... surjectivity of $G_k \rightarrow (G/B)_k$. If k is finite, the latter follows from [16], since B is connected.

2.7. Split solvable groups. — A solvable algebraic k -group H is said to be *split over k* or *k -split* or *k -solvable* if it is connected and possesses a composition series $H = H_0 \supset H_1 \supset \dots \supset H_t = \{e\}$ formed of connected k -subgroups whose successive quotients are k -isomorphic to $\mathbf{G}_a$ or $\mathbf{G}_m$ (this is the “ k -solvable” of [26]). A connected solvable k -group is always split over $\bar{k}$. For a torus, this notion indeed coincides with that introduced in 1.1, in view of 1.9 b). Any extension over k of a k -solvable group by a k -solvable group is k -solvable. Any image of a k -solvable group under a k -morphism is k -solvable [23, Prop. 6]. Let us recall two more fundamental properties of a k -solvable group G :

- (i) If G acts k -morphically on a complete variety V and if $V_k \neq \emptyset$, then V_k contains a fixed point under G [23, p. 35].
- (ii) Every homogeneous k -space of G possesses a rational point over k [22, p. 425].

In particular, if G is a k -subgroup of an algebraic k -group L , then the projection $L_k \rightarrow (L/G)_k$ is surjective.

2.8. Split reductive groups. — Suppose G is reductive, and let T, θ_a, U_a be as in 2.3. Then $\{T, \theta_a\}$ or, by abuse of language, $\{T, U_a\}$ ($a \in \Phi$) is a *splitting datum over k* if T is split over k and if the θ_a are defined over k . The group G is *split over k* if it possesses a splitting datum over k ; it is always split over $\bar{k}$. According to an (unpublished) result of Cartier (which will not be used here), every maximal torus split over k of G is part of a splitting datum (see also [11]). It follows in particular that G is split over k if and only if it possesses a Borel subgroup split over k . One could therefore take this latter condition as the definition of the notion of an algebraic group split over k . However, we will not need it outside of the reductive and solvable cases.

In Lemmas 2.9, 2.10, and 2.11, the group G is reductive, split over k , and the notations of 2.3 refer to a splitting datum over k .

Lemma 0.7 (2.9). *Let V be a subgroup of U normalized by T . Let $t \in T_k$ and $v \in V$ be such that $v.t.v^{-1} \in B_k$. Then $v.t.v^{-1}$ is conjugate to t by an element of V_k .*

Proof. Let $a_1, \dots, a_m$ be the positive roots such that $U_{a_i} \subset V$ arranged in increasing order, and V_i the subgroup generated by the groups U_{a_j} ($1 \leq i \leq j \leq m$). Theorem 1 d) of [10, Exp. 13] shows that the product map is a k -isomorphism of varieties from the product of the U_{a_j} ($j \geq i$) to V_i and, together with 2.5, proves that V_i is normal in V and is the semi-direct product of U_{a_i} with V_{i+1} ($i = 1, \dots, m$; we set $V_{m+1} = \{e\}$).

The proof proceeds by downward induction on the largest integer $r \leq m+1$ such that $v \in V_r$, our assertion being obvious if $r = m+1$. Let us set

$$v = v_r.v', \quad v' = v''.v_r \quad (v', v'' \in V_{r+1}; v_r \in U_{a_r}),$$

□

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and let $x = \theta_{a_r}^{-1}(v_r) \in \mathbf{G}_a$. If $t^{a_r} = 1$, then t centralizes v_r , so $v.t.v^{-1} = v''.t.v''^{-1}$ and we can apply the induction hypothesis. Suppose then $t^{a_r} = c^{-1} \neq 1$. We have

$$t^{-1}.v.t.v^{-1} \in \theta_{a_r}((c-1).x) \cdot V_{r+1},$$

whence $(c-1).x \in k$, so $x \in k$, and $v_r \in U_k$. Consequently,

$$v'.t.v'^{-1} = v_r^{-1}.(v.t.v^{-1}).v_r \in G_k.$$

By virtue of the induction hypothesis, t and $v'.t.v'^{-1}$ are conjugate by an element of V_k ; the same is then true for t and $v.t.v^{-1} = v_r.(v'.t.v'^{-1}).v_r^{-1}$.

The following two lemmas are special cases of propositions that will be established later (nos. 4.21, 5.15, 11.4).

Lemma 0.8 (2.10). *Let V be as in 2.9 and $C = T.V$. Any torus S defined over k of C (resp. split over k of G) is conjugate by an element of C_k (resp. G_k) to a subtorus of T .*

Proof. The torus S has a fixed point in $(G/B)_k$ according to 2.7 (i). In view of 2.3, 2.6, S is therefore conjugate over k to a subtorus of B , and it suffices to prove the assertion concerning C . Suppose k is infinite. There then exists $s \in S_k$ such that $\mathcal{Z}_C(s) = \mathcal{Z}_C(S)$ (1.10). We have the decomposition $s = t.v$ ($t \in T$, $v \in V$) and, since $s \in C_k$ and C is a semi-direct product over k of T and V , we also have $t, v \in C_k$. The elements t and s are conjugate in $V_{\bar{k}}$ [1, §12], hence (2.9) also in V_k , and we can reduce to the case where $s = t$; but then $\mathcal{Z}_C(S) = \mathcal{Z}_C(s) \supset T$, so $S \subset T$. Now let k be finite. The centralizer of S is defined over k , so [23, footnote p. 45] contains a maximal torus T' defined over k . Since $T = \mathcal{Z}_C(T) = \mathcal{N}_C(T)$, the set of elements $v \in V$ such that $v.T.v^{-1} = T'$ is a homogeneous space of T , obviously defined over k , and non-empty [1, §12]. It then contains a rational point over k (2.7 (ii) or [16]). □

Lemma 0.9 (2.11). *Let $N = \mathcal{N}_G(T)$. Then $N = N_k \cdot T$ and $G_k = U_k \cdot N_k \cdot U_k$; two double cosets $U_k \cdot n \cdot U_k$, $U_k \cdot n' \cdot U_k$ corresponding to distinct elements n, n' of N_k are distinct.*

Proof. This lemma is known when k is algebraically closed (2.4). It therefore suffices to prove the first two equalities.

Let $n \in N$. The group ${}^n B$ contains T , so (2.3) is split over k and (2.7 (i)) has a fixed point in $(G/B)_k$. In view of 2.6, 2.10, there then exists $g \in G_k$ such that ${}^g n B = B$ and ${}^g T = T$, whence $g \in N_k$ and $g.n \in B \cap N = T$.

Let $g \in G$. According to the lemma over $\bar{k}$, there exists $n \in N$ such that $g \in U.n.U$. Let w be the image of n in $W(G)$, n' a representative of w in N_k , and V (resp. V') the product of the groups U_a where $a \in \Phi^+$, $w(a) > 0$, (resp. $w(a) < 0$), taken in any order and let $Y = {}^{n'} V' \subset U^-$. We have

$$g.n'^{-1} \in U.T.n'.U.n'^{-1} = U.T.n'.V.V'.n'^{-1} = U.T.Y,$$

□

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whence, in view of 2.3, and the equality $n.V'.n^{-1} = n'.V'.n'^{-1}$,

$$\begin{aligned} g.n'^{-1} &\in (U.T.Y)_k = U_k.T_k.Y_k, \\ g &\in U_k.T_k.n'.U_k \subset U_k.N_k.U_k. \end{aligned}$$

Lemma 0.10 (2.12). *Let G, G' be two isomorphic connected reductive groups, T, T' respective maximal tori of G and G' , α an isomorphism of T onto T' which maps $\Phi(G)$ onto $\Phi(G')$. Let ψ be a set of linearly independent roots of G and $\beta_b : U_b \rightarrow U_{b'}$ ($b \in \psi$, $b' = {}^t\alpha^{-1}(b)$) an isomorphism. Then there exists an isomorphism γ of G onto G' whose restrictions to T and to U_b are respectively α and β_b ($b \in \psi$).*

Proof. By [10, Exp. 24, Cor. 1] we can, without loss of generality, identify G with G' and assume that α is the identity. Let $K \supset k$ be an algebraically closed field of definition for G' and the groups U_b . A K -isomorphism of U_b onto $\mathbf{G}_a$ transforms β_b into an automorphism of $\mathbf{G}_a$, that is, into multiplication by an element $s_b \in K^*$. Since the elements of ψ are independent, there exists $t \in T_K$ such that $t^b = s_b$ ($b \in \psi$). It then suffices to take $\gamma = \text{Int } t$. $\square$

Theorem 0.11 (2.13). *Two connected reductive groups G, G' split over k and isomorphic are isomorphic over k .*

Proof. Let (T, U_a) and $(T', U'_{a'})$ be splitting data over k for G and G' , and K an algebraically closed extension of k over which G and G' are isomorphic. By 2.3 and the conjugacy of maximal tori, there exists a K -isomorphism η of G onto G' which maps (T, U_a) onto $(T', U'_{a'})$. Fix in $\Phi(G)$ and $\Phi(G')$ orders that correspond to each other under ${}^t\eta|_T$. Let U and U' (resp. U^- and U'^-) be the unipotent subgroups of G and G' generated by the groups U_a and $U'_{a'}$ corresponding to positive (resp. negative) roots. For any simple root $a \in \Delta$, let $\beta_a : U_a \rightarrow U'_{a'}$ be a k -isomorphism ($a' = {}^t\eta|_T(a)$) and u_a an element different from 1 of $U_{a,k}$. By 2.11, there exists a unique element $n_a \in \mathcal{N}(T)$ such that $u_a \in U^- \cdot n_a \cdot U^-$, and this element belongs to $\mathcal{N}(T)_k$. Applying this same lemma and 2.4 to the centralizer Z_a of the identity component of the kernel of a (cf. 2.3), we see that the image of n_a in $W(G)$ is the fundamental reflection r_a ; consequently, the subgroup M generated by the n_a is a subgroup of $\mathcal{N}(T)_k$ which maps onto $W(G)$ under the canonical projection.

By 2.12, there exists a K -isomorphism $\gamma : G \rightarrow G'$ which extends $\eta|_T$ and the β_a ($a \in \Delta$). We have $\gamma(u_a) = \beta_a(u_a) \in U'^- \cdot \gamma(n_a) \cdot U'^-$. Since $\beta_a(u_a) \in G'_k$ by hypothesis, 2.11 implies that $\gamma(n_a)$ is also rational over k , so $\gamma(M) = M' \subset G'_k$. Now let $c \in \Phi(G)$. Since M contains at least one representative of each element of $W(G)$, there exist $m \in M$ and $a \in \Delta$ such that $U_c = m \cdot U_a \cdot m^{-1}$. The restriction of γ to U_c is therefore equal to $\text{Int } \gamma(m) \cdot \beta_a \cdot \text{Int } m^{-1}$, hence is defined over k since β_a is and $m, \gamma(m)$ are rational over k . Since the restriction of η to T is defined over k (1.6), it follows that the restriction of γ to the open set $U^- \cdot T \cdot U$ is defined over k , whence the theorem. $\square$

Remark. — Theorem 2.13 is a very special case of a result of Demazure [11].

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2.14. Theorem (Grothendieck [11]). — *Assume G is connected. Then:*

- a) *G has a maximal torus defined over k [11, Exp. XIV, Theor. 1.1].*
- b) *If G is reductive, G splits over a finite separable extension of k .*
- c) *If G is reductive and k is infinite, then G_k is Zariski-dense in G [11, Exp. XIV, 6.5, 6.7].*

When k is perfect, c), for a group not necessarily reductive, and a) are due to Rosenlicht [23]. The same holds for a) if G is solvable [26, Theorem 4] (see also 11.4).

2.15. We indicate here some immediate consequences of 2.14.

- a) *If G is reductive, its center is defined over k .*

Indeed, $\mathcal{Z}(G)$ is k -closed, contained in every maximal torus of G , and therefore in a maximal torus T defined over k by 2.14 a), and is consequently also defined over k_s (1.6).

- b) *Assume G is semi-simple and let N be a connected normal k -subgroup of G . Then N is the almost direct product of the normal k -subgroups of G almost simple over k which it contains, and G is the almost direct product of N and a connected normal k -subgroup N' .*

Indeed, by 2.14 b), the almost simple subgroups of G over $\bar{k}$ are defined over k_s ; they are therefore permuted by the Galois group Γ of k_s/k , and a connected normal subgroup of G is almost simple over k if and only if it is the almost direct product of almost simple factors of G over $\bar{k}$ forming an orbit of Γ .

- c) *Assume G is semi-simple. There exist a k -group G_1 and a k -isogeny $\nu : G_1 \rightarrow G$ determined up to isomorphism by the following conditions: G_1 is the direct product of its normal k -subgroups almost simple over k and ν induces $\bar{k}$ -isomorphisms of these onto the almost simple factors of G .*

Since the almost simple factors of G are defined over k_s , by 2.14 b), our assertion is immediate over k_s . That G_1 and ν are then defined over k follows immediately from base field descent criteria [37].

- d) Assume G is reductive. Let S be a k -torus of G and A a subset of S_k . Then $\mathcal{Z}(A)^0$ and $\mathcal{Z}(S)$ are reductive, defined over k , and S is contained in a maximal torus of G defined over k ⁽¹⁾.

$\mathcal{Z}(A)^0$ and $\mathcal{Z}(S)$ are k -closed. It therefore suffices to carry out the proof when k is separably closed. By 2.14 and 2.10 we can then assume that $S \subset T$, where T is part of a splitting datum of G over k . Since $\mathcal{Z}(S)$ is connected [1, prop. 18.4], it then follows from 2.3 that $\mathcal{Z}(S)$ and $\mathcal{Z}(A)^0$ are generated by T and the groups U_a , where a runs through the roots of G relative to T equal to 1 on S and A respectively. They are defined over k , and also reductive (2.3, remark).

- e) If G possesses a Levi decomposition over k (0.8), then its radical is defined over k .

Let $G = H \cdot R_u(G)$ be a Levi decomposition of G over k , and let Z be the connected center of H . It is defined over k by a), and is the radical of H , hence $R(G) = Z \cdot R_u(H)$ is defined over k .

(¹) For a more direct proof of a more general result, see 10.3, 10.5.

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GROUPES RÉDUCTIFS

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§ 3. ROOTS AND SUBGROUPS NORMALIZED BY A MAXIMAL TORUS

Throughout this section, G denotes a connected reductive group.

3.0. Let M be a free $\mathbf{Z}$ -module of finite type, Φ a finite set of non-zero elements of M , and ψ, η subsets of Φ . We say that ψ normalizes η (in Φ) if, for any $m, n \in \mathbf{N}^+$, $a_1, \dots, a_m \in \psi$, and $b \in \eta$:

$$a = a_1 + a_2 + \dots + a_m + nb \in \Phi \implies a \in \eta. \quad (*)$$

If in addition $\psi \supset \eta$ (resp. $\psi = \eta$), we say that η is an *ideal* of ψ (resp. is *closed* in Φ). The set ψ is *symmetric* if $\psi = -\psi$; *convex* if it contains any linear combination with positive rational coefficients of its elements that belongs to Φ . A convex subset of Φ is obviously closed in Φ .

Recall that if Φ is a root system (2.1), condition (*) is equivalent to this same condition where we set $m = n = 1$. This follows easily from Lemma 19 of [6, § 7, no. 6]. Suppose Φ is symmetric. We set $\psi_s = \psi \cap (-\psi)$, $\psi_u = \psi \cap \mathbb{C}(-\psi)$. These are disjoint subsets of ψ whose union is ψ . If $\Phi = -\Phi$ and if ψ is closed in Φ , then ψ_s and ψ_u are closed and ψ_u is an ideal of ψ . This is clear for ψ_s . If ψ_u were not an ideal, one could find $n \in \mathbf{N}^+$, $a_1, \dots, a_m \in \psi$, and $b \in \psi_u$ such that $a = a_1 + \dots + a_m + nb \in \psi_s$, but then $-b = -a + a_1 + \dots + a_m + (n-1)b \in \psi$, whence $b \in \psi_s$. An intersection of closed subsets of Φ is closed in Φ , which allows us to speak of the smallest closed subset of Φ containing η , which we will call the *closure* of η in Φ .

3.1. Proposition. — Let A be a subring of $\mathbf{R}$ and $0 \rightarrow M' \rightarrow M \xrightarrow{\pi} M'' \rightarrow 0$ an exact sequence of free A -modules of finite type. Then:

- (i) There exists on M one and only one total order compatible with a given total order on M'' and inducing a given total order on M' .
- (ii) Given a finite set Φ of elements of M of which all A -linear combinations with coefficients > 0 are $\neq 0$, there exists a total order on M with respect to which the elements of Φ are > 0 .

Proof. (i) We set $a > b$ if either $\pi(a) > \pi(b)$, or $\pi(a) = \pi(b)$ and $a - b > 0$ for the given order on M' .

(ii) Up to replacing A by its field of fractions, we may assume that A is a field. We proceed by induction on the dimension of M . We may restrict to the case where no element of Φ is a linear combination with coefficients > 0 of other elements of Φ . Let M' be the subspace generated by an element $a \in \Phi$ and π the projection of M onto $M'' = M/M'$. Let Φ'' be the set of non-zero elements of $\pi(\Phi)$. In view of the hypotheses on Φ , it also satisfies the condition imposed on Φ in (ii), whence the existence of a total order on M'' such that Φ'' consists of positive elements. We then take on M the order compatible with the order constructed on M'' and for which $a > 0$. $\square$

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3.2. Roots with respect to a torus. — Let S be a torus acting morphically on G . The weights of S in G are the characters occurring in the representation of S induced on $\mathfrak{g}$. If S is a subtorus of G acting by inner automorphisms, the roots of G relative to S are the non-zero weights of S in G . Their set is denoted by $\Phi(S, G)$, or simply Φ . If S' is a subtorus of S , then $\Phi(S', G)$ is the set of non-zero elements of the image of $\Phi(S, G)$ under restriction. If S is a subtorus of G , and H a closed subgroup of G normalized by S , the weights of S in H are the weights associated with the action of S by inner automorphisms. The non-zero weights of S in H are therefore the roots of G relative to S whose eigenspace has a non-zero intersection with $\mathfrak{h}$.

If S is a maximal torus, the non-zero weights of S in H are also the roots $a \in \Phi(S, G)$ such that $U_a \subset H$. Indeed, if $U_a \subset H$, the Lie algebra $\mathfrak{u}_a$ of U_a is contained in the Lie algebra $\mathfrak{h}$ of H , and so a is a non-zero weight of S in H . The converse follows from 2.3.

In the following lemma, which generalizes Proposition 1, p. 13-02 of [10], we place ourselves in the category of groups with operators with a fixed operator domain P . In particular, all subgroups considered are stable under P , and the homomorphisms commute with P . Recall that by a subquotient of a group, we mean any homomorphic image of a subgroup of this group.

Lemma 0.12 (3.3). *Let H be a nilpotent group (with operators); let $H_1, \dots, H_m$ be subgroups of H with no isomorphic subquotients. Then the following conditions are equivalent:*

- (i) *For every subgroup $A \subset H$, the product map is a bijection from $(A \cap H_1) \times \dots \times (A \cap H_m)$ onto A .*
- (ii) *H possesses a composition series with commutative quotients, each term of which is generated by its intersections with the groups H_i .*
- (iii) *H possesses a composition series $H = L_0 \supset L_1 \supset \dots \supset L_n = \{e\}$ having the property:*
 - a) *for all $i \leq n$, there exists $j(i) \leq m$ such that $L_{i-1} = L_i \cdot (L_{i-1} \cap H_{j(i)})$.*
- (iv) *H possesses a central normal series (L_i) (i.e., L_{i-1}/L_i is in the center of H/L_i for $i = 1, \dots, n$) having the property a).*

Before proceeding to the proof, let us first observe that:

- b) Any composition series (L_i) with commutative quotients, whose terms are generated by their intersections with the H_i , possesses a refinement satisfying a).

To obtain such a refinement, it indeed suffices to insert between L_{i-1} and L_i the subgroups $L_i \cdot (L_{i-1} \cap H_1) \cdots (L_{i-1} \cap H_j)$ ($j = 1, 2, \dots, m-1$). (These are indeed normal subgroups of L_{i-1} since L_{i-1}/L_i is commutative, and they satisfy a) since, by hypothesis, L_{i-1}/L_i is the product of the canonical images of the groups $L_{i-1} \cap H_j$).

(iv) $\Rightarrow$ (i). Let $L_{n-1} = L$. If $n = 1$, there is nothing to prove. If $n = 2$, we can obviously restrict to the case where H_1 and H_2 are non-trivial. Then a) implies that $H = H_1 \times H_2$ and that H_1, H_2 are commutative. Up to quotienting H by $(H_1 \cap A) \times (H_2 \cap A)$,

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one can assume that $H_1 \cap A = H_2 \cap A = \{e\}$. But then the projections of A onto H_1 and H_2 are isomorphic, hence reduced to the identity element.

In the general case, we proceed by induction on n . By applying the induction hypothesis to the image of A in H/L under the canonical projection, we see that

$$A \subset (A \cap (H_1 \cdot L)) \cdot (A \cap (H_2 \cdot L)) \cdots (A \cap (H_m \cdot L)). \quad (1)$$

But, by virtue of a), L is contained in one of the groups H_i , say in H_s , and we have $H_i \cdot L = H_i \times L$ or $H_i \cdot L = H_i$ depending on whether $i \neq s$ or $i = s$, whence again, taking into account what has already been proven:

$$A \cap (H_i \cdot L) \subset (A \cap H_i) \cdot (A \cap L).$$

Since $A \cap L \subset A \cap H_s$, the inclusion (1) becomes

$$A \subset (A \cap H_1) \cdot (A \cap H_2) \cdots (A \cap H_m). \quad (2)$$

Finally, let

$$a_1 \cdot a_2 \cdots a_m = b_1 \cdot b_2 \cdots b_m \quad (a_i, b_i \in A \cap H_i; i = 1, \dots, m).$$

The induction hypothesis, applied to the image of A in H/L , gives $a_i \in b_i \cdot L$, hence $a_i \in b_i \cdot H_s$ for all i , from which $a_i = b_i$ if $i \neq s$, and finally also $a_s = b_s$.

(i) $\Rightarrow$ (ii) $\Rightarrow$ (iii). This follows from the fact that H has at least one composition series with abelian quotients, and from b).

(iii) $\Rightarrow$ (iv). We proceed by induction on n , the case $n = 1$ being trivial. In view of b) it suffices to prove that the terms of the upper central series are generated by their intersections with the groups H_i . By an obvious induction on the length of this series, we see that it suffices to show that the center Z of H is generated by the groups $H_i \cap Z$.

According to a), there exists s such that $H = L_1 \cdot H_s$. Let $B = Z \cdot H_s$. The induction hypothesis, applied to L_1 and to the groups $L_1 \cap H_i$, and the implication (iv) $\Rightarrow$ (i) already established, show that $B \cap L_1$ is generated by its intersections with the H_i . It is therefore the same for $B = H_s \cdot (B \cap L_1)$. A central series of B consisting of B , H_s and a central series of H_s is then formed of groups that are generated by their intersections with the H_i , so B satisfies (iv) in view of b). It then follows from the implication (iv) $\Rightarrow$ (i) that every subgroup of B , in particular Z , is generated by its intersections with the H_i .

Remarks. — 1) the conditions of the preceding lemma are also equivalent to:

(v) Every composition series of G has a refinement satisfying a).

Indeed, (v) $\Rightarrow$ (iii) and, taking into account b), (i) $\Rightarrow$ (v).

2) The group of affine transformations $x \mapsto a \cdot x + b$ of the line (without operators), and the subgroups H_1 of homotheties and H_2 of translations provide an example of a non-nilpotent group and of subgroups satisfying (iii), but not (i).

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Proposition 0.13 (3.4). *Let T be a maximal torus of G , H a closed connected subgroup normalized by T , and ψ the set of roots $a \in \Phi(T, G)$ such that $U_a \subset H$. Let H^* be the group generated by the U_a ($a \in \psi$). Then:*

a) $H = H^* \cdot (T \cap H)^0$.

b) If η is a closed subset of Φ contained in ψ and if H^* is generated by the groups U_a with $a \in \eta$, then $\eta = \psi$.

Proof. a) Let B be a Borel subgroup of H , S a maximal torus of B and U the largest unipotent subgroup of B . The group B is contained in a Borel subgroup B' of $T \cdot H$ and, by conjugation in $T \cdot H$, which leaves H stable, we can assume that $S \subset T$ [1, Cor. 16.6]. On the other hand, we have $B' = T \cdot U$. It follows in particular that H and $T \cdot H$ have the same unipotent subgroups. By the remark in 2.3, the subgroup generated by the unipotent subgroups of $T \cdot H$ is already generated by two maximal unipotent subgroups normalized by T . By 2.3, these latter are generated by the subgroups U_a they contain, thus the subgroup generated by the unipotent subgroups of H is none other than H^* . This implies that $H^* \cdot (T \cap H)^0$ is a subgroup of H which contains a maximal torus S of H and all the unipotent subgroups of H ; it is therefore equal to H (2.3, rem.).

b) Let $\eta_s = \eta \cap (-\eta)$, $\eta_u = \eta \cap C(-\eta)$. These are closed subsets of Φ (3.0). It is known [6, § 7, Prop. 22] that we can fix an ordering on Φ such that $\eta_u \subset \Phi^+$. Let $\eta^\pm = \eta \cap \Phi^\pm$ and $V^\pm$ be the group generated by the U_a ($a \in \eta^\pm$). The set $\eta^\pm$ is also closed in Φ , and 2.5 shows that $V^\pm$ does not contain any group U_b with $b \notin \eta^\pm$.

The set η_s , being closed and symmetric, is also a root system. Let us set $W' = W(\eta_s)$ and show that $M = V^+ \cdot W' \cdot T \cdot V^-$ is a group (we use the convention of No. 2.4). By 2.3, it is generated by T and the groups U_a , where a ranges over η^+ and the opposites of the simple roots of η_s , for the fixed ordering. It therefore suffices to show that M is invariant under left translation by T and by the aforementioned U_a . This is obvious for T , and for U_a if $a \in \eta^+$. Let a then be a simple root of η_s . Proposition 2.5 shows that the subgroup Y of V^+ generated by the U_b ($b \in \eta^+, b \neq a$) is normalized by U_a, U_{-a} , hence in particular that $V^+ = U_a \cdot Y = Y \cdot U_a$, whence

$$U_{-a} \cdot M \subset Y \cdot U_{-a} \cdot U_a \cdot W' \cdot T \cdot V^-.$$

The Bruhat decomposition (2.4) of Z_a gives

$$U_{-a} \cdot U_a \subset T \cdot U_a \cup U_a \cdot r_a \cdot T \cdot U_a, \quad (1)$$

r_a being the fundamental reflection associated to a , whence

$$U_{-a} \cdot V^+ \cdot w' \cdot T \cdot V^- \subset M \cup V^+ \cdot U_{-a} \cdot w \cdot T \cdot V^- \quad (w' \in W'; w = r_a \cdot w'). \quad (2)$$

If $w^{-1}(a) > 0$, then $w^{-1}(-a) \in \eta^-$, so $U_{-a} \cdot w \cdot T \cdot V^- \subset w \cdot T \cdot V^-$ and the last term of (2) is contained in M . Let then $b = w^{-1}(-a) > 0$. We have

$$U_{-a} \cdot w \cdot T \cdot V^- = w \cdot T \cdot U_b \cdot V^-. \quad (3)$$

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Formula (1), with a replaced by $-b$, gives

$$T.U_b.V^- \subset T.V^- \cdot U_{-b} \cdot r_b \cdot T.V^-, \quad (4)$$

whence, by (3),

$$U_{-a}.w.T.V^- \subset w.T.V^- \cup U_{-a}.w.r_b.T.V^-,$$

and, by (2), $U_{-a}M \subset M$.

The set M is therefore indeed a group. The Bruhat decomposition of the quotient of M by its unipotent radical then shows that this quotient has η_s as system of roots. Since V^+ and V^- contain no U_b ($b \notin \eta$), the same holds for M , hence $\eta = \psi$.

Lemma 0.14 (3.5). *Let S be a torus of G , T a maximal torus of G containing S , and H a closed subgroup of G containing $\mathcal{Z}(S)$. Then the subgroup H^* of H generated by the U_a ($a \in \Phi(T, G)$) contained in H , but not in $\mathcal{Z}(S)$, is normalized by $\mathcal{Z}(S)$.*

Let a, b be two elements of $\Phi(T, G)$ such that $U_a \in \mathcal{Z}(S)$, $U_b \notin \mathcal{Z}(S)$, $U_b \subset H$. The group (U_a, U_b) is contained in the subgroup generated by the U_c where c runs through the roots of the form $ma + nb$ ($m, n > 0$) (2.5), and generated by those of these subgroups that it contains (2.3). These latter cannot belong to $\mathcal{Z}(S)$ (since $n \neq 0$), but are in H , hence in H^* , which proves that $(U_a, U_b) \subset H^*$. Consequently, the normalizer of H^* contains every subgroup U_a belonging to $\mathcal{Z}(S)$; since it also contains T , the lemma is proved.

Proposition 0.15 (3.6). *Let H be a connected reductive k -subgroup of maximum rank of G and U a connected unipotent k -subgroup of G normalized by H . Let S be the largest split torus over k of the center of H . Then there exists an order on $X^*(S)$ such that the weights of S in U are > 0 .*

Let us first remark that the center of H is defined over k (2.15), which justifies the definition of S . Let T be a maximal torus of H , hence of G , defined over k , which exists by 2.14, and let ψ be the set of roots a of G with respect to T such that $U_a \subset U$. This is also (3.2) the set of weights of T in U . By 3.1, it therefore suffices, to establish the proposition, to show that every linear combination with integer coefficients > 0 of a non-empty family of elements of ψ has a non-zero restriction to S . Suppose the contrary, and let c be such a linear combination which vanishes on S . The Galois group Γ of k_s/k operates on $X^*(T)$, by permuting the elements of ψ , and on $X_*(T)$, leaving $X_*(S)$ fixed point by point; replacing c if necessary by the sum of its transforms by Γ , we can thus assume c invariant under Γ , hence defined over k . It is then trivial on the identity component Z of the center of H (1.8). On the other hand, the Weyl group $W(H)$ of H also operates on ψ , so the sum c' of the transforms of c by $W(H)$ is also a linear combination with integer coefficients > 0 of elements of ψ , which vanishes on Z . This last fact shows that c' is a linear combination, with rational coefficients, of roots of H with respect to T . Since it is invariant under $W(H)$, it must then be zero. But $T.U$ is part of a Borel subgroup of G , so ψ is contained in the set of positive roots of G for a suitable order, and $c' \neq 0$, whence a contradiction.

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Corollary 0.16 (3.7). *Let U be a connected unipotent k -subgroup of G and S a k -torus of G whose centralizer $\mathcal{Z}(S)$ normalizes U . Then the weights of S in U are all non-zero.*

The group $H = \mathcal{Z}(S)$ is reductive, connected, defined over k (2.15 d)) and of maximum rank. Let T and ψ be as in the preceding proof. The weights of S in U are the restrictions to S of the elements of ψ . If $a \in \psi$ has a zero restriction to S , it is a root of H with respect to T , and it is necessarily zero on the largest central torus of H , in contradiction with 3.6.

3.8. Quasi-closed sets of roots and associated subgroups. — (i) Let S be a torus of G and T a maximal torus of G containing S . A subset μ of $\Phi(T, G) = \Phi$ is *quasi-closed* if the group generated by the U_a ($a \in \mu$) contains no other group U_b ($b \in \Phi$). By 3.4, a closed set is also quasi-closed. (We remark in passing that the converse can only fail in the cases mentioned in the remark to 2.5, where (U_a, U_b) is strictly smaller than $U_{(a,b)}$ ($a, b \in \Phi, a + b \neq 0$)).

A subset ψ of $\Phi(S, G)$ is quasi-closed if the set of roots of G with respect to T whose restriction to S is contained in $\psi \cup \{0\}$ is quasi-closed. Since two maximal tori of G containing S are conjugate in $\mathcal{Z}(S)$, it is clear that this condition will then be satisfied for every maximal torus of $\mathcal{Z}(S)$. Any intersection of quasi-closed subsets is quasi-closed. There is therefore a smallest quasi-closed set containing a subset σ of $\Phi(S, G)$, which will be called the *quasi-closure* of σ . If $\psi \subset \Phi(S, G)$ is quasi-closed, $-\psi$ is also; to show this, it obviously suffices to consider the case where $S = T$, and this then follows from the existence of an automorphism of G preserving T and inducing on T the automorphism $t \mapsto t^{-1}$ ([10, exp. 24, cor. 1]).

(ii) Let ψ be a quasi-closed subset of $\Phi(S, T)$ and μ (resp. ν) the set of elements of $\Phi(T, G)$ whose restriction to S is contained in $\psi \cup \{0\}$ (resp. ψ). We denote by $G_{\psi}^{(S)}$ or G_{ψ} the group generated by T and the U_a ($a \in \mu$), and by $G_{\psi}^{*(S)}$ or G_{ψ}^* the group generated by the U_a ($a \in \nu$). By definition, G_{ψ} contains no other group U_b ($b \notin \mu$), and is therefore generated by $\mathcal{Z}(S)$ and G_{ψ}^* . Moreover, G_{ψ}^* is normalized by $\mathcal{Z}(S)$ according to 3.5, so $G_{\psi} = \mathcal{Z}(S) \cdot G_{\psi}^*$. This also shows that G_{ψ} and G_{ψ}^* do not change if we replace in the preceding definition T by another maximal torus of G containing S .

By definition, we have

$$G_{\mu}^{(T)} = G_{\psi}^{(S)}, \quad G_{\nu'}^{*(T)} = G_{\psi}^{*(S)} \quad (1)$$

where ν' is the quasi-closure of ν (it may be that $\nu' \neq \nu$).

More generally, let S' be a torus containing S and let σ (resp. τ) be the quasi-closure of the set of relative roots with respect to S' whose restriction to S belongs to $\psi \cup \{0\}$ (resp. ψ). We then clearly have

$$G_{\sigma}^{(S')} = G_{\psi}^{(S)}, \quad G_{\tau}^{*(S')} = G_{\psi}^{*(S)}. \quad (2)$$

The weights of S in G_{ψ} (resp. G_{ψ}^*) are the elements of $\psi \cup \{0\}$ (resp. of ψ or $\psi \cup \{0\}$). We also remark that if $\psi = \Phi(S, G)$, then $G_{\psi} = G$ and G_{ψ}^* is the product of almost simple factors of G not centralized by S .

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- (iii) Let ψ, η be quasi-closed subsets of $\Phi(S, G)$. We say that η is a *quasi-ideal* of ψ if G_{η}^* is a normal subgroup of G_{ψ}^* . We shall see (3.9) that an ideal is a quasi-ideal.
- (iv) If G_{ψ}^* is unipotent, we will often denote it by U_{ψ} , and we will say that ψ is *unipotent*. If ψ is quasi-closed and consists of roots > 0 for a suitable order on $X^*(S)$, then it is unipotent, and ψ is quasi-closed unipotent. Indeed, let Φ^+ be the set of roots relative to T that are positive for an order on $X^*(T)$ compatible with the order given on $X^*(S)$ (3.1). We have $\psi \subset \Phi^+$, hence also $\psi' \subset \Phi^+$, and $G_{\psi'}^{*(T)}$ is part of the maximal unipotent group U generated by the U_a ($a \in \Phi^+$). Furthermore, if $a, b \in \psi$, $m, n \in \mathbf{N}^+$ and $m.a + n.b \in \psi'$, then the restriction to S of $m.a + n.b$ cannot be zero, hence is part of ψ and $m.a + n.b \in \psi$. It then follows from 2.5 that ψ is quasi-closed and that S does not have the zero weight in U_{ψ} .

If $S = T$, then conversely every quasi-closed unipotent set consists of positive roots, since $T.U_{\psi}$ is contained in a Borel subgroup of G . This also holds if S is defined over k and contains the maximal k -split torus of the center of $\mathcal{Z}(S)$, by 3.6. But simple examples show that this converse is not true in general.

Proposition 0.17 (3.9). *We keep the notation of 3.8. Let ψ, η be quasi-closed subsets of $\Phi(S, G)$ such that ψ normalizes η . Then G_{ψ}^* normalizes G_{η}^* .*

Proof. Let T be a maximal torus of G containing S and $j : X^*(T) \rightarrow X^*(S)$ the restriction homomorphism. Since $\mathcal{Z}(S)$ normalizes G_{η}^* , it suffices to show that if $a \in j^{-1}(\psi) \cap \Phi(T, G)$ and $b \in j^{-1}(\eta) \cap \Phi(T, G)$, then $(U_a, U_b) \subset G_{\eta}^*$. Suppose first that $m.j(a) + n.j(b) \neq 0$ for all $m, n \in \mathbf{N}^+$; since ψ normalizes η , we then have $m.a + n.b \in \eta$ whenever $m.a + n.b \in \Phi$, and our assertion follows from 2.5. If, on the other hand, there exist $m, n \in \mathbf{N}^+$ such that $m.j(a) + n.j(b) = 0$, then $j(a) = (m+1).j(a) + n.j(b) \in \eta$, so $U_a \in G_{\eta}^*$, from which obviously $(U_a, U_b) \subset G_{\eta}^*$. $\square$

Proposition 0.18 (3.10). *Let S be a torus and ψ, η two quasi-closed subsets of $\Phi(S, G)$ positive for an order given on $X^*(S)$. Let σ be the set of elements of $\Phi(S, G)$ which are the sum of a non-zero linear combination with integer coefficients ≥ 0 of elements of ψ and a non-zero linear combination with integer coefficients ≥ 0 of elements of η . Then σ is closed unipotent, and $(U_{\psi}, U_{\eta}) \subset U_{\sigma}$.*

Proof. It is clear that σ is closed. Let T be a maximal torus of G containing S , and Φ^+ the set of roots of G with respect to T , positive for an order on $X^*(T)$ compatible with the order given on $X^*(S)$. Let ψ', η', σ' be the inverse images of ψ, η, σ in Φ . By 3.8 (iv), we have $U_{\psi} = U_{\psi'}^{(T)}$ and similarly for η, σ . On the other hand, σ' is obtained from ψ', η' in the same way as σ is from ψ and η . The proposition is therefore a consequence of 2.3, 2.5 and the identities $(x, yz) = (x, y) \cdot {}^y(x, z)$ and $(xy, z) = {}^x(y, z) \cdot (x, z)$. $\square$

Proposition 0.19 (3.11). *Let S be a torus of G , ψ a quasi-closed unipotent set of roots relative to S , $\psi_1, \dots, \psi_n$ quasi-closed unipotent subsets of ψ of which ψ is the disjoint union, and H a closed connected unipotent subgroup of U_{ψ} (resp. G_{ψ}^*) normalized by S . Then*

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the product map defines an isomorphism of varieties from $(H \cap U_{\psi_1}) \times \cdots \times (H \cap U_{\psi_n})$ (resp. $(H \cap \mathcal{Z}(S)) \times (H \cap U_{\psi_1}) \times \cdots \times (H \cap U_{\psi_n})$) onto H .

By 3.5, U_ψ is normalized by $\mathcal{Z}(S)$, so it is the unipotent radical of G_ψ , and $S \cdot H \cdot U_\psi$ is a closed solvable subgroup of G_ψ . Let B be a Borel subgroup of G containing $S \cdot H \cdot U_\psi$ and T a maximal torus of B containing S , hence also belonging to $\mathcal{Z}(S)$. The weights of T in the unipotent radical U of B are the positive elements of $\Phi = \Phi(T, G)$ for a suitable order. Let $j : X^*(T) \rightarrow X^*(S)$ be the restriction homomorphism and $\eta_0 = j^{-1}(0) \cap \Phi^+$, $\eta_i = j^{-1}(\psi_i) \cap \Phi$ ($1 \leq i \leq n$), $\eta = j^{-1}(\psi) \cap \Phi$, and $\sigma = \eta \cup \eta_0$. Obviously, η_0 is closed and, by 2.3, $U_{\eta_0} = \mathcal{Z}(S) \cap U$; by 3.8 (iv), $\eta_1, \dots, \eta_n, \eta$ are unipotent quasi-closed subsets of Φ^+ , η is the disjoint union of $\eta_1, \dots, \eta_n$, and we have $U_\psi = U_\eta^{(T)}$, $U_{\psi_i} = U_{\eta_i}^{(T)}$ ($1 \leq i \leq n$). It also follows from 2.3 that σ is the set of weights of T in $V = G_\psi \cap U$, hence that σ is quasi-closed and that $V = U_\sigma$. Let $a_1, \dots, a_q$ (resp. $b_1, \dots, b_r$) be the elements of η (resp. σ) ordered by increasing order. By 2.5, the product map defines an isomorphism of varieties from $U_{a_1} \times \cdots \times U_{a_q}$ (resp. $U_{b_1} \times \cdots \times U_{b_r}$) onto a normal subgroup V_j (resp. Y_j) of U_η (resp. U_σ) for all $j \leq q$ (resp. $j \leq r$). The groups V_j (resp. Y_j) therefore form a composition series of U_η (resp. U_σ) whose elements are generated by their intersections with the groups U_{η_i} ($i = 1, \dots, n$) (resp. $i = 0, \dots, n$). The proposition then follows from 3.3 (where S is taken as the operator group).

Proposition 0.20 (3.12). *Let S be a torus, H a closed connected subgroup of G normalized by S , and ψ a convex set of roots of G relative to S . Suppose that every weight of S in H belongs to ψ (resp. $\psi \cup \{0\}$). Then H is contained in G_ψ^* (resp. G_ψ).*

Proof. Let S' be a maximal torus of $S \cdot H$ containing S . The group $S \cdot H$ is generated by S' and two maximal unipotent subgroups normalized by S' (2.3, remark), so H is generated by $(S' \cap H)^0$ and two unipotent subgroups normalized by S . It therefore suffices to consider the case where H is unipotent.

Let B be a Borel subgroup of G containing $S \cdot H$, T a maximal torus of B containing S , and U the unipotent radical of B . Let Φ^+ be the set of roots of U relative to T , and φ_0 the set of elements of Φ^+ whose restriction to S' is zero. Let $\varphi_0, \varphi_1, \dots, \varphi_n$ be the equivalence classes of the relation obtained on Φ^+ by considering as equivalent two roots whose restrictions to S differ only by a strictly positive coefficient; $\varphi_0, \varphi_1, \dots, \varphi_n$ are convex and (2.3) $\mathcal{Z}(S) \cap U = U_{\varphi_0}$. Let ψ_i be the set of restrictions to S of the elements of φ_i , and

$$U_i = U_{\psi_i} = U_{\varphi_i}^{(T)} \quad (1 \leq i \leq n), \quad U_0 = U_{\varphi_0}^{(T)} = \mathcal{Z}(S) \cap U.$$

It follows from 3.11 that $H = (H \cap \mathcal{Z}(S)) \cdot (H \cap U_1) \cdots (H \cap U_n)$, and that $H \cap \mathcal{Z}(S) \neq \{e\}$ only if S has weight zero in H . Let $i \geq 1$. Then $H \cap U_i \neq \{e\}$ implies $\psi \cap \psi_i \neq \emptyset$, hence, since ψ is convex, $\psi_i \subset \psi$, and $U_i \subset U_\psi$, whence the proposition. $\square$

Theorem 0.21 (3.13). *Let S be a k -torus of G , H a k -closed connected subgroup of G normalized by $\mathcal{Z}(S)$, and ψ a quasi-closed subset of $\Phi(S, G)$. Then:*

- a) H is defined over k , and possesses a unique Levi subgroup L normalized by $\mathcal{Z}(S)$. The groups L and $R_u(H)$ are defined over k .
- b) The groups G_ψ and G_ψ^* (3.8) are defined over k if and only if ψ is (1.7). The sets $\psi_s = \psi \cap (-\psi)$ and $\psi_u = \psi \cap \mathcal{C}(-\psi)$ are quasi-closed, and ψ_u is unipotent. If $H = G_\psi$ (resp. $H = G_\psi^*$), then $L = G_{\psi_s}$ (resp. $L = G_{\psi_s}^*$) and $R_u(H) = U_{\psi_u}$.

Proof of a). It suffices to carry out the proof when k is separably closed. By 2.10 and 2.14, S is contained in a maximal torus T of G belonging to a splitting datum over k . The group $R_u(H)$ (resp. H) is then generated by the root groups U_a that it contains (resp. and $(T \cap H)^0$) and is therefore defined over k .

Let η be the set of $a \in \Phi(T, G)$ such that $U_a \subset H$, and $\eta_s = \eta \cap (-\eta)$, $\eta_u = \eta \cap \mathbb{C}(-\eta)$. Let us show that the set η_u is equal to the set μ of $a \in \Phi(T, G)$ such that $U_a \subset R_u(H)$. Let $a \in \mu$. If $-a$ belonged to η , then the group Z'_a generated by U_a and U_{-a} would normalize $R_u(H)$, its unipotent radical would contain $R_u(H) \cap Z'_a$, hence U_a , and would be $\neq \{e\}$, which is absurd since Z'_a is almost simple of dimension three (2.3). Thus, $-a \notin \eta$, hence $a \in \eta_u$ and $\mu \subset \eta_u$. On the other hand, the quotient of $T \cdot H$ by $R_u(H)$ is reductive, so it has a symmetric set of roots, whence $\mu \supset \eta_u$, and $\mu = \eta_u$, which shows in particular that η_u is quasi-closed unipotent.

The set η_s is quasi-closed since η and $-\eta$ are (3.8(i)) and the group L generated by the U_a ($a \in \eta_s$) and $T' = (T \cap H)^0$ is reductive. By 2.3, 3.4, it contains every reductive subgroup of H normalized by T , and we have $H = L \cdot R_u(H)$. The results recalled in 2.3 also show that the Lie algebras of L and $R_u(H)$ are transverse and that $L \cap R_u(H)$ is connected, hence reduced to $\{e\}$. The group H is consequently a semi-direct product of L and $R_u(H)$, and L is the unique reductive subgroup normalized by T having this property. It remains to show that L is stable under $\mathcal{Z}(S)$. The subgroup L' of H generated by the ${}^x L$ ($x \in \mathcal{Z}(S)$) is reductive, since it is stable under an automorphism of G which induces $t \mapsto t^{-1}$ on T (the existence of which follows from Cor. 1 of [10, exp. 24]), and therefore possesses a set of roots ν with respect to T which is symmetric. We obviously have $\eta_s \subset \nu \subset \eta$, whence $\nu = \eta_s$ and $L' = L$, which completes the proof of a). Let us also note that we have shown that

$$L = (T \cap H)^0 \cdot G_{\eta_s}^{*(T)}, \quad R_u(H) = G_{\eta_u}^{*(T)}. \quad (1)$$

□

Proof of b). If G_ψ (resp. G_ψ^*) is defined over k , the same is obviously true for ψ ; conversely, if ψ is defined over k then G_ψ and G_ψ^* are k -closed, normalized by $\mathcal{Z}(S)$, and therefore defined over k by a).

Let $j : X^*(T) \rightarrow X^*(S)$ be the restriction homomorphism, and let η be as above. If $H = G_\psi$ (resp. $H = G_\psi^*$), then we have $j^{-1}(\psi) \cap \Phi(T, G) \subset \eta \subset j^{-1}(\psi \cup \{0\}) \cap \Phi(T, G)$, whence in particular $j^{-1}(\psi_s) \cap \Phi(T, G) \subset \eta_s$ and $j^{-1}(\psi_u) \cap \Phi(T, G) \subset \eta_u$. In view of (1), it follows that

$$G_{\psi_s} \text{ (resp. } G_{\psi_s}^*) \subset L \quad \text{and} \quad G_{\psi_u}^* \subset R_u(H). \quad (2)$$

□

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But $G_{\Psi_u}^*$ is normalized by G_{Ψ_s} (3.0, 3.9), so $H = G_{\Psi_s} \cdot G_{\Psi_u}^*$ (resp. $G_{\Psi_s}^* \cdot G_{\Psi_u}^*$), and the inclusions (1) are in fact equalities.

3.14. Corollary. — *Let H be a connected k -subgroup of G of maximum rank. Then H has a Levi decomposition (0.8) over k , and two k -Levi subgroups of H are conjugate by a unique element of $R_u(H)_k$. The k -Levi subgroups of H are the centralizers in H of the maximal tori of $R(H)$ which are defined over k .*

H contains a maximal torus defined over k (2.14 a)), so it has a Levi decomposition over k by 3.13. Let $H = L \cdot R_u(H)$ be such a decomposition and T a maximal torus of L . The radical of H is defined over k and equal to $Z \cdot R_u(H)$, where Z is the connected center of L and also a maximal torus of $R(H)$ defined over k (2.15 e)). We have $L \subset \mathcal{Z}_H(Z)$ by construction. On the other hand, $\mathcal{Z}_H(Z) \cap R_u(H)$ is stable under T , and thus generated by the subgroups U_α ($\alpha \in \Phi(T, G)$) that it contains (2.3), which, taking 3.6 into account, shows that $\mathcal{Z}_H(Z) \cap R_u(H) = \{e\}$, whence $L = \mathcal{Z}_H(Z)$, and our last assertion.

Let L' be a second k -Levi subgroup and Z' its connected center. The tori Z and Z' are maximal in $R(H)$, and thus conjugate by an element $u \in R_u(H)_k$ [1, Th. 12.2]. This element is unique, because the normalizer and the centralizer of Z in $R(H)$ coincide [1, 10.2] and we have seen above that the latter reduces to Z . The element u is therefore rational over an inseparable extension of k , and, to prove that it is rational over k^7 , one can assume k is separably closed, and thus G is split over k . In view of 2.10, applied to G and to $T \cdot R_u(H)$, we can assume that T is part of a splitting datum and find $v \in R_u(H)_k$ such that $v \cdot Z' \cdot v^{-1} \subset T$. We then have $v \cdot Z' \cdot v^{-1} \subset L \cap R(H)$, hence $v \cdot Z' \cdot v^{-1} = Z$ and $v \cdot L' \cdot v^{-1} = L$.

3.15. Counter-examples. — We assume here that the characteristic p of k is non-zero. Let V be a vector space over k of dimension two, V_p its p -th symmetric power, and x, y a basis of V_k . We identify V_p with k^{p+1} by taking as a basis the monomials $x^p, y^p, x^{p-1} \cdot y, \dots, x \cdot y^{p-1}$. The group $\text{GL}(V)$ acts canonically on V_p , and we denote by L the image in $\mathbf{GL}_{p+1}$ of the group $\text{SL}(V)$ of elements of determinant one in $\text{GL}(V)$ under this representation.

⁷This also follows from a theorem of Rosenlicht (see 11.4).

Let P be the subgroup of $\mathbf{GL}_{p+1}$ leaving stable the subspace W generated by x^p and y^p , and let $U = R_u(P)$. It is immediate that $U \cong (\mathbf{G}_a)^{2(p-1)}$ and that we have a Levi decomposition $P = M \cdot U$ with $M \cong \mathbf{GL}_2 \times \mathbf{GL}_{p-1}$. The projection $\pi : P \rightarrow M' = P/U$ induces an isomorphism ψ of M onto M' . On the other hand, it is elementary that L leaves W stable, so $L \subset P$. Let $H = L \cdot U$ be the group generated by L and U . We have $L \cap U = \{e\}$, so H is, from the set-theoretic point of view, the semi-direct product of L and U . We can also write $H = L' \cdot U$, where $L' = H \cap M = \psi^{-1}\pi(L)$ is k -isomorphic to L . The Lie algebras of L' and U are transverse (since those of M and U are), so L' is a Levi subgroup of H .

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The subgroups L and L' of H are not conjugate in H , because if they were, one could find a complementary subspace W' of W in V_p stable under L , and an elementary calculation (or indeed [10, Exp. 20, p. 12]) shows that this is not the case.

One can take as a basis of the Lie algebra $\mathfrak{l}$ of L the vectors H, X, Y tangent to the images of the groups of diagonal, upper unipotent triangular, and lower unipotent triangular matrices respectively. An elementary calculation left to the reader shows that H is diagonal, and that:

- a) if $p = 2$, X and Y form a basis of the Lie algebra $\mathfrak{u}$ of U ;
- b) if $p \geq 3$, $\mathfrak{l}$ and $\mathfrak{u}$ are transverse.

(One verifies for example that X and Y have zero diagonal entries and that if $p \geq 3$, $X_{p,p+1} \neq 0$, $X_{p+1,p} = 0$, $Y_{p,p+1} = 0$, $Y_{p+1,p} \neq 0$.) If $p \geq 3$, H is therefore an example of a group possessing two non-conjugate Levi subgroups L, L' . If $p = 2$, H is a subgroup of $\mathbf{GL}_3$, normalized by the maximal torus of diagonal matrices of $\mathbf{GL}_3$, and thus to which 3.13 applies, but which possesses a maximal reductive subgroup L such that $H = L \cdot U$ and whose Lie algebra has a non-zero intersection with that of U .

Lemma 0.22 (3.16). *Let U be a connected unipotent k -group, S a k -torus operating k -morphically on U , and f a non-trivial character of S . Suppose that over k_s , U possesses a vector space structure such that the operation of an element $s \in S_{\bar{k}}$ be the homothety of ratio s^f . Then this is already the case over k , and f is defined over k^8 .*

The set of weights of S in U reduces to f , and is on the other hand stable under the Galois group Γ of k_s/k , so f is defined over k .

The hypotheses imply that we have $\gamma(c \cdot u) = \gamma c \cdot \gamma u$, for all $\gamma \in \Gamma$, $c \in k_s$ and $u \in U_{k_s}$. It therefore suffices to show that U possesses a basis consisting of elements rational over k . It is clear that U possesses a basis consisting of elements rational over a suitable finite Galois extension K of k . Let $\Gamma' = \text{Gal}(K/k)$ and let $(u_1, \dots, u_n)$ be a subset of U_K that is free in U_{k_s} (viewed as a vector space over k_s) and maximal. Then, for all $u \in U_K$ and $c \in K$, the element $\sum_{\gamma \in \Gamma'} \gamma c \cdot \gamma u$ is rational over k , and is thus a linear combination of the u_i ($1 \leq i \leq n$). It is then the same for u by the normal basis theorem, hence $(u_1, \dots, u_n)$ is a basis of U_{k_s} .

Theorem 0.23 (3.17). *Let S be a k -torus of G , ψ a unipotent quasi-closed set of elements of $\Phi(S, G)$ defined over k , and f an element defined over k of ψ such that $\eta = \psi \setminus \{f\}$ is a quasi-ideal of ψ (3.8). Let H be a k -subgroup of U_ψ , normalized by a maximal torus T of G defined over k and containing S . Then the quotient $H/(H \cap U_\eta)$ has a vector space structure defined over k such that the automorphism induced by $\text{Int } s$ ($s \in S_{\bar{k}}$) is the homothety of ratio s^f .*

By 3.16, we can assume that $k = k_s$, and thus that G is split over k and that $T \dots$

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⁸This lemma is also an easy consequence of the lemma on p. 109 of [26]. It is also true when $f = 0$ [26, th. 3, p. 107].

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is part of a splitting datum. There exists a Borel subgroup of G containing $T.U_\psi$, and this subgroup is generated by the root groups U_a , where a runs through the set Φ^+ of roots > 0 for a suitable order, and is therefore defined over k . Let μ (resp. ν) be the set of elements of Φ^+ whose restriction to S is contained in ψ (resp. η). Then (2.3, 3.11) the product map induces a k -isomorphism α of varieties from the product of the U_a , $a \in \mu$ (resp. U_b , $b \in \nu$), onto U_ψ (resp. U_η) and a bijective morphism of the product of the $U_a \cap H$, $a \in \mu$ (resp. $U_b \cap H$, $b \in \nu$) onto $U_\psi \cap H$ (resp. $U_\eta \cap H$), where the elements of μ (resp. ν) are ordered in an arbitrary way. But $U_a \cap H$ is equal to $\{e\}$ or to U_a (2.3) and α commutes with the action of T . This map therefore induces a k -isomorphism of varieties from the product of the U_a ($a \in \mu, a \notin \nu, U_a \subset H$) onto $H/(H \cap U_\eta)$ which commutes with T , whence the theorem.

Corollary 0.24 (3.18). *Any connected unipotent k -subgroup H of G normalized by a maximal torus T of G defined over k is split over k . If σ is a unipotent quasi-closed subset of $\Phi(S, G)$ defined over k , then U_σ is a split unipotent k -group over k .*

Proof. By 3.6, there exists an order on $X^*(T_d)$ such that the weights of T_d in H are > 0 . Let $a_1, \dots, a_m$ be the elements of $\Phi^+(T_d, G)$, arranged in increasing order. For any i between 1 and m , $\{a_i, \dots, a_m\}$ is a quasi-ideal of $\{a_1, \dots, a_m\}$. The first part of the corollary then follows from the theorem by descending induction on the smallest integer i such that $H \subset U_{\{a_i, \dots, a_m\}}$.

The group U_σ is defined over k (3.13), normalized by $\mathcal{Z}(S)$ (3.8 (ii)), hence also by a maximal torus of G defined over k (2.15 d)). The second assertion of the corollary is therefore a special case of the first. $\square$

Remark 1. Suppose S is split over k , and σ consists of positive elements of $\Phi(S, G)$ for a suitable order on $X^*(S)$. Let $b_1, \dots, b_m$ be the elements of σ arranged in increasing order. For any $i \geq 1$, the set $\sigma_i = \{b_i, \dots, b_m\}$ is a quasi-closed subset of $\Phi(S, G)$, because it is the intersection of σ with the set of roots $\geq b_i$, which is clearly closed. Since S is split over k , every element of $\Phi(S, G)$ is defined over k , so $U_i = U_{\sigma_i}$ is a unipotent k -group. By 3.10, U_{i+1} is a normal subgroup of U_i , and by 3.17, U_i/U_{i+1} admits a structure of a vector space over k , such that the inner automorphism by $s \in S_k$ induces a homothety with ratio s^{b_i} on it. Therefore, in this case there exists a composition series over k , with successive quotients that are vector spaces over k on which S acts by homotheties, which refines 3.18.

The positivity assumption made on σ is automatically satisfied if $\sigma = (c)$ is the set of positive integer multiples of an element $c \in \Phi(S, G)$, or, by 3.6, if S is the largest split torus over k in the center of $\mathcal{Z}(S)$, hence, even more particularly, if S is a maximal split torus of G over k .

Corollary 0.25 (3.19). *Let S be a torus of G , f a non-trivial character of S , and α an isomorphism of $\mathbf{G}_a$ onto a subgroup H of G normalized by S satisfying $s \cdot \alpha(t) \cdot s^{-1} = \alpha(s^f \cdot t)$ ($s \in S, t \in \mathbf{G}_a$). Then H is contained in the subgroup $U_{(f)}^{(S)}$, where (f) denotes the set of elements of $\Phi(S, G)$ of the form $n \cdot f$ ($n \in \mathbf{N}^+$).*

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Let ψ be the set of roots relative to S of the form $c \cdot f$ with $c > 0$. This set is convex, hence $H \subset U_\psi$ by 3.12.

Let η be the smallest quasi-closed subset of ψ such that $H \subset U_\eta$.

Suppose that $\eta \neq \{f\}$, and let c be the smallest non-integer such that $c \cdot f \in \eta$. The set $\mu = \eta - \{c \cdot f\}$ is a quasi-ideal of η . Let π be the canonical projection of U_η onto the vector space $U_\eta/U_\mu = V$, and let β be the morphism of varieties from $\mathbf{G}_m$ to V which maps $t \in \mathbf{G}_m$ to $\pi(\alpha(t))$. By 3.17, we have

$$\beta(s^f \cdot t) = s^{c \cdot f} \cdot \beta(t) \quad (s \in S, t \in \mathbf{G}_m). \quad (1)$$

Since $H \not\subset U_\mu$, β is not zero, but then (1) and the fact that β is a morphism imply that c is an integer, whence a contradiction.

Corollary 0.26 (3.20). *Let H be a connected k -subgroup of G , normalized by a maximal torus T of G defined over k . If k is infinite, then H_k is Zariski-dense in H .*

The group H admits a Levi decomposition $H = L \cdot R_u(H)$ over k by 3.13. The group $R_u(H)$ is normalized by T , hence split over k by 3.18, hence isomorphic over k to an affine space [26, p. 101], which implies that $R_u(H)_k$ is Zariski-dense in $R_u(H)$. Finally, L_k is Zariski-dense in L by 2.14.

Lemma 0.27 (3.21). *Let S be a torus of G , ψ a convex subset of $\Phi(S, G)$ and A the identity component of the kernel of the characters belonging to ψ_s .*

- a) ψ_s contains every element of $\Phi(S, G)$ which vanishes on A .
- b) If S is a maximal torus of G , every element $w \in W(S, G)$ which acts trivially on $X_*(A)$ is a product of reflections r_b ($b \in \psi_s$).

Proof. a) Let $c \in \Phi(S, G)$ vanishing on A . There exist therefore elements of ψ_s of which c is a linear combination with rational coefficients. Since ψ_s is symmetric, we can assume these coefficients are > 0 ; on the other hand, ψ_s is convex, since ψ is, hence $c \in \psi_s$.

- b) By [6], w is a product of reflections r_b , where b is a root vanishing on $X_*(A)$, hence a character of S vanishing on A , hence an element of ψ_s by a).

□

Proposition 0.28 (3.22). *Let S be a torus of G and ψ, η two quasi-closed subsets of $\Phi(S, G)$. Then:*

- a) $\mathfrak{g}_\psi \cap \mathfrak{g}_\eta = \mathfrak{g}_{\psi \cap \eta}$ and $(G_\psi \cap G_\eta)^0 = G_{\psi \cap \eta}$.
- b) If $\psi \cap \eta$ is convex, $G_\psi \cap G_\eta = G_{\psi \cap \eta}$.
- c) If ψ is unipotent, $U_\psi \cap G_\eta = U_\psi \cap G_\eta^* = U_{\psi \cap \eta}$.

Let T be a maximal torus of G containing S and $j : X^*(T) \rightarrow X^*(S)$ the restriction homomorphism.

Proof. a) A root subgroup U_a ($a \in \Phi(T, G)$) with respect to T is contained in G_ψ if and only if a is a weight of T in G_ψ (3.8), hence if and only if $j(a) \in \psi \cup \{0\}$. This also holds for η or $\psi \cap \eta$, whence a).

- b) It suffices to consider the case where k is algebraically closed, hence where T is contained in

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of a splitting datum over k of G . On the other hand, if $\mu \subset \Phi(S, G)$ is convex, then $\Phi(T, G) \cap j^{-1}(\mu \cup \{0\})$ is also convex, and obviously

$$j^{-1}((\psi \cap \eta) \cup \{0\}) = j^{-1}(\psi \cup \{0\}) \cap j^{-1}(\eta \cup \{0\}).$$

We can therefore assume that $S = T$.

Let $x \in G_\psi \cap G_\eta$. We must show that $x \in G_{\psi \cap \eta}$. The torus xTx^{-1} being conjugate to T in $(G_\psi \cap G_\eta)^0$, we can, in view of a), restrict ourselves to the case where $x \in \mathcal{N}(T)$. Let $A = T_{\psi_s}$ and $B = T_{\eta_s}$ (1.2). According to 3.13, 3.14, $G_{\psi_s} = \mathcal{Z}(A)$ and $G_{\eta_s} = \mathcal{Z}(B)$ are Levi subgroups of G_ψ and G_η respectively, whose element w of the Weyl group $W(T, G)$ of G defined by x belongs to $W(T, G_{\psi_s}) \cap W(T, G_{\eta_s})$; it centralizes $X_*(A)$, $X_*(B)$ and consequently also $X_*(C)$, where C is the identity component of the intersection of the kernels of the elements of $\psi_s \cap \eta_s = (\psi \cap \eta)_s$. Our assertion then follows from 3.21 b).

c) Here also, we can assume k algebraically closed and $S = T$, and c) then follows from the properties of U recalled in 2.3.

Corollary 0.29 (3.23). *Let S be defined over k and let ψ, η be two quasi-closed subsets of $\Phi(S, G)$ defined over k .*

- (i) $G_{\psi, k}^* \subset G_{\eta, k}$ if and only if $\psi \subset \eta$.
- (ii) $G_{\psi, k}^* = G_{\eta, k}^*$ (resp. $G_{\psi, k} = G_{\eta, k}$) if and only if $\psi = \eta$.

The groups G_ψ , G_η , G_ψ^* , G_η^* are defined over k according to 3.13. Assertion (ii) is an immediate consequence of (i) and it is clear that $\psi \subset \eta$ implies $G_{\psi, k}^* \subset G_{\eta, k}$. It remains to show that $G_{\psi, k}^* \subset G_{\eta, k}$ implies $\psi \subset \eta$.

Let $a \in \Phi(S, G)$ and (a) the set of positive integer multiples of a contained in $\Phi(S, G)$. It is closed, unipotent (3.8 (iv)), so $\mu = (a) \cap \psi$ is quasi-closed and unipotent. Since ψ is a union of subsets of this type, we can assume that it is unipotent. In this case, $U_\psi \cap G_\eta = U_{\psi \cap \eta}$ according to 3.22. The groups $U_{\psi \cap \eta}$ and U_ψ are unipotent split over k (3.18), so the underlying variety of U_ψ is k -isomorphic to the product of $U_{\psi \cap \eta}$ by a vector space over k [26, cor. 1, 2 to th. 1]. We cannot then have $U_{\psi, k} \subset U_{\psi \cap \eta, k}$ unless $U_\psi = U_{\psi \cap \eta}$, hence $\psi \subset \eta$.

Proposition 0.30 (3.24). *Let S be a split k -torus and ψ a convex subset of $\Phi(S, G)$. Then, there exist convex unipotent subsets $\psi_1, \dots, \psi_m$ of $\Phi(S, G)$ such that $\Phi(S, G)$ is the disjoint union of $\psi, \psi_1, \dots, \psi_m$ and that the product map defines a k -isomorphism of varieties of $G_\psi \times U_{\psi_1} \times \dots \times U_{\psi_m}$ onto an open subset of G . If $\psi \cup (-\psi) = \Phi(S, G)$, we can take $m = 1$ (hence $\psi_1 = \Phi(S, G) - \psi$).*

Let $X = X^*(S) \otimes \mathbf{Q}$, and let Y be the vector subspace of X generated by ψ_s . If a positive linear combination with rational coefficients of elements of ψ_u belongs to Y , it is identically zero; indeed, from the relation

$$\sum_{i=1}^r c_i a_i \in Y \quad (a_i \in \psi_u, c_i \in \mathbf{Q}, c_i \geq 0)$$

we obtain

$$-c_1 a_1 = \sum_{i=2}^r c_i a_i + \sum_{j=1}^t d_j b_j \quad (b_j \in \psi_s, d_j \geq 0)$$

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and if c_1 were not zero, $-a_1$ would have to belong to ψ , which is absurd. By virtue of 3.1, there exists in X/Y an order such that the canonical images in X/Y of the elements of ψ_u are positive. Let ψ' be the set of all elements of $\Phi(S, G)$ whose canonical image in X/Y is positive for the order in question. It is clear that ψ' is convex, that $\psi'_s = \psi_s$, and that $\psi_u \subset \psi'_u$, these two sets being equal if $\psi \cup (-\psi) = \Phi(S, G)$. Let $\psi_1, \dots, \psi_{m-1}$ be the minimal convex subsets of ψ'_u not contained in ψ_u . The set ψ'_u is the disjoint union of $\psi_u, \psi_1, \dots, \psi_{m-1}$; consequently (3.11) the product map is an isomorphism of varieties of $U_{\psi_1} \times U_{\psi_1} \times \dots \times U_{\psi_{m-1}}$ onto $U_{\psi'_u}$, hence (3.13) of $G_{\psi'} \times U_{\psi_1} \times \dots \times U_{\psi_{m-1}}$ onto $G_{\psi'}$. Finally, if we set $\psi_m = -\psi'_u$, the product map is an isomorphism of varieties of $G_{\psi'} \times U_{\psi_m}$ onto an open subset of G . Indeed, $G_{\psi'}$ and U_{ψ_m} are normalized by any maximal torus contained in $\mathcal{Z}(S)$ and it follows from 2.3 that their Lie algebras are linearly independent and of sum equal to $\mathfrak{g}$; moreover, $G_{\psi'} \cap U_{\psi_m} = \{e\}$ by virtue of 3.22 c).

3.25. Corollary. — *The variety G/G_{ψ} is rational over k . The fibration of G by G_{ψ} has a local k -section, and the projection $G_k \rightarrow (G/G_{\psi})_k$ is surjective.*

The composition of the product map and the projection $\pi : G \rightarrow G/G_{\psi}$ induces a k -isomorphism of varieties of $U_{\psi_1} \times \dots \times U_{\psi_m}$ onto an open subset V of G/G_{ψ} , and the variety underlying U_{ψ_i} is k -isomorphic to an affine space [26, p. 101], whence the first assertion. The local k -section is defined on V as the inverse of $\pi : U_{\psi_1} \dots U_{\psi_m} \rightarrow V$, and the last assertion follows from 2.6, 2.14.

§ 4. PARABOLIC SUBGROUPS

In this whole section, G is reductive, connected.

4.1. A closed subgroup P of an algebraic group H is *parabolic* if it contains a Borel subgroup of H . In view of [1, § 16], this is equivalent to requiring that G/P be a complete variety or a projective variety.

4.2. Standard parabolic subgroups. — (i) We fix a maximal torus T of G and an order on the set Φ of roots of G with respect to T , and we use the notation of 2.3. For any subset θ of Δ , we denote by $[\theta]$ the set of roots which are linear combinations with integer coefficients of elements of θ , and we set

$$\pi_{\theta} = [\theta] \cup \Phi^+, \quad \pi_{\theta}^- = [\theta] \cup \Phi^-, \quad \alpha_{\theta} = \mathcal{C}[\theta] \cap \Phi^+, \quad \alpha_{\theta}^- = \mathcal{C}[\theta] \cap \Phi^-. \quad (1)$$

Obviously

$$\pi_{\theta}^- = -\pi_{\theta}, \quad [\theta] = \pi_{\theta} \cap \pi_{\theta}^- = (\pi_{\theta})_s, \quad (\pi_{\theta})_u = \alpha_{\theta}, \quad (\pi_{\theta}^-)_u = \alpha_{\theta}^- = -\alpha_{\theta}. \quad (2)$$

The sets π_{θ} and π_{θ}^- are the only closed subsets of Φ containing respectively Φ^+ and Φ^- [6, § 7, no. 7, Prop. 20].

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(ii) We shall write $P_{\theta}, P_{\theta}^-, Z_{\theta}, V_{\theta}, V_{\theta}^-$ respectively for $G_{\pi_{\theta}}, G_{\pi_{\theta}^-}, \mathcal{Z}(T_{\theta}), U_{\alpha_{\theta}}, U_{\alpha_{\theta}^-}$ (cf. 3.8; according to the convention of 1.2, T_{θ} denotes the identity component of the intersection of the kernels of the characters $a \in \theta$). We have the Levi decompositions

$$P_{\theta} = Z_{\theta} \cdot V_{\theta}, \quad P_{\theta}^- = Z_{\theta} \cdot V_{\theta}^-. \quad (3)$$

The subgroups P_{θ} ($\theta \subset \Delta$) are the *standard parabolic subgroups* (associated with T and Φ^+). Obviously $P_{\emptyset} = B$ and $P_{\Delta} = G$.

(iii) Let $W_{\theta} = W(Z_{\theta})$. This is the subgroup of $W(G)$ generated by the fundamental reflections r_b ($b \in \theta$). According to 2.4, and with the notation convention made there, we have

$$Z_{\theta} = (B \cap Z_{\theta}) \cdot W_{\theta} \cdot (B \cap Z_{\theta}),$$

whence

$$P_{\theta} = B \cdot W_{\theta} \cdot B, \quad P_{\theta}^- = B^- \cdot W_{\theta} \cdot B^-. \quad (4)$$

The group $V_{\theta}^- \cap P_{\theta}$ is stable under T , contains no group U_a ($a \in \Phi$), hence is reduced to the identity element (2.3). On the other hand, the Lie algebra of G is the direct sum of the algebras of V_{θ}^- and P_{θ} . Consequently,

the product map induces an isomorphism of varieties from $V_{\bar{\theta}}^- \times P_{\bar{\theta}}$ onto an open subset of G , and the fibration of G by $P_{\bar{\theta}}$ has a local section (cf. also 3.24, 3.25).

4.3. From 4.2 and [33, Th. 3], it follows that the subgroups P_{θ} ($\theta \in \Delta$) are the only subgroups of G containing B , and that every parabolic subgroup of G is connected, equal to its normalizer, and conjugate to one and only one standard parabolic subgroup. (The fact that the axioms of [33] are satisfied follows immediately from the calculations of [9], which will moreover be found again later (5.16).) It follows that if two parabolic subgroups P and P' of G are conjugate, and if $P \cap P'$ is parabolic, then $P = P'$.

4.4. Proposition. — *Let P, P' be two parabolic subgroups of G and L a Levi subgroup of P .*

- a) *If $P \subset P'$, the transporter $T(P, P') = \{x \in G \mid {}^x P \subset P'\}$ of P into P' is equal to P' .*
- b) *$(P \cap P') \cdot R_u(P)$ is a parabolic subgroup of G . It is equal to P if and only if P' contains a Levi subgroup of P . It is a Borel subgroup of G if P' is one.*
- c) *The parabolic subgroups of G contained in P are the semidirect products of the parabolic subgroups of L by $R_u(P)$. Two parabolic subgroups of G contained in P are conjugate in G if and only if their intersections with L are conjugate in L .*

Proof. a) If ${}^x P \subset P'$, then ${}^x P' \cap P' \supset {}^x P$, hence ${}^x P' = P'$, and $x \in P'$ by 4.3.

- b) The intersection $P \cap P'$ contains a maximal torus of G (2.4). After conjugation by a suitable element of G , we can therefore assume, in the notation of 4.2, that $P = P_{\theta}$ and $P \cap P' \supset T$. Let ψ be the set of roots of P' with respect to T . It contains at least one element of each pair $\{a, -a\}$ ($a \in \Phi$); the same is therefore true for $\eta = (\pi_{\theta} \cap \psi) \cup \alpha_{\theta}$. But this set is closed in Φ , because $\pi_{\theta} \cap \psi$ is closed and α_{θ} is an ideal in π_{θ} . By Prop. 20 of [6, §7, n° 7], η contains the set of positive roots

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of G for a suitable order. As η is also the set of roots of $(P \cap P') \cdot R_u(P)$ with respect to T , it follows that $(P \cap P') \cdot R_u(P)$ contains a Borel subgroup of G (2.3), whence the first assertion of b).

If P' contains a Levi subgroup of P , the same holds for $P \cap P'$, hence $(P \cap P') \cdot R_u(P) = P$. Conversely, this equality implies that the elements $a \in [\theta]$ are all roots of $P \cap P'$ with respect to T , hence $P \cap P' \supset Z_{\theta}$. Suppose finally that P' is a Borel subgroup of G . Then the set ψ of roots of P' contains only one element of each pair $(a, -a)$ ($a \in \Phi$). The same is therefore true for the set η considered above, which is thus the set of positive roots for a suitable order [6, loc. cit.], and $(P \cap P') \cdot R_u(P)$ is then the corresponding Borel subgroup.

c) Every Borel subgroup of P contains $R_u(P)$ and projects onto a Borel subgroup of $P/R_u(P)$ by [1, th. 22.1]. Consequently, the Borel subgroups of P are the semi-direct products of the Borel subgroups of L with $R_u(P)$, whence the first part of c). If two parabolic subgroups R, R' contained in P are conjugate in G , they are so in P by a), hence their images in $P/R_u(P)$ are also conjugate, which is equivalent to saying that $L \cap R$ and $L \cap R'$ are conjugate in L . The converse is obvious.

Corollary 0.31 (4.5). *$P \cap P'$ is connected. If P' is conjugate to P and contains $R_u(P)$, then $P = P'$.*

The group $(P \cap P') \cdot R_u(P)$ is parabolic, hence connected, hence equal to $(P \cap P')^0 \cdot R_u(P)$, which shows that $P \cap P'$ is generated by $(P \cap P')^0$ and $H = (P \cap P') \cap R_u(P)$. But H is conjugate to a subgroup of U stable under T ; it is therefore connected (2.3), whence the first assertion.

If $P' \supset R_u(P)$, then $P \cap P'$ contains $(P \cap P') \cdot R_u(P)$, which is parabolic (4.4), and the second assertion follows from 4.3.

Lemma 0.32 (4.6). *Suppose G split over k . Let P, P' be two k -parabolic subgroups of G , and (T, U_a) a splitting datum over k of G . Then there exist $g \in G_k$ and $\theta \in \Delta$ such that ${}^g P = P_{\theta}$ and ${}^g P' \supset T$. In particular, $P \cap P'$ contains a maximal torus conjugate to T over k , and $R(P), R_u(P)$ are split over k .*

There exists a unique θ such that P is conjugate to P_{θ} over $\bar{k}$ (4.3). We will show first that P is conjugate to P_{θ} over k . The unipotent radical V_{θ} of P_{θ} is contained in a unique conjugate of P_{θ} (4.5), hence admits in G/P a unique fixed point M , which is then rational over k by 2.7 (i), since V_{θ} is split over k . The map $g \mapsto g \cdot M$ is then a separable k -morphism of G onto G/P , constant on the classes $x \cdot P_{\theta}$; it therefore induces a k -isomorphism of varieties φ from G/P_{θ} onto G/P . It is immediate that $\varphi^{-1}(P)$ is the fixed point M' of P in G/P_{θ} , which is therefore rational over k . But the fibration of G by P_{θ} has a local section with image V_{θ}^- (4.2), which is defined over k , and the projection $G_k \rightarrow (G/P_{\theta})_k$ is surjective (2.6, 2.14). If $x \in G_k$ maps to M' , then ${}^x P_{\theta} = P$.

This shows that we can reduce to the case where $P = P_{\theta}$, and that there exists $x \in G_k$

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such that $P'' = {}^x P \supset B$. But by (2.11) we can write $x = a \cdot n^{-1} \cdot b$ with $a, b \in B_k$ and $n \in \mathcal{N}(T)_k$. We then have ${}^b P = P$ and ${}^b P' = {}^n P'' \supset T$, which proves the lemma.

Proposition 0.33 (4.7). *Let P, P' be two k -parabolic subgroups of G . Then $P \cap P'$ contains a maximal torus of G defined over k . Every connected k -closed subgroup of G normalized by $P \cap P'$ is defined over k . In particular, $P \cap P'$, $R(P \cap P')$, $R_u(P \cap P')$ are defined over k . The group $P \cap P'$ possesses k -Levi subgroups and any two of them are conjugate by a unique element of $R_u(P \cap P')_k$.*

G is split over k_s (2.14), so $P \cap P'$ is defined over k_s (4.5, 4.6). Since it is k -closed, it is also defined over k . On the other hand, it is of maximum rank (4.6), so it contains a maximal torus of G defined over k (2.14). The other assertions are then consequences of 3.13 and 3.14.

4.8. Opposite parabolic subgroups. — Two parabolic subgroups of G are *opposite* if their intersection is a Levi subgroup of each of them.

It is clear that $P_\emptyset^-$ is the only parabolic subgroup opposite to $P_\emptyset$ containing $Z_\emptyset$. Consequently, in view of 4.3 and the conjugacy of Levi decompositions in a parabolic subgroup (4.7), given a parabolic subgroup P , the map $\alpha : P' \mapsto P \cap P'$ is a bijection from the set of parabolic subgroups opposite to P to the set of Levi subgroups of P . It then follows from 4.7 that *if P is a k -parabolic subgroup, G contains a k -parabolic subgroup opposite to P , and any two such subgroups are conjugate by a unique element of $R_u(P)_k$* . In particular, P and P' are opposite if and only if there exist $g \in G$ and $\theta \subset \Delta$ such that ${}^g P = P_\theta$ and ${}^g P' = P_\theta^-$.

4.9. Two conjugacy classes $\mathcal{P}, \mathcal{P}'$ of parabolic subgroups are said to be *opposite* if there exist parabolic subgroups $P \in \mathcal{P}, P' \in \mathcal{P}'$ which are opposite. In view of 4.8, there exists one and only one class opposite to a given class, and any parabolic subgroup opposite to an element of $\mathcal{P}$ (resp. $\mathcal{P}'$) belongs to $\mathcal{P}'$ (resp. $\mathcal{P}$). If $\mathcal{P} = \mathcal{P}'$, we say that $\mathcal{P}$ is *self-opposite*.

Fix an order on $\Phi(G)$ and let i be the opposition involution of Φ (2.1). If $n \in \mathcal{N}(T)$ represents the element of W which maps Φ^+ onto Φ^- , then

$${}^n P_\theta = P_{i(\theta)}^-, \quad (\theta \subset \Delta);$$

consequently, if $P_\theta \subset \mathcal{P}$, then the opposite class contains $P_{i(\theta)}^-$. If the symmetry $a \mapsto -a$ ($a \in \Phi$) belongs to $W(G)$, then i is the identity, so every conjugacy class of parabolic subgroups is self-opposite.

Proposition 0.34 (4.10). *Let P, P' be two parabolic subgroups of G and $\mathcal{P}, \mathcal{P}'$ their conjugacy classes. Then the following conditions are equivalent:*

- a) P and P' are opposite.
- b) $\mathcal{P}$ and $\mathcal{P}'$ are opposite and P, P' contain two opposite Borel subgroups of G .
- c) $(P \cap P') \cdot R_u(P) = P$, $(P \cap P') \cdot R_u(P') = P'$ and P, P' contain two opposite Borel subgroups of G .

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- d) The product map is an isomorphism of varieties from $P \times R_u(P')$ onto an open subset of G .
- e) $P \cap R_u(P') = P' \cap R_u(P) = \{e\}$.

If P and P' are opposite, then (4.8) there exist $g \in G$ and $\theta \subset \Delta$ such that ${}^g P = P_\theta$ and ${}^g P' = P_\theta^-$, hence a) $\Rightarrow$ b), c), e) and, taking 4.2 into account, a) $\Rightarrow$ d).

Suppose now that P and P' contain two opposite Borel subgroups. In view of 4.8, we can, after conjugation by a suitable element of G , assume that $P \supset B$ and $P' \supset B^-$, hence (4.3) that $P = P_\theta$ and $P' = P_\eta^-$ ($\theta, \eta \subset \Delta$). But P' is the unique element of $\mathcal{P}'$ containing B^- (4.3); if $\mathcal{P}'$ is opposite to $\mathcal{P}$, we must therefore have $\eta = \theta$, which proves that b) $\Rightarrow$ a). On the other hand, if c) is true, then $P \cap P' \supset Z_\theta, Z_\eta$ and Z_θ, Z_η are Levi subgroups of $P \cap P'$. Since they contain T , they are equal (3.13), hence c) $\Rightarrow$ a).

Let us now show that d) $\Rightarrow$ a). We may assume that $P = P_\theta$ and $P' \supset T$ (4.6). Then $R_u(P') \subset V_\theta^-$ and $\dim R_u(P') = \dim G - \dim P = \dim V_\theta^-$, hence $R_u(P') = V_\theta^-$ and $P' = P_\theta^-$.

It remains to prove that e) $\Rightarrow$ a). We can again assume that $P = P_\theta$ and $P' \supset T$. The equality $P \cap R_u(P') = \{e\}$ implies $R_u(P') \subset V_\theta^-$. On the other hand, the equality $P' \cap V_\theta = \{e\}$ shows that the Levi subgroup L of P' containing T is contained in Z_θ . The set $\Phi(T, L)$ is obviously symmetric; if it were strictly smaller than $[\theta]$, or if $R_u(P') \neq V_\theta^-$, then one could find a root α such that $U_\alpha \not\subset P'$ and $U_{-\alpha} \not\subset P'$, which is absurd, hence $P' = P_\theta^-$.

Proposition 0.35 (4.11). *Let P, P' be opposite parabolic subgroups of G . Then the subgroup generated by $R_u(P)$ and $R_u(P')$ is normal in G .*

Proof. We may assume that $P = P_\theta$ and $P' = P_\theta^-$ (4.8). The subgroup M generated by $R_u(P) = V_\theta$ and $R_u(P') = V_\theta^-$ is closed and connected [10, Exp. 3, th. 2], normalized by V_θ, V_θ^- , and Z_θ , hence by P_θ and P_θ^- . Consequently (2.3), the normalizer of M is open in G , and therefore equal to G . $\square$

Lemma 0.36 (4.12). *Let P, P' be two parabolic subgroups of G . Then the set M of elements $g \in G$ such that gP and P' contain opposite Borel subgroups is an open subset of G of the form $P' \cdot x \cdot P$ ($x \in G$); it has a rational point over k if k is infinite or if P and P' are defined over k .*

Proof. We first show that M is an open subset of the form $P' \cdot x \cdot P$ ($x \in G$). For this, we can replace P and P' by conjugate subgroups, and thus (4.3) assume that P, P' contain respectively B and B^- . According to 4.8, $x \in M$ if and only if there exists $g \in G$ such that ${}^gB \subset P$ and ${}^gB^- \subset P'$, which means (4.4 a)) that $g^{-1} \cdot x \in P$ and $g \in P'$, hence $x \in P' \cdot P$. The set $P' \cdot P$ is open since $B^- \cdot B$ is open (2.3).

If k is infinite, then G_k is Zariski dense in G (2.14), hence $M_k \neq \emptyset$. Now suppose k is finite and P, P' are defined over k . Then P and P' contain Borel subgroups defined over k , and two Borel subgroups of G defined over k are conjugate over k ([23, p. 45], [5, 4.10]). It therefore suffices to show that if B is a Borel subgroup of G defined over k , then there exists a Borel subgroup of G defined over k and opposite to B , but this follows from 4.8. $\square$

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4.13. Theorem. — *Let P, P' be k -parabolic subgroups of G .*

- a) *The fibration of G by P has a local section defined over k ; the projection $G_k \rightarrow (G/P)_k$ is surjective.*
- b) *If P and P' are minimal (among the k -parabolic subgroups), they are conjugate over k .*
- c) *If P and P' are conjugate over an extension of k , they are also conjugate over k .*

Proof. a) By 4.8, there exists a k -parabolic subgroup P'' opposite to P ; its unipotent radical is the image of a local k -section of the fibration of G by P , by 4.10; the second part of a) then follows from 2.6 and 2.14.

b) If k is finite, P and P' are k -Borel subgroups of G [23, p. 45], hence are conjugate over k [5, 4.10].

Suppose then that k is infinite, and remark first that if Q and Q' are minimal k -parabolic subgroups containing opposite Borel subgroups, then Q and Q' are opposite. Indeed, $(Q \cap Q') \cdot R_u(Q)$ and $(Q \cap Q') \cdot R_u(Q')$ are parabolic subgroups (4.4) defined over k (4.7), hence equal to Q and Q' respectively, and condition 4.10 c) is satisfied.

Let M (resp. M') be the set of elements $g \in G$ such that gP (resp. ${}^gP'$) and P contain opposite Borel subgroups; M and M' are non-empty open subsets of G (4.12). The same is therefore true for $M \cap M'$, and since G_k is dense in G , there exists $x \in M_k \cap M'_k$. By the remark made previously, xP and ${}^xP'$ are both opposite to P . Being defined over k , they are then conjugate by an element of $R_u(P)_k$ by 4.8, whence b).

c) P' is the unique subgroup conjugate to P containing $R_u(P')$ by 4.5, hence P' has a unique fixed point A on G/P , which is necessarily k -closed. But (2.14) G is split over k_s ; hence (4.6) the unipotent radical of P' is also split over k_s , and (2.7 (i)) has a fixed point in G/P rational over k_s . Consequently, A is also rational over k_s , hence rational over k . By a), we can find $x \in G_k$ who is mapped to A by the natural projection. We then have ${}^xP = P'$. $\square$

Remarks. — 1) The preceding proof shows that if k is infinite, two minimal k -parabolic subgroups have a common opposite defined over k . We will see later that this is still true over any field for a k -parabolic group P and an arbitrary element of the conjugacy class of P (6.27).

2) In view of 4.15, 4.13 a) is a special case of 3.25.

4.14. Corollary. — *The conjugacy class $\mathcal{P}$ of parabolic subgroups containing the minimal k -parabolic subgroups is self-opposite.*

Proof. Indeed, the opposite class contains an element defined over k (4.8), of the same dimension as the minimal k -parabolic subgroups, hence minimal, and conjugate to an element of $\mathcal{P}$ by 4.13. $\square$

4.15. Theorem. — a) *Let S be a split torus over k of G and Φ^+ (resp. Φ^-) the set of roots of G with respect to S which are > 0 (resp. < 0) for a given order on $X^*(S)$. Then G_{Φ^+}*

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and G_{Φ^-} are two opposite k -parabolic subgroups having $\mathcal{Z}(S)$ as a common k -Levi subgroup.

b) *Let P be a k -parabolic subgroup of G and S a maximal k -split torus of $\mathcal{R}(P)$. Then there exists an ordering on $\Phi(S, G)$ such that $P = G_{\Phi^+}$. The k -Levi subgroups of G are the centralizers of the maximal k -split tori of $\mathcal{R}(P)$.*

a) The groups G_{Φ^+} and G_{Φ^-} are defined over k since Φ^+ and Φ^- are. Let T be a maximal torus of G defined over k and containing S (2.15 d)). Let us fix an ordering on $X^*(T)$ compatible with the given ordering on $X^*(S)$ (3.1). Then $G_{\Phi^+} = G_{\eta}^{(T)} = G_{\eta}$, $G_{\Phi^-} = G_{-\eta}^{(T)}$, where η is the set of elements of $\Phi(T, G)$ whose restriction to S

belongs to $\Phi^+ \cup \{0\}$. Obviously, $\eta \supset \Phi(T, G)^+$, so $G_\eta, G_{-\eta}$ are parabolic, and (4.2) there exists a subset θ of the set of simple roots of G with respect to T such that $G_\eta = P_\theta$, hence $G_{-\eta} = P_{\bar{\theta}}$.

b) The group $(\mathcal{Z}(S) \cap P)^0$ is defined over k . This follows from 10.5, or can be seen as follows: P contains a maximal torus T' of G defined over k (2.14), which is therefore part of a splitting datum of G over k_s (2.14, 2.10); S is conjugate in P over k_s to a subtorus of T' (2.10), hence $(\mathcal{Z}(S) \cap P)^0$ is conjugate over k_s to a group normalized by T' , k_s -closed, hence defined over k_s (3.13). Consequently, $(\mathcal{Z}(S) \cap P)^0$ is k -closed and defined over k_s , hence defined over k .

Thanks to 2.14 a), we can find a maximal torus T of G defined over k such that $S \subset T \subset \mathcal{Z}(S) \cap P$. Let $S' = \mathcal{R}(P) \cap T$. This is a maximal torus of $\mathcal{R}(P)$, and a subgroup of T which is k -closed, hence (1.6) defined over k . Its centralizer $\mathcal{Z}(S')$ is then defined over k (2.15 d)), and is a Levi subgroup of P whose connected center is S' (4.2, 3.14). By construction, S is the largest k -split torus of S' , so (3.6) the weights of S in $\mathcal{R}_u(P)$ are > 0 for a suitable ordering on $X^*(S)$. Let us place on $X^*(T)$ an ordering compatible with the one just fixed on $X^*(S)$. Given the structure of parabolic groups (4.2, 4.3), we have $\Phi(T, G) = \mu \cup \nu \cup (-\nu)$ where μ is the set of roots trivial on S' and ν is the set of weights of T in $\mathcal{R}_u(P)$. Consequently, a root is trivial on S' if and only if it is trivial on S , and $\mu \cup \nu$ is the set of roots whose restriction to S is ≥ 0 . We then have $P = G_{\Phi^+}$ by definition (3.8) and $\mathcal{Z}(S) = \mathcal{Z}(S')$. Thus, the centralizer of a maximal k -split torus of $\mathcal{R}(P)$ is a k -Levi subgroup of P . Since the k -Levi subgroups of P are conjugate over k (3.14), this completes the proof of the theorem.

Corollary 0.37 (4.16). Let P be a k -parabolic subgroup of G . Then the following conditions are equivalent:

- (i) P is minimal.
 - (ii) $\mathcal{R}(P)$ contains a maximal k -split torus of G .
 - (iii) The k -Levi subgroups of P are the centralizers of the maximal k -split tori of G contained in $\mathcal{R}(P)$.
 - (iv) Any maximal k -split torus of G contained in P is contained in $\mathcal{R}(P)$.
- (i) $\Rightarrow$ (ii). Let S be a maximal k -split torus of $\mathcal{R}(P)$ and S' a split torus

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maximal over k of G containing S . Let us choose in $\Phi = \Phi(S, G)$ an order such that $P = G_{\Phi^+}$, which is possible by 4.15 b), and in $\Psi = \Phi(S', G)$ an order compatible with this one. The group $P' = G_{\Psi^+}^{(S')}$ is parabolic, admits $\mathcal{Z}(S')$ as a Levi subgroup, is defined over k (4.15 a)) and is contained in P . These groups are therefore equal, and $\mathcal{Z}(S')$ is a Levi subgroup of P , so $S' \subset R(P)$ and finally $S' = S$.

(ii) $\Rightarrow$ (iii). This follows from 4.15 b) and 4.7.

(iii) $\Rightarrow$ (iv) Let S (resp. S') be a maximal k -split torus of G contained in P (resp. $R(P)$). We have $P = \mathcal{Z}(S') \cdot R_u(P)$. The projection S'' of S onto the factor $\mathcal{Z}(S')$ of this decomposition is a k -split torus (1.9); the same is therefore true for $S' \cdot S''$ (1.4). Since S' is maximal, this means that $S'' \subset S'$, whence $S \subset S' \cdot R_u(P) = R(P)$.

(iv) $\Rightarrow$ (i). Let P' be a minimal parabolic k -subgroup contained in P , and S a maximal k -split torus of G contained in $R(P')$; by virtue of the implications (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii) already established, such a torus S exists and $\mathcal{Z}(S) \subset P'$. On the other hand, $S \subset R(P)$ and we have (4.15) $P = \mathcal{Z}(S) \cdot R_u(P) \subset P'$, whence $P = P'$.

Corollary 0.38 (4.17). *The group G possesses a proper parabolic k -subgroup if and only if it contains a non-central k -split torus.*

This is an obvious consequence of 4.15 a) and 4.16.

Corollary 0.39 (4.18). *The intersection of two parabolic k -subgroups P, P' of G contains the centralizer of a maximal k -split torus of G .*

We may assume P and P' are minimal. By 4.7, $P \cap P'$ contains a maximal torus T of G defined over k . The group $T \cap R(P)$ is a maximal torus of $R(P)$, defined over k by 1.6. By virtue of 4.2, 4.3 and 2.15 d), $\mathcal{Z}(T \cap R(P))$ is a Levi k -subgroup of P , hence (4.6) $T \cap R(P)$ contains a maximal k -split torus of G . The latter is then necessarily equal to the largest k -split torus T_d of T . Similarly, $T_d \subset R(P')$, and the inclusion $\mathcal{Z}(T_d) \subset P \cap P'$ is a consequence of 4.16.

Corollary 0.40 (4.19). *Let $P \subset P'$ be parabolic k -subgroups of G . Then the maximal k -split tori of $R(P')$ are the intersections of $R(P')$ with the maximal k -split tori of $R(P)$.*

Since $R(P) \supset R(P')$, this is true for at least one maximal k -split torus of $R(P')$, and thus for all of them by 4.15, the conjugation over k of Levi k -subgroups (3.14), and the fact that $R(P')$ is normal in $R(P)$.

Corollary 0.41 (4.20). *Let P, P' be parabolic k -subgroups of G . Then $P = P'$ if and only if $P_k = P'_k$.*

If k is infinite, this follows from the fact that P_k and P'_k are dense in P and P' (2.14, 3.20), but the proof below holds without any assumption on k .

Let Q, Q' be minimal parabolic k -subgroups of G contained in P and P' respectively. In view of 4.16, 4.18, we can find a maximal k -split torus S of G such that $\mathcal{Z}(S)$ is a common Levi k -subgroup of Q and Q' . Let $V = R(P) \cap S$. By 4.19 and 4.15, there exists a quasi-closed subset α of $\Phi(V, G)$ such that $P = G_\alpha^{(V)}$. If μ is

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the set of elements of $\Phi(S, G)$ whose restriction to V belongs to $\alpha \cup \{0\}$, then μ is quasi-closed and $P = G_\mu^{(S)}$ (3.8). Similarly, there exists a quasi-closed subset ν of $\Phi(S, G)$ such that $P' = G_\nu^{(S)}$. Our assertion is now a consequence of 3.23.

Theorem 0.42 (4.21). *The maximal k -split tori of G are conjugate over k .*

Proof. Let S, S' be two maximal k -split tori of G . By 4.15 and 4.16, there exist two minimal parabolic k -subgroups P and P' such that $\mathcal{Z}(S)$ and $\mathcal{Z}(S')$ are k -Levi subgroups of P and P' respectively. In view of 4.13, P and P' are conjugate over k , and by 4.7, the k -Levi subgroups of P are conjugate over k . Consequently, there exists $g \in G_k$ such that ${}^g\mathcal{Z}(S) = \mathcal{Z}(S')$. But then gS and S' coincide with the largest k -split torus of the center Z of $\mathcal{Z}(S')$ (recall that Z^0 is defined over k by 2.15 a)). $\square$

Remark 2. The theorem holds more generally for a connected algebraic k -group whose unipotent radical is split over k (11.6).

Corollary 0.43 (4.22). *Let S, S' be maximal k -split tori of G and let A, A' be subsets of S_k and S'_k , or subtori of S and S' respectively. Let $x \in G_k$ such that ${}^xA = A'$. Then there exists $y \in G_k$ such that ${}^yS = S'$ and ${}^ya = {}^xa$ ($a \in A$).*

Proof. Let $S'' = {}^xS$. The groups S' and S'' are maximal k -split tori of the group $\mathcal{Z}(A')^0$, which is defined over k (1.6, 2.15 d)). By the theorem, we can find $z \in \mathcal{Z}(A')^0_k$ such that ${}^zS'' = S'$. Then $y = z \cdot x$ satisfies our conditions. $\square$

Definition 0.1 (4.23). The dimension (common by 4.21) of the maximal k -split tori of G is the k -rank of G and is denoted by $r_k(G)$. The semisimple k -rank of G is the k -rank of the derived group of G . If k is algebraically closed, the prefix or subscript k is generally omitted. The group G is anisotropic over k if its k -rank is zero, or, what amounts to the same by 4.17, if it contains no proper parabolic k -subgroup and if its connected center is anisotropic over k .

4.24. Suppose that G is the identity component of the orthogonal group $\mathbf{O}(Q)$ of a quadratic form Q on a k -vector space V . Then G is anisotropic over k if and only if V_k contains no non-zero isotropic vector, in other words if Q does not represent zero on k .

Indeed, if V_k contains a non-zero isotropic vector, one can classically choose coordinates in V_k such that $Q = c \cdot x_1 \cdot x_2 + Q'(x_3, \dots, x_n)$ ($c \in k^*$), and G contains the transformations $(x_1 \mapsto \lambda \cdot x_1, x_2 \mapsto \lambda^{-1} \cdot x_2, x_i \mapsto x_i \ (i \geq 3))$ ($\lambda \in k^*$) which define a k -split torus of G .

Conversely, if G contains a k -split torus $S \neq \{e\}$, then, diagonalizing S over k , we see that there exist $x \in V_k \setminus \{0\}$ and an integer $m \geq 1$ such that $Q(x) = Q(\lambda^m \cdot x) = \lambda^{2m} \cdot Q(x)$ for all $\lambda \in k$, whence $Q(x) = 0$.

Similarly, we see that if Q is non-degenerate, the k -rank of G is equal to the index of Q , that is, to the dimension of the maximal isotropic subspaces (of course, rational over k) of Q .

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Proposition 0.44 (4.25). *Let G' be a reductive k -group and $f : G \rightarrow G'$ a separable k -isogeny. Then the maximal k -split tori of G' (resp. of G) are the images under f (resp. the identity components of the inverse images) of the maximal k -split tori of G (resp. G'). In particular, G and G' have the same k -rank and the same semisimple k -rank.*

(The notion of separable morphism is recalled in 10.3; here, this is equivalent to requiring that f induces an isomorphism of Lie algebras.)

Proof. Since the image of a k -split torus under a k -morphism is a k -split torus (1.3), it suffices, in view of 4.21, to show that if S' is a maximal k -split torus of G' , then $S = (f^{-1}(S'))^0$ is a k -split torus of G .

Let T' be a maximal torus of G' defined over k and containing S' (2.15 d)). Then $T = (f^{-1}(T'))^0$ is a maximal torus of G [1, th. 22.1]. It contains consequently the kernel of f , so T is the entire inverse image of T' . Since f is separable, T is an irreducible rational cycle over k [22, Prop. 1, Cor.], hence T is a k -subgroup [39, Prop. 1, p. 208]; 1.8 then shows that S is the largest k -split torus of T . $\square$

Remark 3 (4.26). The preceding proposition is not true without restriction on f , as shown by the following example.

Suppose k is non-perfect of characteristic two, and let t be an element of k which is not a square in k , Q the quadratic form in three variables given by $Q(x, y, z) = x^2 + t \cdot y \cdot z$, and $G = \mathbf{O}(Q)$ the orthogonal group of Q . The form Q does not represent zero over k , so G is anisotropic over k (4.24). On the other hand, the bilinear form associated to Q admits the line $y = z = 0$ as kernel. We deduce immediately that the elements of $G_{\bar{k}}$ are the matrices of the form

$$M = \begin{pmatrix} I & (t \cdot a \cdot b)^{1/2} & (t \cdot c \cdot d)^{1/2} \\ 0 & a & c \\ 0 & b & d \end{pmatrix}, \quad (a \cdot d - b \cdot c = 1),$$

hence that $M \mapsto \begin{pmatrix} a & c \\ b & d \end{pmatrix}$ is a k -isogeny of G onto $\mathbf{SL}_2 = G'$. However, $r_k(G) = 0$ and $r_k(G') = 1$.

Proposition 0.45 (4.27). *Let $G_1, \dots, G_n$ be connected normal k -subgroups of which G is the almost direct product (0.6), S a maximal k -split torus of G , and $S_i = (S \cap G_i)^0$ ($1 \leq i \leq n$). Then S is the almost direct product of the S_i and S_i is a maximal k -split torus of G_i ($1 \leq i \leq n$). In particular, $r_k(G) = \sum_i r_k(G_i)$.*

Proof. Thanks to 4.25, we can reduce to the case where G is the direct product of the G_i . The projection S'_i of S onto G_i ($1 \leq i \leq n$) is a k -split torus (1.3), so $\prod S'_i = S$, and $S'_i = S_i$ ($1 \leq i \leq n$). On the other hand, if S''_i is a k -split torus of G_i containing S_i , then $S \cdot S''_i$ is a k -split torus (1.3, 1.9), hence equal to S , whence $S''_i = S_i$, and S_i is indeed maximal in G_i . $\square$

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4.28. Corollary. — *G possesses a largest connected normal k -subgroup H which is anisotropic over k . The group G is the almost direct product of this subgroup, of its normal almost simple k -subgroups over k of k -rank > 0 , and of its largest central k -split torus.*

The subgroup in question is generated by the connected normal almost simple k -subgroups over k which are anisotropic over k of $\mathcal{D}G$ (see 2.15 b)) and by the largest k -anisotropic subtorus of $\mathcal{Z}(G)$ (2.15 a), 1.8).

We conclude this section with two results which will be used in § 8.

4.29. Proposition. — *Let $\{H_i\}_{i \in I}$ be a set of connected k -subgroups of G , P a minimal parabolic k -subgroup of G , and H the subgroup generated by $R_u(P)$ and the H_i ($i \in I$). Suppose that for all $i \in I$, the group H_i is normalized by a parabolic k -subgroup P_i . Then $\mathcal{N}(H) = P'$ is a parabolic k -subgroup containing P , and $R_u(P') = R_u(H)$.*

The group $(P \cap P_i) \cdot R_u(P)$ is a parabolic k -subgroup contained in P , hence equal to P , and P_i contains a Levi k -subgroup L_i of P (4.4, 4.7). The group F_i generated by H_i and $R_u(P)$ is normalized by L_i , hence by $P = L_i \cdot R_u(P)$ and consequently $\mathcal{N}(H) \supset P$. The group $\mathcal{N}(H)$, being k -closed, is then defined over k (4.7). Since $P' \supset P$, we have $R_u(P') \subset R_u(P) \subset H$; since moreover $R_u(P')$ is normalized by $H \subset P'$, we also have $R_u(P') \subset R_u(H)$. On the other hand, $R_u(H)$ is normalized by P' , hence $R_u(H) \subset R_u(P')$.

4.30. Corollary. — *Let $\{P_i\}_{i \in I}$ be a set of minimal parabolic k -subgroups. Then the subgroup H generated by the groups $R_u(P_i)$ is normal in the group generated by the P_i ($i \in I$).*

Indeed, 4.29 shows that P_i normalizes H for any $i \in I$.

§ 5. ROOTS, WEYL GROUPS AND RELATIVE CELLULAR DECOMPOSITIONS

Throughout this section, G is a connected reductive group, and S is a maximal k -split torus of G .

5.1. Relative Weyl group and k -roots. — The roots of G with respect to S will be called k -roots or roots relative to k of G (with respect to S if this precision seems necessary). We shall write ${}_k\Phi(G)$ or ${}_k\Phi$ for $\Phi(S, G)$. The relative Weyl group of G relative to k is the quotient $\mathcal{N}(S)/\mathcal{Z}(S)$ and is denoted by ${}_kW$, or

${}_k W(G)$, ${}_k W(S, G)$. Since $\mathcal{Z}(S)$ is the identity component of $\mathcal{N}(S)$, this group is finite. It operates canonically on $X^*(S)$ (leaving ${}_k \Phi$ stable), on $X_*(S)$ (by permuting the kernels of the k -roots), hence also on $X^*(S) \otimes \mathbf{A}$ and $X_*(S) \otimes \mathbf{A}$, where $\mathbf{A}$ is any ring. An inner product on $X^*(S) \otimes \mathbf{R}$ or $X_*(S) \otimes \mathbf{R}$ invariant under ${}_k W$ is called admissible, and will be denoted in general by $(\ , \)$. Since ${}_k W$ is finite, such an inner product always exists. We will often identify $X^*(S) \otimes \mathbf{R}$ and $X_*(S) \otimes \mathbf{R}$ by means of an admissible inner product.

If k is algebraically closed, or else if G is split over k , then ${}_k W$ and ${}_k \Phi$ are identified with the usual Weyl group and roots, and the mention of k will be omitted.

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By 4.21, the root sets $\Phi(S, G)$ and the relative Weyl groups associated to two maximal k -split tori are isomorphic.

5.2. Given $a \in {}_k \Phi(S, G)$, we will denote by ${}_k U_a$ or $U_{(a)}$ the group denoted by $G_{(a)}^{*(S)}$ in 3.8, where (a) is the set of k -roots which are positive integer multiples of a . This group is unipotent (3.8 (iv)), defined over k (3.13).

Let $w \in {}_k W$, n a representative of w in $\mathcal{N}(S)$, and ψ a quasi-closed subset of ${}_k \Phi$. In the notation of 3.8, it is clear that

$${}^n G_\psi = G_{w(\psi)}, \quad {}^n G_\psi^* = G_{w(\psi)}^*. \quad (1)$$

The groups ${}^n G_\psi$ and ${}^n G_\psi^*$ therefore depend only on the image w of n in ${}_k W$ and will also be denoted by ${}^w G_\psi$ and ${}^w G_\psi^*$. These groups are defined over k (3.13).

Let A be a subset of $\mathcal{N}(S)$ (resp. $\mathcal{N}(S)_k$) and A' its canonical image in ${}_k W(G)$. Classes of elements of G (resp. G_k) such as $P.A$, $P.A.P$ (resp. $P_k.A$, $P_k.A.P_k$) which depend only on A' will also be denoted by $P.A'$, $P.A'.P$ (resp. $P_k.A'$, $P_k.A'.P_k$).

Theorem 0.46 (5.3). *The rank of ${}_k \Phi(G)$ is equal to the semi-simple k -rank of G (4.23). The group ${}_k W(G)$, viewed as a group of automorphisms of $X_*(S) \otimes \mathbf{R}$, equipped with an admissible scalar product, is generated by the reflections with respect to the hyperplanes annihilating a k -root. Each connected component of $\mathcal{N}(S)$ meets G_k .*

Let S_a ($a \in {}_k \Phi$) be the identity component of the kernel of a , and let Z be the intersection of the groups S_a . The group Z^0 is the identity component of $\mathcal{Z}(G) \cap S$, because the adjoint representation of G has $\mathcal{Z}(G)$ as kernel [23, lemma 1, p. 39]. (This also follows from the fact that if $z \in Z$, then all roots of G with respect to a maximal torus of G containing S are trivial on z , and from 2.3.) Since $r_k(G) = r_k(\mathcal{Z}(G)) + r_k(\mathcal{D}G)$ by 2.2 (ii) and 4.27, this establishes the first assertion.

Let us now show that if $r_k(\mathcal{D}G) \neq 0$, then $\mathcal{N}(S)_k \neq \mathcal{Z}(S)_k$. By 4.15, there exist two opposite k -parabolic subgroups P and P' having $\mathcal{Z}(S)$ as a common Levi subgroup. By 4.16, these groups are minimal among the k -parabolic subgroups, hence we can find $x \in G_k$ such that ${}^x P = P'$ (4.13). The groups ${}^x(\mathcal{Z}(S))$ and $\mathcal{Z}(S)$ are two k -Levi subgroups of P' , so by (4.7), up to multiplying x by an element of $R_u(P')_k$, we can assume that x normalizes $\mathcal{Z}(S)$. It must then also normalize S , which is the largest k -split torus in the center of $\mathcal{Z}(S)$; on the other hand, it does not act trivially on ${}_k \Phi$ by 4.15, hence $x \in \mathcal{N}(S)_k$ and $x \notin \mathcal{Z}(S)_k$.

Let us now prove that ${}_k W$ contains the group W' generated by the reflections r_a with respect to the hyperplanes $X_*(S_a) \otimes \mathbf{R}$ of $X_*(S) \otimes \mathbf{R}$ ($a \in {}_k \Phi$).

Let $M = \mathcal{Z}(S_a)$. It is a connected reductive k -group (2.15 d)). It contains the unipotent group $U_{(a)}$ (5.2), so $\mathcal{Z}(S) \neq M$, and S is not central in M . By 4.27, and the fact that S_a is of codimension one in S , this implies that $r_k(\mathcal{D}M) = 1$ and that $V = (S \cap \mathcal{D}M)^0$ is a maximal k -split torus of $\mathcal{D}M$. By what has already been proved, we can find in $(\mathcal{D}M)_k$ an element x which normalizes but does not centralize V . Since $\dim V = 1$, we must have $(\text{Int } x)(t) = t^{-1}$ ($t \in V$). On the other hand, x centralizes S_a , therefore

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normalizes S and induces a non-trivial involutive automorphism of $X_*(S) \otimes \mathbf{R}$ leaving $X_*(S_a) \otimes \mathbf{R}$ pointwise fixed. This is necessarily a symmetry with respect to this hyperplane for any admissible scalar product, hence $r_a \in {}_k W$ and $W' \subset {}_k W$; moreover, every connected component of $\mathcal{N}(S)$ projecting into W' contains an element of G_k .

It remains to show that ${}_k W = W'$. Given the classical properties of groups generated by reflections [6], it suffices for this to prove that if $w \in {}_k W$ stabilizes a Weyl chamber C of W' in $X_*(S) \otimes \mathbf{R}$, then w is the identity. There exists a point $D \in C$ fixed by w . One can therefore find an order on $X^*(S)$ such that if $a \in X^*(S)$ is non-zero on D , then $a > 0$ if and only if $a(D) > 0$. Since the roots are non-zero on D , the element w permutes

the set ψ of k -roots > 0 , hence any $n \in \mathcal{N}(S)$ representing w normalizes the group $G_\psi = P$, which is a minimal k -parabolic group by 4.15, 4.16. Since P is its own normalizer (4.3), we have $n \in P$ and it suffices to show that

$$\mathcal{N}(S) \cap P = \mathcal{Z}(S),$$

which is immediate: indeed, we can write $n = u \cdot v$ ($u \in \mathcal{Z}(S)$, $v \in R_u(P)$) (4.16), whence $v \in R_u(P) \cap \mathcal{N}(S)$, and it is known that in a connected solvable group, the normalizer of a torus is equal to its centralizer [1, prop. 10.2].

5.4. Corollary. — *The largest connected normal k -subgroup M anisotropic over k of $\mathcal{Z}(S)$ is normal in $\mathcal{N}(S)$.*

The group M is obviously normalized by $\mathcal{N}(S)_k$. On the other hand (4.28), $\mathcal{Z}(S)$ is the almost direct product of S and M , hence M is normal in $\mathcal{Z}(S)$, and consequently in $\mathcal{N}(S) = \mathcal{N}(S)_k \cdot \mathcal{Z}(S)$.

Remark. — Slightly more generally, this shows that any normal k -subgroup of $\mathcal{Z}(S)$ invariant under any k -automorphism of $\mathcal{Z}(S)$ is normal in $\mathcal{N}(S)$.

5.5. Corollary. — *Let K be an extension of k , T a maximal K -split torus of G containing S . Let $\mathcal{N}(S, T) = \mathcal{N}(S) \cap \mathcal{N}(T)$ and ${}_K W_k = \mathcal{N}(S, T)/\mathcal{Z}(T)$. Then $\mathcal{N}(S) = \mathcal{N}(S, T) \cdot \mathcal{Z}(S)$, and ${}_k W$ identifies with the restriction of ${}_K W_k$ to S .*

To prove the first equality, it suffices, by the theorem, to show that $\mathcal{N}(S)_k \subset \mathcal{N}(S, T) \cdot \mathcal{Z}(S)$, which follows from 4.22. The second assertion is just another way of stating the first.

5.6. Corollary. — *Let $f : G \rightarrow G'$ be a separable k -isogeny from G to a connected reductive k -group G' , and let $S' = f(S)$. Then f induces an isogeny from $\mathcal{N}(S)$ to $\mathcal{N}(S')$, an isomorphism from ${}_k W(G)$ to ${}_k W(G')$, and $f^* : X^*(S') \rightarrow X^*(S)$ induces a bijection from ${}_k \Phi(G')$ to ${}_k \Phi(G)$.*

The group S' is a maximal k -split torus of G' by 4.25, and f induces an isomorphism of the Lie algebras, whence the last part of the corollary. The other assertions then follow from the theorem.

5.7. Proposition. — *Let ψ be a subset of ${}_k \Phi(G)$ containing all rational multiples of its elements that belong to ${}_k \Phi(G)$, w an element of ${}_k W(G)$ which is a product of reflections r_a ($a \in \psi$), and $c \in X^*(S)$. Then $w(c) - c$ is a linear combination with integer coefficients of elements of ψ .*

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Let T be a maximal torus of G containing S and $j : X^*(T) \rightarrow X^*(S)$ the restriction homomorphism. Let $\eta = j^{-1}(\psi) \cap \Phi(T, G)$. By 5.5, $W = W(T, G)$ contains at least one element w_a whose restriction to S is r_a ($a \in {}_k \Phi$). Such an element acts trivially on the hyperplane V_a of $X_*(S) \otimes \mathbf{R}$ annihilating a , and therefore [6] is a product of reflections $r_b \in W$, where b belongs to the set of roots that vanish on V_a . These latter roots belong to η , given the assumption made on ψ , hence W contains an element w' whose restriction to S is w , and which is a product of reflections r_b ($b \in \eta$). Since on the other hand S is a direct factor in T [1, § 7], j is surjective. We are thus reduced to the case where $S = T$ is a maximal torus of G and where ${}_k W = W$, ${}_k \Phi = \Phi(G)$, for which our assertion is known [6, § 7, no. 9, prop. 27].

Corollary 0.47 (5.8). *Let $(\cdot, \cdot)$ be an admissible scalar product on $X^*(S) \otimes \mathbf{R}$, $a \in {}_k \Phi(G)$ and $c \in X^*(S)$. Then $2(a, c)/(a, a)$ is an integer. In particular, ${}_k \Phi(G)$ is a root system in the sense of 2.1 in $(X^*(S), N)$ where N denotes the set of characters of S that are trivial on $S \cap \mathcal{D}G$.*

The restriction $X^*(S) \rightarrow X^*(S')$, where $S' = (S \cap \mathcal{D}G)^0$, is an isomorphism of ${}_k \Phi(G)$ onto ${}_k \Phi(\mathcal{D}G)$, and the rank of ${}_k \Phi(\mathcal{D}G)$ is equal to $r_k(\mathcal{D}G)$ by 5.3, hence ${}_k \Phi(G)$ generates in $X^*(S) \otimes \mathbf{Q}$ a complementary subspace of $N \otimes \mathbf{Q}$. On the other hand, we have

$$r_a(c) = c - 2 \cdot (a, c) \cdot (a, a)^{-1} \cdot a,$$

hence $2(a, c) \cdot (a, a)^{-1} \in \mathbf{Z}$ by 5.7. The other conditions imposed on a root system are satisfied in view of 5.3.

Corollary 0.48 (5.9). *If P is a minimal k -parabolic subgroup containing S , then $P = G_\psi^{(S)}$, where ψ is the set of positive k -roots for a suitable order, and conversely. The group $\text{Int}_G \mathcal{N}(S)$ induces a simply transitive permutation group, isomorphic to ${}_k W$, on the minimal k -parabolic subgroups containing S .*

The first assertion follows from 4.15, 4.16, the second from 5.3, 5.8 and from the simple transitivity of the Weyl group on the Weyl chambers.

Proposition 0.49 (5.10). *Let $G_1, \dots, G_n$ be connected normal k -subgroups of G of which G is the almost direct product and $S_i = (S \cap G_i)^0$ ($1 \leq i \leq n$). Then, for any $a \in {}_k \Phi(G)$, there is one and only one value of the index i such that the restriction of a to S_i is non-zero; the group ${}_k \mathbf{U}_a$ is contained in the corresponding group G_i . If Φ_i is the set of non-zero k -roots on S_i , then Φ is the direct sum of the Φ_i and the restriction homomorphism $X^*(S) \rightarrow X^*(S_i)$ induces an isomorphism of Φ_i onto ${}_k \Phi(G_i)$.*

Let T be a maximal torus of G containing S and $b \in \Phi(T, G)$ a root whose restriction to S is a . For any i , we have the following implications

$$a|_{S_i} \neq 0 \Rightarrow b|_{S_i} \neq 0 \Rightarrow U_b \subset G_i \Rightarrow b|_{T \cap G_j} = 0 \ (j \neq i) \Rightarrow a|_{S_j} = 0 \ (j \neq i) \Rightarrow a|_{S_i} \neq 0$$

the second and third of which follow from 2.3 and the others are obvious. These assertions are therefore all equivalent, hence the first part of the conclusion. The second follows immediately.

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5.11. Corollary. — The connected normal k -subgroups of G almost simple over k and non-anisotropic over k are the groups G_ψ^* , where ψ runs through the irreducible direct factors of ${}_k\Phi(G)$. In particular, if G is almost simple over k , ${}_k\Phi$ is an irreducible root system. Every connected normal k -subgroup of G is a product of a group G_ψ^* , where ψ is an ideal of Φ , a split torus over k , and an anisotropic group over k , which are uniquely determined.

Let ψ be an irreducible direct factor of ${}_k\Phi$. It is an ideal, so G_ψ^* is a connected normal k -subgroup of G ; Prop. 5.10 shows that G_ψ^* is almost simple over k and that every connected normal k -group almost simple over k , of non-zero k -rank, is obtained in this way. The last assertion then follows from 4.28.

5.12. Standard parabolic k -subgroups. — We fix an order on ${}_k\Phi$ and we denote by ${}_k\Delta$ the set of corresponding simple k -roots. To any subset θ of ${}_k\Delta$, we can associate as in 4.2 subsets $\pi_\theta, \pi_\theta^-,$ etc. of ${}_k\Phi$ and subgroups $P_\theta, P_\theta^-,$ etc. which are defined over k by 3.13; paragraphs 4.2 (i), (ii) remain valid, once T and Φ are replaced by S and ${}_k\Phi$. To avoid confusion with the corresponding notions over $\bar{k}$, we will often write ${}_kP_\theta, {}_kP_\theta^-, {}_kZ_\theta, {}_kV_\theta, {}_kV_\theta^-$ for the groups which, in the notation of 3.8, would be denoted by $G_{\pi_\theta}^{(S)}, G_{\pi_\theta^-}^{(S)}, \mathcal{Z}(S_\theta), U_{\pi_\theta^-}^{(S)}$ and $U_{\pi_\theta}^{(S)}$. The groups ${}_kP_\theta$ ($\theta \subset {}_k\Delta$) are the standard parabolic k -subgroups of G . The group ${}_kP_\theta$ is minimal if $\theta = \emptyset$, equal to G if $\theta = {}_k\Delta$. The unipotent radical ${}_kV_\emptyset$ of ${}_kP_\emptyset$ will also be denoted by ${}_kU$. Up to No. 5.20 inclusive, we will write P for ${}_kP_\emptyset$.

5.13. Let $w \in {}_kW$ and let $\mu_w = {}_k\Phi^+ \cap w({}_k\Phi^-)$ and $\nu_w = {}_k\Phi^+ \cap w({}_k\Phi^+)$. These are convex sets of k -roots of which ${}_k\Phi^+$ is the disjoint union. We set

$${}_kU'_w = U'_{\mu_w} = U_{\mu_w}^{(S)}, \quad {}_kU''_w = U''_{\nu_w} = U_{\nu_w}^{(S)}. \quad (1)$$

By 3.11, the product map is a k -isomorphism of varieties from $U'_w \times U''_w$ onto ${}_kU$.

More generally, let $s \in {}_kW$,

$$\mu_{s,w} = s({}_k\Phi^+) \cap w({}_k\Phi^-), \quad \nu_{s,w} = s({}_k\Phi^+) \cap w({}_k\Phi^+).$$

These are convex sets of k -roots, of which $s({}_k\Phi^+)$ is the disjoint union. We have

$$U_{\mu_{s,w}}^{(S)} = {}^s({}_kU) \cap w({}_kU^-), \quad U_{\nu_{s,w}}^{(S)} = {}^s({}_kU) \cap w({}_kU), \quad (2)$$

and the product map is an isomorphism of varieties from $U_{\mu_{s,w}}^{(S)} \times U_{\nu_{s,w}}^{(S)}$ onto ${}^s({}_kU)$.

Proposition 0.50 (5.14). *The standard parabolic k -subgroups are the only k -subgroups containing $P = {}_kP_\emptyset$. Every parabolic k -subgroup of G is conjugate over k to a unique standard parabolic k -subgroup; G is generated by P and the subgroups $U_{(-\alpha)}$ ($\alpha \in {}_k\Delta$).*

Let P' be a parabolic k -subgroup containing P . By 4.16, 4.19, the intersection $S' = R(P') \cap S$ is a maximal split torus over k for $R(P')$, and, by 4.15, one can write $P' = G_\psi^{(S')}$, where ψ is the set of elements of $\Phi(S', G)$ which are > 0 for a suitable order. We therefore also have $P' = G_\eta^{(S)}$, where η is the set of k -roots whose restriction to S' is either zero or contained in ψ . Since ψ is closed in $\Phi(S', G)$, the set η

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is closed in ${}_k\Phi$, contains ${}_k\Phi^+$ since $P' \supset P$, and the existence of $\theta \subset {}_k\Delta$ such that $\eta = \pi_\theta$ then follows from [6, § 7, n° 7, prop. 20].

Let Q be a k -parabolic subgroup of G . In view of 4.13 and what has just been proved, Q is conjugate over k to at least one standard k -parabolic subgroup. The uniqueness of the latter is a consequence of 4.3. Finally, the preceding shows that the subgroup generated by P and the groups $U_{(-\alpha)}$ ($\alpha \in {}_k\Delta$) is the group P_θ ($\theta = {}_k\Delta$), hence is equal to G .

5.15. Theorem. — *Let $N = \mathcal{N}(S)$ and $U = {}_kU$. Let $s \in {}_kW$ and $n, n' \in N$. Then $G_k = {}^sU_k \cdot N_k \cdot U_k$. We have $n = n'$ if and only if ${}^sU \cdot n \cdot U = {}^sU \cdot n' \cdot U$. The set ${}^sU \cdot n \cdot U$ is locally closed in G and the product map is*

an isomorphism of varieties from $({}^sU \cap {}^sU^-) \times \{n\} \times U$ onto ${}^sU \cdot n \cdot U$, which is defined over k if $n \in N_k$. The group G_k is the disjoint union of the double cosets $P_k \cdot w \cdot P_k$ ($w \in {}_k W$).

Using the fact that N_k contains an element whose canonical image in ${}_k W$ is s , we immediately reduce to the case where $s = 1$.

Let $g \in G_k$. According to 4.18, the group $P \cap g^{-1} \cdot P \cdot g$ contains the centralizer $\mathcal{Z}(S')$ of a maximal k -split torus S' of G . The groups $\mathcal{Z}(S)$ and $\mathcal{Z}(S')$ are two k -Levi subgroups of P , hence (3.14) are conjugate by an element of $R_u(P)_k$, which shows the existence of $u \in P_k$ such that $u \cdot g^{-1} \cdot P \cdot g \cdot u^{-1} \supset \mathcal{Z}(S)$. Using 5.9, we see that we can then find $v \in N_k$ such that $v \cdot u \cdot g^{-1} \in \mathcal{N}(P)_k$; since P is equal to its normalizer (4.3), this proves that $g \in P_k \cdot N_k \cdot P_k$, hence also that $g \in U_k \cdot N_k \cdot U_k$, in view of the existence of the Levi decomposition $P = \mathcal{Z}(S) \cdot U$.

The set $U \cdot n \cdot U$ is an orbit of $U \times U$ operating on G by left and right translations, hence is open in its closure [1, § 15.2], i.e., is locally closed. Let w be the canonical image of n in ${}_k W$. We have, in the notation of 5.13,

$$U \cdot n \cdot U = U'_w \cdot U''_w \cdot n \cdot U = U'_w \cdot n \cdot U, \quad (1)$$

since $n^{-1} \cdot U''_w \cdot n \subset U$, and thus the product map defines a surjective morphism $\varphi : U'_w \times \{n\} \times U \rightarrow U \cdot n \cdot U$. As $n^{-1} \cdot U'_w \cdot n \subset U^-$, the composition of φ and left translation by n^{-1} is the restriction to a closed subset of the product map $U^- \times U \rightarrow U^- \cdot U$. As the latter is an isomorphism (4.10), it follows that φ is an isomorphism; moreover, the groups U'_w and U being defined over k , φ is also defined over k if $n \in N_k$. Let us now prove that $U \cdot n \cdot U = U \cdot n' \cdot U$ implies $n = n'$, in other words that $N \cap U \cdot n \cdot U = \{n\}$. In view of (1), this is further equivalent to $N \cap n^{-1} \cdot U'_w \cdot n \cdot U = \{e\}$, which is a consequence of

$$N \cap U^- \cdot U = \{e\}. \quad (2)$$

It therefore suffices to establish (2). Let $n = v \cdot u$ ($n \in N$, $v \in U^-$, $u \in U$). For any $s \in S$, we obviously have

$$\left(v^{-1} \cdot {}^s v \cdot {}^{s^{-1}} v^{-1} \right) \cdot ({}^n s \cdot s^{-1}) \cdot (s \cdot u \cdot s^{-1} \cdot u^{-1}) = e.^9$$

But the factors defined by the parentheses belong respectively to U^- , S , U , and must therefore all be equal to 1 by 4.10. This implies that $u, v \in \mathcal{Z}(S)$, whence $u = v = 1$ and $n = 1$.

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The first part of the theorem and the decomposition $P_k = \mathcal{Z}(S)_k \cdot U_k$ show that G_k is the union of the double cosets $P_k \cdot w \cdot P_k$. It remains to show that the latter are disjoint. Let $w' \in P_k \cdot w \cdot P_k$ and let n', n be representatives of w' and w in N_k . Then there exist $u, v \in U_k$ and $a, b \in \mathcal{Z}(S)_k$ such that $n' = u \cdot a \cdot n \cdot b \cdot v$, whence $n' = a \cdot n \cdot b$ and $w' = w$.

Proposition 0.51 (5.16). *Let $w \in {}_k W$ and r be a simple reflection. We have:*

- (i) $r \cdot P_k \cdot \{w, r \cdot w\} \cdot P_k = P_k \cdot \{w, r \cdot w\} \cdot P_k$.
- (ii) $r \cdot P_k \cdot r \neq P_k$.

Let a be the k -simple root such that $r = r_a$. Up to replacing w by $r \cdot w$, we can assume that $w^{-1}(a) \in {}_k \Phi^+$. The group U is the semidirect product of the unipotent radical V_a of the standard parabolic k -group $P_{\{a\}}$ and of the group $U_{\{a\}}$, and this latter is the intersection of U with $\mathcal{Z}(S_a)$. We therefore have

$$r \cdot P_k \cdot \{w, r \cdot w\} \cdot P_k = \mathcal{Z}(S)_k \cdot r \cdot U_k \cdot \{w, r \cdot w\} \cdot P_k = \mathcal{Z}(S)_k \cdot r \cdot V_{a,k} \cdot U_{\{a\},k} \cdot \{w, r \cdot w\} \cdot P_k,$$

$$r \cdot P_k \cdot \{w, r \cdot w\} \cdot P_k = \mathcal{Z}(S)_k \cdot V_{a,k} \cdot r \cdot U_{\{a\},k} \cdot \{w, r \cdot w\} \cdot P_k.$$

Applying 5.15 to $\mathcal{Z}(S_a)$, we deduce

$$r \cdot P_k \cdot \{w, r \cdot w\} \cdot P_k \subset P_k \cdot \mathcal{Z}(S_a)_k \cdot w \cdot P_k = P_k \cdot U_{\{a\},k} \cdot \{1, r\} \cdot U_{\{a\},k} \cdot w \cdot P_k$$

hence, since $w^{-1}(a) > 0$,

$$r \cdot P_k \cdot \{w, r \cdot w\} \cdot P_k \subset P_k \cdot \{r \cdot w, w\} \cdot P_k.$$

Multiplying both sides on the left by r , we also obtain the opposite inclusion, whence (i).

(ii) is a consequence of 3.23 and of the fact that $r({}_k \Phi^+) \neq {}_k \Phi^+$.

⁹[[UNCERTAIN: $\left(v^{-1} \cdot {}^s v \cdot {}^{s^{-1}} v^{-1} \right) \cdot ({}^n s \cdot s^{-1}) \cdot (s \cdot u \cdot s^{-1} \cdot u^{-1}) = e$. — CONFIDENCE: 92% — REASON: The superscripts in the equation are small and slightly faint in the print, but this represents the most mathematically consistent transcription.]]

Proposition 0.52 (5.17). *Let θ be a subset of ${}_k\Delta$ and W_θ the subgroup of ${}_kW$ generated by the simple reflections corresponding to elements of θ . Then $P_{\theta,k} = P_k \cdot W_\theta \cdot P_k$.*

The relative Weyl group of Z_θ (cf. 5.12) is W_θ . By virtue of 5.15, we therefore have

$$P_{\theta,k} = Z_{\theta,k} \cdot V_{\theta,k} = (U \cap Z_\theta)_k \cdot W_\theta \cdot (U \cap Z_\theta)_k \cdot V_{\theta,k},$$

whence

$$P_{\theta,k} = P_k \cdot P_{\theta,k} \cdot P_k = P_k \cdot W_\theta \cdot P_k.$$

Corollary 0.53 (5.18). (i) *The groups $P_{\theta,k}$ are the only subgroups of G_k containing P_k .*

(ii) *Let $\theta, \theta' \subset {}_k\Delta$ and $g \in G_k$. Then ${}^gP_{\theta',k} \subset P_{\theta,k}$ if and only if $\theta' \subset \theta$ and $g \in P_{\theta,k}$. In particular, $P_{\theta,k}$ is its own normalizer in G_k .*

5.15 and 5.16 show that the axioms of [33] are satisfied; 5.18 then follows from 5.17 and Ths. 2, 3, 4 of [33].

Corollary 0.54 (5.19). *Let Q be a parabolic k -subgroup of G . Then Q_k is the normalizer in G_k of $R_u(Q)_k$.*

In view of 5.14, we may assume that $Q = P_\theta$ ($\theta \subset {}_k\Delta$). The normalizer M of $R_u(Q)_k$ then contains $P_{\theta,k}$, hence (by 5.18) is equal to a group $P_{\theta',k}$ ($\theta \subset \theta' \subset {}_k\Delta$). Suppose...

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that $\theta' \neq \emptyset$. Let a be an element of θ' not contained in θ , and n a representative in N_k of the fundamental reflection r_a . We have $U_{(-a),k} \subset R_u(P_\theta)_k$, whence also

$$U_{(-a),k} = {}^nU_{(a),k} \subset R_u(P_\theta)_k \subset P_k,$$

which is absurd, hence $\theta' = \emptyset$ and $M = Q_k$.

Corollary 0.55 (5.20). *Let θ, θ' be two subsets of ${}_k\Delta$. Then there exists between the double cosets $P_{\theta,k} \cdot g \cdot P_{\theta',k}$ ($g \in G_k$) of G_k and the double cosets $W_\theta \cdot w \cdot W_{\theta'}$ of ${}_kW$ ($w \in {}_kW$) a bijective correspondence characterized by the relation*

$$P_{\theta,k} \cdot g \cdot P_{\theta',k} = P_k \cdot W_\theta \cdot w \cdot W_{\theta'} \cdot P_k.$$

Let $g \in G_k$. Let n be the element of N_k such that $g \in U_k \cdot n \cdot U_k$ (5.15), and w the canonical image of n in ${}_kW$. Using 5.16 (i) and 5.17, we have

$$P_{\theta,k} \cdot g \cdot P_{\theta',k} = P_k \cdot W_\theta \cdot P_k \cdot w \cdot P_k \cdot W_{\theta'} \cdot P_k \subset P_k \cdot W_\theta \cdot w \cdot W_{\theta'} \cdot P_k \subset P_{\theta,k} \cdot w \cdot P_{\theta',k} = P_{\theta,k} \cdot g \cdot P_{\theta',k}.$$

It then follows from the last assertion of 5.15 that two distinct double cosets $W_\theta \cdot w \cdot W_{\theta'}$, $W_\theta \cdot w' \cdot W_{\theta'}$ correspond to distinct double cosets of G_k modulo $P_{\theta,k}$, $P_{\theta',k}$.

We will end this section with some simple remarks on the conjugacy classes of parabolic subgroups.

Proposition 0.56 (5.21). *Let $\mathcal{P}, \mathcal{Q}$ be two conjugacy classes of parabolic subgroups of G and $P, P' \in \mathcal{P}$, $Q, Q' \in \mathcal{Q}$ such that $P \cap Q$ and $P' \cap Q'$ are parabolic subgroups. Then:*

- (i) *If $P \subset Q$, every element of $\mathcal{P}$ is contained in a unique element of $\mathcal{Q}$.*
- (ii) *The groups $P \cap Q$ and $P' \cap Q'$ are conjugate.*
- (iii) *The parabolic subgroups R, R' generated by P, Q and P', Q' respectively are conjugate.*

Assertion (i) is contained in 4.3.

Let us resume the notation of 4.2. We can find $g, g' \in G$ such that

$${}^g(P \cap Q) \supset B, \quad {}^{g'}(P' \cap Q') \supset B;$$

according to 4.3, there then exist $\theta, \psi \subset \Delta$ such that

$${}^gP = {}^{g'}P' = P_\theta, \quad {}^gQ = {}^{g'}Q' = P_\psi.$$

It is then clear that

$$\begin{aligned} {}^g(P \cap Q) &= {}^{g'}(P' \cap Q') = P_{\theta \cap \psi}, \\ {}^gR &= {}^{g'}R' = P_{\theta \cup \psi}, \end{aligned}$$

whence the proposition.

Definition 0.2 (5.22). Let $\mathcal{P}, \mathcal{Q}$ be two conjugacy classes of parabolic subgroups of G . We write $\mathcal{P} \prec \mathcal{Q}$ if every element of $\mathcal{P}$ is contained in an element of $\mathcal{Q}$.

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of $\mathcal{Q}$. The preceding proposition shows that it is sufficient for this that there exists an element of $\mathcal{P}$ contained in an element of $\mathcal{Q}$.

According to 5.21, the parabolic subgroups of the form $P \cap Q$ ($P \in \mathcal{P}, Q \in \mathcal{Q}$) (resp. generated by two subgroups $P \in \mathcal{P}, Q \in \mathcal{Q}$ such that $P \cap Q$ is parabolic) form a single conjugacy class, which will be denoted by $\mathcal{P} \wedge \mathcal{Q}$ (resp. $\mathcal{P} \vee \mathcal{Q}$).

5.23. We will denote by ${}_k\mathcal{P}_\theta$ the conjugacy class of the standard k -parabolic group ${}_kP_\theta$ ($\theta \subset {}_k\Delta$) (5.12). We obviously have

$${}_k\mathcal{P}_\theta \subset {}_k\mathcal{P}_\psi \iff \theta \subset \psi \quad (\theta, \psi \subset {}_k\Delta).$$

If $\mathcal{P}$ and $\mathcal{Q}$ each contain an element defined over k , the same holds for $\mathcal{P} \wedge \mathcal{Q}$ and $\mathcal{P} \vee \mathcal{Q}$ in view of 4.7, 4.13. According to 5.14, there exist $\theta, \psi \subset {}_k\Delta$ such that $\mathcal{P} = {}_k\mathcal{P}_\theta$ and $\mathcal{Q} = {}_k\mathcal{P}_\psi$. We obviously have:

$${}_k\mathcal{P}_\theta \wedge {}_k\mathcal{P}_\psi = {}_k\mathcal{P}_{\theta \cap \psi}, \quad {}_k\mathcal{P}_\theta \vee {}_k\mathcal{P}_\psi = {}_k\mathcal{P}_{\theta \cup \psi} \quad (\psi, \theta \subset {}_k\Delta).$$

Finally, we see exactly as in 4.9 that if i is the opposition involution of ${}_k\Phi$, then the class opposite to ${}_k\mathcal{P}_\theta$ is ${}_k\mathcal{P}_{i(\theta)}$.

5.24. The group $\Gamma = \text{Gal}(k_s/k)$ acts in the usual way on the set of k_s -subvarieties of G and visibly leaves invariant the set of k_s -parabolic subgroups of G . In the following, we will have to consider the following condition, imposed on a conjugacy class $\mathcal{P}$ of parabolic subgroups:

- (i) *The set $\mathcal{P}_{k_s}$ of elements of $\mathcal{P}$ defined over k_s is stable under Γ ,*

and we will say that $\mathcal{P}$ is defined over k if (i) is satisfied. Although this is not necessary in this work, we will give another interpretation of it here.

A parabolic group is equal to its normalizer, so given $P_0 \in \mathcal{P}$, we can identify $\mathcal{P}$ with G/P_0 by the map φ_0 which associates to $P \in \mathcal{P}$ the unique fixed point of P in G/P_0 . The map φ_0 commutes with G , which acts on $\mathcal{P}$ by inner automorphisms and on G/P_0 by left translations, whence a homogeneous space structure (for an algebraic group) on $\mathcal{P}$, which visibly does not depend on the choice of the origin P_0 . It is always this structure that will be referred to below.

If $\mathcal{P}$ contains an element P defined over an extension K of k , then K is a field of definition of the homogeneous space $\mathcal{P}$ (i.e., of $\mathcal{P}$ and of the morphism $G \times \mathcal{P} \rightarrow \mathcal{P}$ defining the action of G) as can be seen by taking P as the origin. In particular (2.14, 4.3), the homogeneous space $\mathcal{P}$ is defined over k_s . This being so, condition (i) is equivalent to:

- (ii) *The homogeneous space $\mathcal{P}$ admits k as a field of definition.*

Indeed, (i) $\Rightarrow$ (ii) by the criteria of descent of the base field (see [37], or also [5, 2.12, 4.10]). The converse is immediate.

Let us remark that it can happen that $\mathcal{P}$ is defined over k , but has no element defined over k . For example, the conjugacy class of Borel subgroups of G is always defined over k .

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§ 6. DESCENT OF THE BASE FIELD

6.1. Notations. — In this section, K always denotes an extension of k , $\Gamma = \text{Aut}(K/k)$ the group of k -automorphisms of K , G a connected reductive group, S a maximal k -split torus of G , and T a maximal K -split torus of G containing S . The groups $X^*(S)$ and $X^*(T)$ are endowed with compatible orderings (i.e., the restriction to S of a positive character of T is positive), and ${}_k\Delta, {}_K\Delta$ denote the corresponding sets of simple relative roots. Note however that the hypothesis $S \subset T$ (and *a fortiori* the compatibility of orderings) does not intervene before no. 6.6.

6.2. An action of Γ on $X^*(T)$. — Let γ be an element of Γ . By virtue of 4.21, 5.9, there exists an inner automorphism α by an element of G_K transforming the torus ${}^\gamma T$ into T and the set of simple roots ${}^\gamma({}_K\Delta)$ relative to ${}^\gamma T$ into ${}_K\Delta$. The automorphism of $X^*(T)$ induced by the composite $\alpha \circ \gamma$ is independent of the chosen inner automorphism α (two such α differing on the left by an inner automorphism centralizing T) and will be

denoted by ${}_K\Delta\gamma$, or, by abuse of notation, $\Delta\gamma$. This automorphism obviously leaves ${}_K\Delta$ and ${}_K\Phi$ invariant, hence also the diagram of ${}_K\Delta$ (2.1). We have the formula

$$({}_K\Delta\gamma(c))(\alpha({}^\gamma t)) = \gamma(c(t)) \quad (c \in X^*(T), t \in T_K, \gamma \in \Gamma). \quad (1)$$

Let P be a K -parabolic subgroup of G containing T . It is immediate, from 5.12, 5.14, that the restriction homomorphism $X^*(P) \rightarrow X^*(T)$ identifies $X^*(P)_K$ with a subgroup of $X^*(T)$, which is of finite index if P is minimal. We can therefore consider $\Delta\gamma$ as operating on $X^*(P)_K$. If β is an inner automorphism of G transforming ${}^\gamma P$ onto P , this action satisfies

$$(\Delta\gamma(c))(\beta({}^\gamma p)) = \gamma(c(p)) \quad (c \in X^*(P)_K, p \in P_K, \gamma \in \Gamma), \quad (2)$$

as can be seen from (1), by noting that $P = \mathcal{N}(P)$ and that P acts trivially on its characters by inner automorphisms.

Let θ be a subset of ${}_K\Delta$. We have the obvious relation

$$\gamma({}_K P_\theta) = {}_K P_{\theta'} \quad (\theta' = \Delta\gamma(\theta)). \quad (3)$$

This constitutes a new definition of $\Delta\gamma$, if we add the fact that, for every character $c \in X^*(T)$ which vanishes on $(T \cap \mathcal{D}G)^0$, we have $\Delta\gamma(c) = {}^\gamma c$.

The group Γ operates on $X^*(T)$ via the ${}_K\Delta\gamma$. When T is invariant under Γ (in particular, when T is defined over k), it is obviously appropriate to distinguish this action from the one usually considered, notably above (1.7), and which we shall sometimes call the *usual* action of Γ on $X^*(T)$. The connection between these two actions can be characterized as follows: for every $\gamma \in \Gamma$, there exists a unique element w of the Weyl group ${}_K W$ such that $w(\gamma({}_K\Delta)) = {}_K\Delta$, and we then have ${}_K\Delta\gamma = w \circ \gamma$. We further remark that if P is defined over k , we have, by (2),

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$$\Delta\gamma(c) = {}^\gamma c \quad (c \in X^*(P)_K). \quad (4)$$

In particular, c is then defined over k if and only if it is fixed by all $\Delta\gamma$ ($\gamma \in \Gamma$).

It follows from 4.21 and 5.9 that the Γ -module obtained by making Γ act on $X^*(T)$ via $\Delta\gamma$ does not depend, up to isomorphism, on T or on ${}_K\Delta$. It is therefore an invariant of the group G (relative to the two fields k, K). On the other hand, if T and T' are two maximal K -split tori, both invariant under Γ , the usual Γ -modules $X^*(T)$ and $X^*(T')$ are not necessarily isomorphic¹⁰.

6.3. Descent and parabolic subgroups. — Let θ be a subset of ${}_K\Delta$. The relation ${}_K P_\theta = {}_K P_{\theta'}$ defines a subset θ' of ${}_K\Delta$ which will be denoted by ${}_K\eta_k(\theta)$. It follows immediately from 4.9 and 5.23 that ${}_K\eta_k$, which is therefore a map from the set of subsets of ${}_K\Delta$ to the set of subsets of ${}_K\Delta$, is compatible with intersection, union, and opposition. This being so, there exists a map ${}_K\rho_k : {}_K\Delta \rightarrow {}_K\Delta \cup \{0\}$, characterized by the relation

$${}_K\eta_k(\theta) = {}_K\rho_k^{-1}(0) \cup \bigcup_{a \in \theta} {}_K\rho_k^{-1}(a).$$

The meaning of this map will appear in the sequel (6.8). We will also denote

$${}_K\Delta_k^0 = {}_K\eta_k(\emptyset) = {}_K\rho_k^{-1}(0),$$

the subset of ${}_K\Delta$ corresponding to the conjugacy class of minimal parabolic subgroups defined over k . In all the notations introduced above, the double index K, k will eventually be omitted if no confusion is likely to result.

A subset θ of ${}_K\Delta$ is said to be *apparent over k* if it is the image under η of a subset of ${}_K\Delta$, that is, if ${}_K P_\theta$ is conjugate to a parabolic subgroup defined over k , and *defined over k* if the conjugacy class of ${}_K P_\theta$ is defined over k (cf. 5.24). It follows immediately from 4.7 that

(1) *A subset θ of ${}_K\Delta$ is apparent over k if and only if it is defined over k and contains ${}_K\Delta_k^0$.*

The set of subsets apparent over k and the set of subsets defined over k are invariant under Γ (for the action defined in no. 6.2) and under the opposition involution.

When T is defined over k , the notion of a set of characters defined over k , introduced in no. 1.7, does not coincide with the one introduced here in the case where the characters in question are simple roots. In this

¹⁰An example of J.-P. Serre shows moreover that there may be no maximal K -split torus stable under Γ .

section, the expression “defined over k ,” when speaking of a set of simple roots, will always be understood in the sense given to it here.

6.4. Galois characterization of subsets of ${}_K\Delta$ defined over k . — Let Δ be the system of simple roots of G . We have the canonical identifications $\Delta = {}_{k_s}\Delta = {}_K\Delta$. The group $\Gamma(k) = \text{Gal}(k_s/k)$ acts on ${}_{k_s}\Delta$, hence on Δ . Let us denote by η_k the map ${}_{k_s}\eta_k$, considered as a map from the set of subsets of ${}_k\Delta$ to the set of subsets of Δ . By an immediate Galois descent, we see that the subsets of ${}_{k_s}\Delta = \Delta$ defined over k are the subsets invariant under $\Gamma(k)$. Consequently,

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(1) A subset of ${}_k\Delta$ is defined over k if and only if its transform under η_K is invariant under $\Gamma(k)$.

When K is a normal extension of k , $\Gamma \cong \Gamma(k)/\Gamma(K)$, so $\Gamma(k)$ operates on ${}_K\Delta$ via Γ . It is clear that the action of $\Gamma(k)$ commutes with the map η_K , so:

(2) When K is a normal extension of k , a subset of ${}_k\Delta$ is defined over k if and only if it is invariant under Γ . In particular (by 6.3, (1)), the images under $\kappa\rho_k^{-1}$ of the elements of ${}_k\Delta$ are the orbits of Γ in ${}_K\Delta$ not contained in ${}_K\Delta_k^0$.

When K is a radical extension of k , every subset of ${}_k\Delta$ is defined over k . More generally, every conjugacy class of parabolic subgroups defined over K is also defined over k (5.24). On the other hand, it is not always true that every subset of ${}_k\Delta$ is apparent over k (that is, that the maximal k -split tori are also maximal K -split) as shown by the example of the group $\mathbf{O}_3^+(\mathbf{Q})$ corresponding to an anisotropic, non-degenerate quadratic form $\mathbf{Q}$ of defect 1 over a field k of characteristic 2, K being a radical extension of k on which $\mathbf{Q}$ acquires non-trivial zeros.

6.5. k -forms, index, classification. — Let $\overline{G}$ be an algebraic group defined over K . The group G is called a k -form of $\overline{G}$ if it is isomorphic to $\overline{G}$ over K . Let us identify $\overline{G}$ with G , over K , so as to be able to consider T and ${}_K\Delta$ as a split torus and a set of simple K -roots of $\overline{G}$. We call the *index* of the k -form G the datum consisting of ${}_K\Delta_k^0$ and of the set of subsets of ${}_k\Delta$ defined over k (from which we also deduce, by 6.3, (1), the set of apparent subsets). If the extension K/k is normal, the *Galois index* consists of the datum of ${}_K\Delta_k^0$ and of the action of Γ on ${}_K\Delta$.

From the sole datum of the diagram of ${}_K\Delta$, one can deduce, independently of the group $\overline{G}$ and of the fields K and k , conditions which must be satisfied by a set of subsets of ${}_K\Delta$, closed under intersection and union, and an element Δ^0 of this set, in order to be able to be the index of a form. Some conditions already follow immediately from 6.3 and 6.4. We know in particular that the index must be invariant under the opposition involution. Let us note that this last condition is, by itself, very restrictive (cf. e.g. [32], no. 5), if we take into account the fact that the restriction of the index to a subset Δ' of ${}_K\Delta$ apparent over k is also a k -form index (it is the index of the k -Levi subgroups of a k -parabolic subgroup belonging to ${}_k\mathfrak{P}_{\Delta'}$). Other conditions imposed on possible indices result from Theorem 6.13 below. Elements of classification can be found in [34, 35, 36].

Let us also point out the existence of a “Witt theorem”, allowing one to characterize a k -form of a given group over $\bar{k}$ by its Galois index and by the k -structure (with specification of the isomorphism over $\bar{k}$) of the centralizer $\mathcal{Z}(S)$ of a maximal k -split torus (cf. [28], [32]; the proof of [28] extends to a non-perfect field if use is made of 2.14).

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Lemma 0.57 (6.6). *Let T and T' be two maximal split tori over K containing S , let there be given in $X^*(T)$ and $X^*(T')$ compatible orders with the same order in $X^*(S)$, and let ${}_K\Delta$ and ${}_K\Delta'$ be the corresponding sets of simple roots. Then, any element $g \in G$ such that $\text{Int } g$ transforms T into T' and ${}_K\Delta$ into ${}_K\Delta'$ belongs to $\mathcal{Z}(S)$.*

Proof. Let P (resp. Q ; resp. Q') be the minimal standard parabolic subgroup defined over k (resp. K ; resp. K) corresponding to the order chosen in $X^*(S)$ (resp. $X^*(T)$; resp. $X^*(T')$). We have $P \supset Q$, $P \supset Q'$ and ${}^gQ = Q'$. By virtue of 4.3, it follows that ${}^gP = P$, hence $g \in P$ and $g = g' \cdot g''$ with $g' \in R_u(P)$ and $g'' \in \mathcal{Z}(S)$. For all $s \in S$, we have $(g, s) = (g', s) \in R_u(P)$, and $(g, s) = {}^gs \cdot s^{-1} \in T' \cdot s^{-1} \subset \mathcal{Z}(S)$, hence $(g, s) = e$ and $g \in \mathcal{Z}(S)$. $\square$

Proposition 0.58 (6.7). *Suppose T is invariant under Γ . Then, for all $c \in X^*(T)$ and all $\gamma \in \Gamma$, the difference $\Delta_\gamma(c) - \gamma(c)$ is a linear combination with integer coefficients of roots vanishing on S .*

Proof. Let w be the element of ${}_K W$ such that $w(\gamma({}_K \Delta)) = {}_K \Delta$. The transform by γ of the order given in $X^*(T)$ is compatible with the order given in $X^*(S)$, and the set of simple roots corresponding to it is $\gamma({}_K \Delta)$. By virtue of the preceding lemma, w thus centralizes S , and $[6]$ is a product of reflections relative to roots vanishing on S . The proposition is then a consequence of 5.7 and of the fact that ${}^\Delta \gamma = w \circ \gamma$. $\square$

Proposition 0.59 (6.8). *Let $j = X^*(T) \rightarrow X^*(S)$ be the restriction homomorphism. Then $j(a) = {}_K \rho_K(a) \in {}_K \Delta \cup \{0\}$ ($a \in {}_K \Delta$).*

Let $k_1 \subset k_2 \subset k_3$ be three fields. We have the obvious transitivity relation ${}_{k_2} \rho_{k_1} \circ {}_{k_3} \rho_{k_2} = {}_{k_3} \rho_{k_1}$ (by setting $\rho(0) = 0$). It follows that:

- (1) si the proposition is true for $k = k_1$, $K = k_3$ and for $k = k_2$, $K = k_3$, it is also true for $k = k_1$, $K = k_2$;
- (2) if the proposition is true for $k = k_1$, $K = k_2$, and if a maximal split torus over k_2 is also maximal split over k_3 , the proposition is true for $k = k_1$, $K = k_3$.

By applying (1) with k_3 algebraically closed, we reduce to the case where K is algebraically closed and then, by applying (2) with $k_2 = k_s$, to the case where we have $K = k_s$, which we shall assume from now on. Let $b \in {}_K \Delta$, $a = \rho(b)$, $a' = b|_S$, (x) be the set of roots of the form $n \cdot x$ with $n \in \mathbf{N}$, and ${}_K P_\theta$ ($\theta \in {}_K \Delta$) the standard parabolic subgroups with respect to S and ${}_K \Delta$ (5.12). It follows from the definition of ρ that $a = 0$ if and only if $U_{(-b)} \subset {}_K P_\emptyset$, and that if $a \neq 0$, $U_{(-b)} \subset {}_K P_a$. On the other hand, if $a' \neq 0$, $U_{(-b)} \subset U_{(-a')}$. Taking 3.22 c) into account, we thus have the implications

$$a' \neq 0 \iff U_{(-b)} \not\subset {}_K P_\emptyset \iff a \neq 0 \implies U_{(-b)} \subset {}_K P_a \implies a' \in \{a, 2a, 0\}.$$

It follows that, in all cases, $a' = a$ or $a' = 2a$.

From Lemma 6.6, it follows that two elements of the same orbit of Γ in ${}_K \Delta$ (with respect to the action defined in no. 6.2) have the same restriction to S . By virtue of 6.4 (2), we thus have, for all $a \in {}_K \Delta$, $\rho^{-1}(a)|_S = \{a\}$ or $\{2a\}$. But every K -root is a linear combination with integer coefficients of elements of ${}_K \Delta$, hence every k -root is a combination

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... linear with integer coefficients of elements of ${}_K \Delta|_S$; consequently, $\rho^{-1}(a)|_S = \{a\}$ for all a .

Corollary 0.60 (6.9). *Let S' be the connected component of the identity element of the subgroup of T defined by the equations $a = 1$ for all $a \in \Delta^0$, and $b = c$ for all $b, c \in {}_K \Delta$ such that $\rho(b) = \rho(c)$ (that is, of the kernel of the set of characters $\Delta^0 \cup \{b - c \mid b, c \in {}_K \Delta, \rho(b) = \rho(c)\}$). Then $(S \cap DG)^0 = (S' \cap DG)^0$.*

This follows from the preceding proposition and an obvious dimension calculation.

6.10. Weyl Groups. — Let us set

$$\mathcal{N}(S, T) = \mathcal{N}(S) \cap \mathcal{N}(T), \quad \mathcal{N}^0(S, T) = \mathcal{Z}(S) \cap \mathcal{N}(T).$$

We will denote by ${}_k W_k = \mathcal{N}(S, T)/\mathcal{Z}(T)$ and ${}_k W_k^0 = \mathcal{N}^0(S, T)/\mathcal{Z}(T)$ the respective canonical images of $\mathcal{N}(S, T)$ and $\mathcal{N}^0(S, T)$ in ${}_k W$. In particular, ${}_k W_k^0$ is the k -Weyl group of $\mathcal{Z}(S)$, generated by the fundamental reflections corresponding to the elements of Δ^0 . It is a normal subgroup of ${}_k W_k$, and we have (cf. 5.5) a canonical isomorphism ${}_k W = {}_k W_k / {}_k W_k^0$.

We will denote by R (resp. ${}_k R$) the sublattice of $X^*(T)$ (resp. $X^*(S)$) generated by the K -roots (resp. k -roots) and we set $X = R \otimes \mathbf{R}$ and $Y = {}_k R \otimes \mathbf{R}$, the groups R and ${}_k R$ being identified with sublattices of X and Y . The restriction map $j : R \rightarrow {}_k R$ extends to a linear map $X \rightarrow Y$, which coincides with ρ on ${}_K \Delta$, and whose kernel N is generated by Δ^0 and by the differences $b - c$ of roots $b, c \in {}_K \Delta$ such that $\rho(b) = \rho(c)$ (cf. 6.9). Having chosen an admissible scalar product in X (5.1), we can identify Y with $N^\perp$, which we shall do. Then, ${}_k W$ being viewed as a group of linear transformations of X , ${}_k W_k$ (resp. ${}_k W_k^0$) is the normalizer (resp. centralizer) of Y in ${}_k W$, and ${}_k W$ is the restriction of ${}_k W_k$ to Y . In particular, the restriction to Y of the chosen scalar product on X is admissible. Admissible scalar products on X and Y which are in the relation thus described will be said to be *compatible*.

If the elements of ${}_K \Delta$ are all defined over k in the sense of 6.3 (for example, if the extension K/k is normal and if Γ operates trivially on ${}_K \Delta$), ${}_k W_k$ is the normalizer of ${}_k W_k^0$ in ${}_k W$; indeed, Y is then the space of all fixed points of ${}_k W_k^0$ in X .

6.11. Suppose that T is defined over k and that the extension K/k is normal. Then, $N^\perp$ is the space of fixed points of the *usual* action of Γ on X . It follows that in the identification of Y with $N^\perp$, the canonical image $j(a)$ in Y of an element $a \in X$ (for example, the restriction to S of a character of T trivial on $\mathcal{Z}(G)$) is identified with the orbit

$$a^0 = (1/|\Gamma|) \cdot \sum_{\gamma \in \Gamma} \gamma a,$$

of a under the action of Γ (when Γ is infinite, it must be replaced, in the preceding formula, by a finite quotient through which it operates on X and whose order is denoted by $|\Gamma|$). In particular, we have, for compatible scalar products, and for any $a, b \in X$,

$$(j(a), j(b)) = (a^0, b^0) = (a, b^0) = (a^0, b). \quad (1)$$

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Finally, ${}_k W_k$ is then the group of all $w \in {}_k W$ such that ${}^\gamma w \equiv w \pmod{{}_k W_k^0}$ for all $\gamma \in \Gamma$, where the left exponent is relative to the action of Γ on ${}_k W$ induced by the usual action of Γ on X .

6.12. The following theorem allows us to determine the diagram of ${}_k \Delta$ from the diagram of ${}_K \Delta$ and the set of $\rho^{-1}(a)$. To simplify its statement, we will use the following notation. Let $M = ((m_{ab}))$ ($a, b \in {}_K \Delta$) be a matrix whose rows and columns are indexed by ${}_K \Delta$. Then, $\rho(M)$ will denote the matrix $((m'_{cd}))$ ($c, d \in {}_k \Delta$), whose rows and columns are indexed by ${}_k \Delta$ and whose coefficients are given by

$$m'_{cd} = \sum_{a \in \rho^{-1}(c)} \sum_{b \in \rho^{-1}(d)} m_{ab}.$$

6.13. Theorem. — (i) *The spaces $X = \mathbf{R} \otimes \mathbf{R}$ and $Y = {}_k \mathbf{R} \otimes \mathbf{R}$ being equipped with compatible inner products, if ${}_K M$ (resp. ${}_k M$) denotes the matrix of inner products of the simple K -roots (resp. of the simple k -roots), we have ${}_k M = (\rho({}_K M^{-1}))^{-1}$.*

(ii) *Let $a \in {}_k \Delta$. Then, the largest integer n_a ($n_a = 1, 2$) such that $n_a \cdot a$ is a k -root is the maximum of the sum of the coefficients of the elements of $\rho^{-1}(a)$ in the K -roots which are linear combinations of elements of $\rho^{-1}(a) \cup \Delta^0$.*

Let $x = K$ or k , and let $\tilde{m}_{xy}$ be the coefficients of the matrix ${}_x M^{-1}$. For all $x \in {}_x \Delta$, let us set $x^* = \sum_y \tilde{m}_{xy} \cdot y$. We have $(x^*, y) = \delta_{xy}$, that is to say, $\{x^*\}$ is the dual basis of ${}_x \Delta$ (in X or in Y , depending on whether $x = K$ or k).

As in no. 6.10, we will identify Y with $N^\perp$, where N denotes the kernel of the canonical map from X to Y . Then, for all $c \in {}_k \Delta$, c^* is defined by the relations

$$(c^*, a) = \delta_{c, \rho(a)} \quad \text{for all } a \in {}_K \Delta. \quad (1)$$

Indeed, the element of X thus defined, which we will temporarily denote by c' , is orthogonal to all differences $a - b$ of simple K -roots such that $\rho(a) = \rho(b)$, as well as to all elements of Δ^0 , and thus also to N . Consequently, it belongs to Y and we have, for all $a \in {}_K \Delta$, $(c', \rho(a)) = (c', a) = \delta_{c, \rho(a)}$, whence $c' = c^*$. From (1), it follows that $c^* = \sum_{a \in \rho^{-1}(c)} a^*$, that is to say,

$$\sum_{d \in {}_k \Delta} {}_k \tilde{m}_{cd} \cdot d = \sum_{a \in \rho^{-1}(c)} \sum_{b \in {}_K \Delta} {}_K \tilde{m}_{ab} \cdot b.$$

Taking the inner product of both sides with d^* , we obtain, again taking (1) into account,

$${}_k \tilde{m}_{cd} = \sum_{a \in \rho^{-1}(c)} \sum_{b \in \rho^{-1}(d)} {}_K \tilde{m}_{ab},$$

which proves (i).

(ii) is an immediate consequence of 6.8.

The preceding theorem allows us in particular to calculate, from ${}_K \Delta$ and the $\rho^{-1}(a)$, the order of the product of two fundamental reflections in ${}_k W$ (or, what amounts to the same, the angle between two simple k -roots). However, this can also be obtained using a simpler formula, which we shall establish now.

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Proposition 0.61 (6.14). *Let a, b be two k -simple roots and let φ_0 (resp. φ_a , resp. φ_b , resp. φ_{ab}) be the cardinality of the reduced root system having as its set of simple roots Δ^0 (resp. $\eta(a) = \Delta^0 \cup \rho^{-1}(a)$; resp. $\eta(b)$; resp. $\eta(a, b)$). Then, the order m_{ab} of the product of the corresponding fundamental reflections in ${}_k W$ is given by*

$$m_{ab} = 2(\varphi_{ab} - \varphi_0) / (\varphi_a + \varphi_b - 2\varphi_0).$$

In this proof, the roots in question are only ever given up to a strictly positive proportionality factor. In particular, the equality $a = b$ means $a = r \cdot b$ ($r \in \mathbf{R}$, $r > 0$).

Let ψ be the system of k -roots which are linear combinations of a and b , $W(\psi)$ its Weyl group, and $2m_{ab}$ the order of $W(\psi)$; for every k -root x , let $f(x)$ be the cardinality of the set of K -roots whose restriction to S is x . For the restriction to S of a K -root c to be zero (resp. proportional to a ; resp. to b ; resp. to be a linear combination

of a and b), it is necessary and sufficient (6.8) that c be a linear combination of simple roots belonging to Δ^0 (resp. $\eta(a)$; resp. $\eta(b)$; resp. $\eta(a, b)$). We therefore have

$$f(a) = \frac{1}{2}(\varphi_a - \varphi_0), \quad f(b) = \frac{1}{2}(\varphi_b - \varphi_0); \quad (1)$$

$$\sum_{x \in \psi} f(x) = \varphi_{ab} - \varphi_0. \quad (2)$$

On the other hand, it is clear that for all $w \in {}_k W$, $f(w(x)) = f(x)$. Now, it is known that every element of ψ is, in exactly two ways, the image of a simple root (a or b) under an element of $W(\psi)$. Consequently,

$$2 \sum_{x \in \psi} f(x) = 2m_{ab} \cdot f(a) + 2m_{ab} \cdot f(b). \quad (3)$$

The formula in the statement is an immediate consequence of (1), (2), (3).

Proposition 0.62 (6.15). *Let a, b be two distinct k -simple roots and m_{ab} the order of the product of the corresponding fundamental reflections. Then, the following three conditions are equivalent:*

- (i) *The pair a, b is connected; in other words, $m_{ab} > 2$.*
- (ii) *For every K -root $a' \in \rho^{-1}(a)$, there exists a K -root $b' \in \rho^{-1}(b)$ and a subset θ of Δ^0 such that $\theta \cup \{a', b'\}$ is connected.*
- (iii) *There exist K -roots $a' \in \rho^{-1}(a)$ and $b' \in \rho^{-1}(b)$, and a subset θ of Δ^0 , such that $\theta \cup \{a', b'\}$ is connected.*

We may, without loss of generality, assume that ${}_k \Delta = \{a, b\}$: it suffices to replace G by the subgroup $G_{\Phi'}$, where Φ' is the system of k -roots which are linear combinations of a and b .

As before, we denote by (x) the set of roots of the form $n \cdot x$ ($n \in \mathbf{N}$). Let ψ (resp. ψ') be the smallest ideal of the system ${}_k \Phi$ (resp. of the system ${}_K \Phi$) containing b (resp. $\rho^{-1}(b)$). The smallest normal subgroup of G containing all $U_{(b')}$ with $b' \in \rho^{-1}(b)$ is also the smallest normal subgroup containing $U_{(b)}$. This subgroup, which we shall denote by H , is defined over k , since it is k -closed and defined over k_s (2.14 a)).

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It then follows from 5.11 that $\mathbf{H} = \mathbf{G}_{\Psi}^* = \mathbf{G}_{\Psi'}^*$. This being so, we have the following immediate implications:

$$(i) \Rightarrow a \in \Psi \Rightarrow \mathbf{H} \supset \mathbf{U}_{(a)} \Rightarrow \Psi' \supset \rho^{-1}(a) \Rightarrow (ii) \Rightarrow (iii) \Rightarrow \Psi' \cap \rho^{-1}(a) \neq \emptyset \Rightarrow \mathbf{H} \cap \mathbf{U}_{(a)} \neq \{e\} \Rightarrow (i).$$

Corollary 0.63 (6.16). *Let $\theta = \{a_1, \dots, a_t\}$ be a connected subset of ${}_k \Delta$ and $a'_1 \in \rho^{-1}(a_1)$. Then there exist $a'_i \in \rho^{-1}(a_i)$ ($2 \leq i \leq t$) and a subset ψ of Δ^0 such that $\{a'_1, \dots, a'_t\} \cup \psi$ is connected.*

This follows from 6.15 by induction on t .

6.17. The functor $R_{K/k}$. — In this section and the following four, K is a finite separable extension of degree d of k , contained in k_s , and $\sigma_1 = \text{id}, \sigma_2, \dots, \sigma_d$ are the distinct monomorphisms of K into k_s . We refer to [38, Chap. I], (see also [5, 2.8]), for the definition of the restriction of scalars functor $R_{K/k}$, which goes from the category of K -groups (or quasi-projective K -varieties) to that of k -groups (or quasi-projective k -varieties). If $\mathbf{H}$ is a connected K -group and $\mathbf{H}' = R_{K/k} \mathbf{H}$, then there exists a K -morphism $\mu : \mathbf{H}' \rightarrow \mathbf{H}$ such that

$$\mu^0 = (\sigma_1 \mu, \dots, \sigma_d \mu) : \mathbf{H}' \xrightarrow{\sim} \sigma_1 \mathbf{H} \times \dots \times \sigma_d \mathbf{H} \quad (1)$$

is a k_s -isomorphism, and this characterizes the pair $(\mathbf{H}', \mu)$ up to k -isomorphism.

Every K -morphism $\alpha : \mathbf{G} \rightarrow \mathbf{H}$ can be written in one and only one way in the form $\mu \circ \beta$, where $\beta : \mathbf{G} \rightarrow \mathbf{H}'$ is a k -morphism.

$(\mu^0)^{-1}$ maps $\mathbf{H} = \sigma_1 \mathbf{H}$ onto a subgroup of $\mathbf{H}'$ which is defined over k_s and visibly stable under every element of $\text{Gal}(k_s/K)$, hence in fact defined over K , whence the existence of a K -morphism $\nu : \mathbf{H} \rightarrow \mathbf{H}'$ such that $\nu^0 = (\nu, \sigma_2 \nu, \dots, \sigma_d \nu)$ is the inverse of μ^0 [3, § 1].

Proposition 0.64 (6.18). *We keep the notation of 6.17. Then $R_{K/k}$ defines a bijection from the connected K -subgroups of $\mathbf{H}$ to the set of connected k -subgroups $\mathbf{L}'$ of $\mathbf{H}'$ such that $\mu^0(\mathbf{L}')$ is the product of its intersections with the $\sigma_i \mathbf{H}$. If $\mathbf{L}' = R_{K/k}(\mathbf{L})$, and if $\mathcal{Z}(\mathbf{L})^0$ (resp. $\mathcal{N}(\mathbf{L})^0$) is defined over K , then $R_{K/k} \mathcal{Z}(\mathbf{L})^0 = \mathcal{Z}(\mathbf{L}')^0$ (resp. $R_{K/k} \mathcal{N}(\mathbf{L})^0 = \mathcal{N}(\mathbf{L}')^0$).*

Proof. It is clear that if $\mathbf{L}$ is a connected K -subgroup of $\mathbf{H}$, then $\mathbf{L}' = R_{K/k} \mathbf{L}$ has the indicated property. Conversely, let $\mathbf{L}'$ be a connected k -subgroup of $\mathbf{H}'$ such that $\mu^0(\mathbf{L}')$ is the product of its projections onto the $\sigma_i \mathbf{H}$. What was said above shows that $\mu^0(\mathbf{L}') \cap \sigma_1 \mathbf{H} = \mathbf{L}$ and the projection η of $\mathbf{L}'$ onto $\mathbf{L}$ are defined over K , which immediately implies that $(\mathbf{L}', \eta)$ satisfies the characteristic property (1) of 6.17.

The second assertion is an immediate consequence of the first. $\square$

Corollary 0.65 (6.19). *The functor $R_{K/k}$ defines a bijection from the Cartan subgroups defined over K (resp. maximal tori defined over K , resp. parabolic K -subgroups) of $\mathbf{H}$ to the Cartan subgroups defined over k (resp. maximal tori defined over k , resp. parabolic k -subgroups) of $\mathbf{H}'$.*

Let us first remark that the functor $R_{K/k}$ preserves the property of being a torus, or of being commutative, or nilpotent. We will also use the fact that the parabolic subgroups of a direct product are the direct products of parabolic subgroups of

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factors; this results, for example, from 4.2, 4.3, after an obvious reduction to the semisimple case.

Let L be a K -subgroup of H . If L is a Cartan subgroup (resp. a maximal torus; resp. a parabolic subgroup), the same is true of the subgroup ${}^{\sigma_i}L$ of ${}^{\sigma_i}H$, and thus also of the subgroup $R_{K/k}L$ of H' . Conversely, if L' is a Cartan subgroup (resp. a maximal torus; resp. a parabolic subgroup) of H' , then $\mu^0(L')$ is the direct product of its intersections with the ${}^{\sigma_i}H$; if, moreover, L' is defined over k , there exists (by 6.18) a K -subgroup L of H such that $L' = R_{K/k}L$, and L is a Cartan subgroup of H , since it is nilpotent and equal to its connected normalizer (resp. is a maximal torus of H for reasons of dimension; resp. is a parabolic subgroup of H because $R_{K/k}(H/L) = H'/L'$ is a projective variety, and the same is therefore true of H/L).

6.20. Let T be a torus defined over K and $T' = R_{K/k}T$. Then the map $b \mapsto \sum_i {}^{\sigma_i}(b \circ \mu)$ defines an isomorphism α from $X^*(T)_K$ onto $X^*(T')_k$ [21, § 1.4]; in particular (1.8), T_d and T'_d have the same dimension. We have a natural homomorphism $X^*(T)_K \rightarrow X^*(T'_d)$, the composition of α and the restriction $X^*(T') \rightarrow X^*(T'_d)$, which is injective with finite cokernel, but is not bijective in general.

Suppose T is split over K . Then the projection of $\mu^0(T')$ onto ${}^{\sigma_i}T$ induces an isomorphism from T'_d onto ${}^{\sigma_i}T$ for all i , whence an isomorphism $\beta_i : X^*({}^{\sigma_i}T) \rightarrow X^*(T'_d)$; it suffices to verify this for $T = \mathbf{G}_m$, for which it is elementary. In particular, β_1 , which we also denote by β , is an isomorphism of $X^*(T) = X^*(T)_K$ onto $X^*(T'_d) = X^*(T'_d)_k$.

6.21. (i) Suppose now that H is reductive. From 6.19, 6.20, we obtain: $r_K(H) = r_k(H')$; if A is a maximal split torus of H over K , then the largest split torus A'_d of $A' = R_{K/k}A$ over k is a maximal split torus of H' over k ; the canonical isomorphism $\beta : X^*(A) \rightarrow X^*(A'_d)$ of 6.20 induces a bijection of ${}_K\Phi(H)$ onto ${}_k\Phi(H')$, and $\mathcal{Z}(A') = \mathcal{Z}(A'_d)$.

Let ψ be a quasi-closed subset of ${}_K\Phi(H)$ (3.8). The group H_ψ can be characterized as the subgroup of H containing $\mathcal{Z}(A)$ and whose Lie algebra is the sum of that of $\mathcal{Z}(A)$ and the maximal eigenspaces of A in $\mathfrak{h}$ corresponding to the elements of ψ . In view of 6.20 and the equality $\mathcal{Z}(A') = \mathcal{Z}(A'_d)$, it follows that

$$R_{K/k}H_\psi = H'_{\beta(\psi)}. \quad (1)$$

One sees in an analogous way that if η is quasi-closed, consisting of positive roots for a suitable order, then $R_{K/k}H_\eta^* = H'_{\beta(\eta)}^*$. Since H_ψ^* is generated by groups of this type, we also have

$$R_{K/k}H_\psi^* = H'_{\beta(\psi)}^*. \quad (2)$$

It follows in particular from (1) that if we fix on $X^*(A)$ and $X^*(A'_d)$ orders corresponding to each other by β , then, in the notation of 5.12,

$$R_{K/k}P_\theta = {}_kP_{\beta(\theta)}, \quad (3)$$

θ being a subset of the set of simple k -roots of H .

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(ii) If H is semisimple and almost simple over K , then H' is semisimple and almost simple over k . Indeed, the groups ${}^{\sigma_i}H$ are then the almost simple factors of H' and they are transitively permuted by $\Gamma = \text{Gal}(k_s/k)$. Every semisimple and almost simple group G over k is k -isogenous to a group $R_{k'/k}H$ where k' is a suitably chosen extension of k , and where H is a k' -group, almost simple over $\bar{k}$. Indeed, up to replacing it by a k -isogenous group, we can assume that G is $\bar{k}$ -isomorphic to a direct product of almost simple subgroups. Let H be one of them and let $k' \subset \bar{k}$ be its smallest field of definition containing k . This is a separable extension of finite degree of k (2.14), which we can therefore assume to be contained in k_s . The simple factors of G are permuted by Γ , necessarily transitively. We can therefore write them in the form ${}^{\nu_1}H, \dots, {}^{\nu_d}H$ where $\nu_1, \dots, \nu_d$ form a system of representatives of the left cosets of $\text{Gal}(k_s/k)$ modulo $\text{Gal}(k_s/k')$. The restrictions of the ν_i to k' are obviously the isomorphisms of k' into k_s . On the other hand, the product of the ${}^{\nu_i}H$ different from H is defined over k_s , stable under $\text{Gal}(k_s/k')$, hence also defined over k' , and thus the projection π of G onto H is defined over k' . Consequently, the pair (G, π) satisfies property (1) of 6.17 and $G \cong R_{k'/k}H$.

Proposition 0.66 (6.22). *Let θ_i ($i = 1, 2$) be two subsets of ${}_k\Delta$, $P_i = {}_kP_{\theta_i}$ the corresponding standard k -parabolic subgroups (5.12), ${}_kW_i$ (resp. ${}_K W_i$) the subgroup of ${}_kW$ (resp. ${}_K W$) generated by the fundamental reflections corresponding to the elements of θ_i (resp. ${}^K\eta_k(\theta_i)$ (6.3)) and $w \in {}_kW$. Then, the double coset $P_{1,K} \cdot w \cdot P_{2,K} \subset G_K$*

(5.20) has rational points over k if and only if the corresponding double coset ${}_kW_1 \cdot w \cdot {}_kW_2$ has a non-empty intersection with ${}_kW_k$ (6.10). When this is the case, this intersection is the preimage under the canonical map ${}_kW_k \rightarrow {}_kW$ of a double coset ${}_kW_1 \cdot w' \cdot {}_kW_2$, with $w' \in {}_kW$, and the double coset $P_{1,k} \cdot w' \cdot P_{2,k}$ is the set of rational points over k of $P_{1,K} \cdot w \cdot P_{2,K}$.

Proof. Let x' be any element of ${}_kW$, x a representative of x' in ${}_kW_k$, and n a representative of x' in $\mathcal{N}(S)_k$. Since T and ${}^\sigma T$ are two maximal split tori over K of $\mathcal{Z}(S)$, there exists $z \in \mathcal{Z}(S)_K$ such that ${}^\sigma T = {}^z T$, that is to say such that $n' = n^{-1} \cdot z \in \mathcal{N}(S, T)_K$. The canonical image x'' of n' in ${}_KW_k$ is congruent to $x \pmod{{}_KW_k^0}$. Since $\mathcal{Z}(T)_K \subset P_{i,K}$, we therefore have

$$P_{1,k} \cdot x' \cdot P_{2,k} = P_{1,k} \cdot n \cdot P_{2,k} \subset P_{1,K} \cdot n \cdot P_{2,K} = P_{1,K} \cdot n' \cdot P_{2,K} = P_{1,K} \cdot x'' \cdot P_{2,K} = P_{1,K} \cdot x \cdot P_{2,K}.$$

The first part of the statement follows from this, taking 5.20 into account.

To establish the second part, it now suffices to show that if $\bar{x}' \in {}_kW$ is such that $P_{1,K} \cdot \bar{x}' \cdot P_{2,K} = P_{1,K} \cdot x' \cdot P_{2,K}$, then $\bar{x}' \in {}_kW_1 \cdot x' \cdot {}_kW_2$. Let $P = {}_kP_\emptyset$, for any parabolic subgroup $Q \supset \mathcal{Z}(S)$, let ${}^u Q$ denote its unique opposite containing $\mathcal{Z}(S)$, and let P' be a k -parabolic subgroup such that $\mathcal{Z}(S) \subset P' \subset ({}^{x'} P_2 \cap P_1) \cdot R_u(P_1)$, and $P'' = ({}^{x'^{-1}} P' \cap P_2) \cdot R_u(P_2)$. By 4.4, 5.9, there exist $y' \in {}_kW_1$ and $y'' \in {}_kW_2$ such that ${}^{y'} P' = P$ and ${}^{y''} P'' = P$. It follows from the construction of P', P'' that $P' \cdot x' \cdot P''$ is dense in $P_1 \cdot x' \cdot P_2$, hence that $P \cdot y' \cdot x' \cdot y'' \cdot P$ is dense in $P_1 \cdot x' \cdot P_2$. Similarly, there exist $\bar{y}' \in {}_kW_1$ and $\bar{y}'' \in {}_kW_2$ such that $P \cdot \bar{y}' \cdot \bar{x}' \cdot \bar{y}'' \cdot P$ is dense in $P_1 \cdot \bar{x}' \cdot P_2$. But then $y' \cdot x' \cdot y'' = \bar{y}' \cdot \bar{x}' \cdot \bar{y}''$, by virtue of 5.15, and $\bar{x}' \in {}_kW_1 \cdot x' \cdot {}_kW_2$. $\square$

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6.23. Corollary. — Let P and P' be two parabolic subgroups of G defined over k . Then, two distinct double cosets of P_k, P'_k in G_k are contained in distinct double cosets of P_K, P'_K in G_K .

6.24. Remarks. — a) Proposition 6.22 takes a particularly simple form when $P_1 = P_2 = {}_kP_0$, a minimal k -parabolic subgroup, which we will again denote by P . In this case, the double cosets of P_k in G_K correspond to the double cosets of ${}_kW_k^0$ in ${}_kW$, and the proposition asserts that the double cosets of P_K that possess a rational point in k are those which correspond to the double cosets of ${}_kW_k^0$ in ${}_kW_k$; the latter are in fact simple cosets, since ${}_kW_k^0$ is normal in ${}_kW_k$. The set of rational points of the double coset of P_K associated to a given coset of ${}_kW_k^0$ in ${}_kW_k$ is then the double coset of P_k in G_k corresponding to the element of ${}_kW$ representing the latter.

b) When $K = \bar{k}$ (or, if preferred, when K is the universal domain), Proposition 6.22 gives the condition for a “geometric” double coset $P_1 \cdot g \cdot P_2$ to possess rational points over k . Note that such a double coset can be defined over k without possessing rational points.

6.25. Proposition. — Let P and P^- be two opposite minimal k -parabolic subgroups, and let U and U^- be their unipotent radicals. We then have $G_K = U_K \cdot U_K^- \cdot P_K$. In particular, $G = U \cdot U^- \cdot P$.

In the following, we assume, which is permissible (4.8, 4.13, 4.16), that $P \cap P^- = \mathcal{Z}(S)$. Before proving this proposition, we shall establish two corollaries.

6.26. Corollary. — Every element of ${}_kW$ has a representative in the set $U_k \cdot U_k^- \cdot U_k$. Indeed, it follows from 6.25 that $G_k = U_k \cdot U_k^- \cdot P_k = U_k \cdot U_k^- \cdot U_k \cdot \mathcal{Z}(S)_k$.

6.27. Corollary. — Given two conjugate parabolic subgroups Q, Q' , one of which is defined over k , there exists a parabolic subgroup defined over k which is opposite to both of them.

There is no loss of generality in assuming, in the notation of 5.12 and 6.25, that $Q = {}_kP_\theta \supset P$ and that ${}_kP_\theta^- = P^-$ (4.8, 5.14). Let $g \in G$ be such that $Q' = {}^g Q$. In view of 6.25, we can write $g = u \cdot u' \cdot p$ with $u \in U_k$, $u' \in U^-$ and $p \in P$. Then, the parabolic subgroup ${}^u({}_kP_\theta^-) = {}^{u \cdot u'}({}_kP_\theta^-)$ is simultaneously opposite to ${}^u({}_kP_\theta) = Q$ and to ${}^{u \cdot u'}({}_kP_\theta) = Q'$.

One could have just as easily deduced Proposition 6.25 from Corollary 6.27. This is, moreover, essentially what is expressed by the following lemma, the first step in the proof of 6.25.

6.28. Lemma. — With the notation being that of 6.25, assume that, for every parabolic subgroup P' conjugate to P , there exists a parabolic subgroup defined over k and simultaneously opposite to P and to P' . Then, $G_K = U_K \cdot U_K^- \cdot P_K$ for every extension K of k .

Let $g \in G_K$ and let Q be a parabolic subgroup defined over k opposite to P and to ${}^g P$. Then (4.13) there exists $u \in U_k$ such that ${}^u Q = P^-$. The group ${}^{u \cdot g} P$ is opposite to P^- and defined over K , so there exists a $u' \in U_K^-$ such that ${}^{u' \cdot u \cdot g} P = P$. But then,

$$u' \cdot u \cdot g \in P_K \quad \text{and} \quad g \in u^{-1} \cdot u'^{-1} \cdot P_K \subset U_K \cdot U_K^- \cdot P_K.$$

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6.29. Lemma. — Suppose that k is equal to the finite field $\mathbf{F}_{q^2}$ with q^2 elements, and that G is k -isogenous to SU_3 . Then, G has $(q^3 + 1)$ k -Borel subgroups and the number of k -Borel subgroups not opposite to any given k -Borel subgroup is less than $2(q + 1)$.

Let $K = \mathbf{F}_{q^2}$ and let γ be the element of $\mathrm{Gal}(K/k)$ distinct from the identity. The group G is split over K and the variety of Borel subgroups of G (5.24) is defined over K and can be identified with the flag variety (pairs consisting of a point and a line containing it) of a projective plane M defined over K (in the sense of algebraic geometry), and whose set M_K of rational points over K is therefore a projective plane over K (in the usual sense). For any point $x \in M_K$, the transform under γ of the stabilizer of x leaves invariant one and only one line which we denote by D_x . The correspondence $x \mapsto D_x$ is a “Hermitian polarity” in M_K (if $G = \mathrm{SU}_3(f)$, this is the polarity associated with the Hermitian form f). The identification mentioned above associates to every Borel subgroup B the flag F that it stabilizes, and F is defined over k if and only if B is. It is immediate that the flags defined over k are the flags of the form (x, D_x) , with $x \in M_K$. Their number is equal to that of the points $x \in M_K$ which are isotropic (i.e. which belong to their polar line D_x), and thus to $q^3 + 1$.

Let (y, D) be a flag of M , and let B be its stabilizer. For the isotropy group of a flag (x, D_x) rational over k not to be opposite to B , it is necessary and sufficient that we have either $x \in D$ or $y \in D_x$. Since any line contains 0, 1 or $q + 1$ isotropic rational points over K , the number of flags (x, D_x) satisfying the first condition is equal to 0, 1 or $q + 1$. The same holds for the number of flags satisfying the second condition; indeed, if $y \in M_K$, this latter condition can be written as $x \in D_y$, and otherwise, y is contained in at most one line defined over K .

6.30. Proof of Proposition 6.25.

a) k is infinite. — The conjugacy class $\mathcal{P}$ of minimal k -parabolic subgroups is self-opposite (4.14), hence two elements of this class are opposite if and only if they contain opposite Borel subgroups (4.10). It then follows from 4.12 that the hypotheses of 6.28 are satisfied.

b) *Reduction to the case of relative semi-simple rank 1.* — We propose to show that the proposition is true in general if it is true for groups of relative semi-simple rank 1. In the calculations that follow, the subscripts K are omitted (one can, moreover, if one wishes, reduce immediately to the case where K is the universal domain, either by using the relation $(U^-.P)_K = U_K^-.P_K$, or by combining the proof of Corollary 6.27 and Lemma 6.28). Let us set $G^* = U_k.U^-.P$; let a be any simple k -root, and let us denote by (a) (resp. $(a)^*$) the set of positive roots proportional (resp. non-proportional) to a . The proposition being assumed true for the group $G' = G_{(a) \cup \{-a\}}$, we have

$$\begin{aligned} G^*.U_{-(a)} &= U_k.U^-. \mathcal{Z}(S).U_{(a)^*}.U_{-(a)} = U_k.U_{-(a)^*}.U_{-(a)}. \mathcal{Z}(S).U_{(a)^*}.U_{-(a)^*} = \\ &= U_k.U_{-(a)^*}.G'.U_{(a)^*} = U_k.U_{-(a)^*}.U_{(a),k}.U_{-(a)}. \mathcal{Z}(S).U_{(a)}.U_{(a)^*} = \\ &= U_k.U_{(a),k}.U_{-(a)^*}.U_{-(a)}. \mathcal{Z}(S).U_{(a)}.U_{(a)^*} = U_k.U^-.P = G^*. \end{aligned}$$

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Now $G^* \cdot P = G^*$. But P and the groups $U_{-(a)}$ corresponding to all simple roots generate G (5.14). Consequently, $G^* \cdot G = G^*$, and $G^* = G$.

c) k is finite and $r_k(\mathcal{D}G) = 1$.

Let $k = \mathbf{F}_q$. We want to show that G satisfies the hypothesis of Lemma 6.28. This does not change if we replace G by $\mathcal{D}G$. We will therefore assume G is semi-simple and $r_k(G) = 1$; it is then almost simple over k since there does not exist any non-trivial anisotropic semi-simple group over a finite field. Let $G_0, \dots, G_{n-1}$ be the quasi-simple normal subgroups of G and let k' be the smallest field of definition common to the G_i . The group $\mathrm{Gal}(k'/k)$ permutes the G_i transitively, and since it is cyclic, it is of order n and permutes the G_i cyclically. In particular, we have $k' = \mathbf{F}_{q^n}$. Let γ be a generator of $\mathrm{Gal}(k'/k)$. We will assume that G_i is the transform of G_0 by γ^i , which we will also denote by G' , under γ^i . The group G is k -isogenous to $R_{k'/k}(G')$ (6.21 (ii)), so the k' -rank of G' is equal to 1 (6.21 (i)).

Every Borel subgroup B of G is a product $B_0 \cdot B_1 \dots B_{n-1}$ of Borel subgroups of $G_0, G_1, \dots, G_{n-1}$ respectively. According to 6.19, the number of k -Borel subgroups of G is equal to that of the k' -Borel subgroups of G' .

Let $B = B_0 \dots B_{n-1}$ be a Borel subgroup of G . Another Borel subgroup $B' = B'_0 \cdot B'_1 \dots B'_{n-1}$ is opposite to it if and only if B'_i is opposite to B_i for all i . On the other hand, it follows from 4.18, 5.9 that in a reductive k -group, of semi-simple k -rank equal to one, two distinct minimal parabolic k -subgroups are opposite. Consequently, to establish our assertion it will suffice to show that if N_i denotes the number of Borel subgroups of G_i defined over k' and not opposite to B_i , we have $\sum_i N_i < N - 1$ for any B . Now it follows from the classification that G' is, up to k' -isogeny, the group $\mathbf{SL}_2$ or the group $\mathbf{SU}_3$. In the first case, we have $N = q^n + 1$ and $0 \leq N_i \leq 1$, whence $\sum_i N_i \leq n < N - 1$. In the second case, it follows from Lemma 6.29 that $N = q^{3n} + 1$ and $N_i \leq 2(q^n + 1)$, whence $\sum_i N_i \leq 2n(q^n + 1) < N - 1$, which completes the proof.

Remark. — Prop. 6.25 answers a question of Grothendieck and Demazure. It was also proved by Steinberg and Demazure.

§ 7. A MAXIMAL SPLIT SUBGROUP

Lemma 0.67. 7.1. *Let there be given a root system of rank 2, and an order on it. Let a, b be the simple roots, r_a, r_b the corresponding fundamental reflections and W the Weyl group. For any integer $i \geq 1$, let $c_i = a$ or b depending on whether i is even or odd, and let $w_i \in W$ be defined inductively by the relations $w_1 = r_b$ and $w_i = r_{c_i} \cdot w_{i-1}$ for $i \geq 2$. Let m (resp. m' ; resp. m'') be the smallest value of the integer $i \geq 2$ such that $(r_a \cdot r_b)^i = e$ (resp. such that $w_{i-1}(a)$ is a simple root; resp. such that $w_i(a)$ is a negative root). Then, $m = m' = m''$ and $w_{m-1}(a) = c_m$.*

The root $w_{m''-1}(a)$ is positive and its transform by $r_{c_{m''}}$ is negative. By...

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consequently, $w_{m''-1}(a)$ can only be the simple root $c_{m''}$ [6, § 7, n° 6, Prop. 17]. In particular, $m' \leq m''$.

Let us set $w_{m'-1}(a) = c$ where c is therefore equal to a or b . We have $w_{m'-1} \cdot r_a \cdot w_{m'-1}^{-1} = r_c$. Depending on whether $c = c_{m'-1}$ or $c_{m'}$, this latter relation can also be written as $(r_a \cdot r_b)^{m'-1} = e$ or $(r_a \cdot r_b)^{m'} = e$. It follows that $m \leq m'$.

Finally, the relation $(r_a \cdot r_b)^m = e$ can be written as $w_{m-1} \cdot r_a \cdot w_{m-1}^{-1} = r_{c_m}$, or even $w_{m-1}(a) = \pm c_m$. Since $m \leq m''$, the $+$ sign must be chosen. But then, $w_m(a) = r_{c_m}(c_m) = -c_m$, hence $m'' \leq m$, which completes the proof.

Before stating the theorem of this section, let us recall that in a root system Φ , the set (d) of positive integer multiples of a root is equal to $\{d\}$ or $\{d, 2d\}$ and that d is said to be non-multipliable in the first case. The set Φ' of non-multipliable roots is a reduced root system.

Theorem 0.68 (7.2). *Suppose G is connected reductive. Let S be a maximal k -split torus of G , $\Phi = \Phi(S, G)$ the system of k -roots of G , Φ' that of the non-multipliable roots of Φ , and Δ the set of simple roots of Φ' for a given order on $X^*(S)$. For each $a \in \Delta$, let E_a be a one-dimensional k -subgroup of $U_{(a)}$ normalized by S , and let V be the group generated by the E_a . Then G has a unique connected reductive k -split subgroup F containing $S \cdot V$. The torus S is a maximal torus of F and $\Phi' = \Phi(S, F)$. In particular, $W(F) \cong W(G)$. For all $d \in \Phi$, the intersection $U_{(d)} \cap F$ is the root group $U_{d'}$ of F corresponding to the root d' of Φ' contained in (d) .*

For any element $b \in \Phi'$, we have $(b) = \{b\}$, so the group $U_{(b)}$ admits a structure of vector space over k such that $\text{Int } s$ ($s \in S$) is the homothety of ratio s^b (3.18, rem.). Consequently, any one-dimensional k -subgroup of $U_{(b)}$ stable under S is k -isomorphic to the additive group $\mathbf{G}_a$. This holds in particular for the groups E_a in the statement.

Let v be any element of $E_{a,k}$, distinct from the identity element. According to Theorem 5.15, applied to the reductive group $G_{\{a, -a\}}$ (3.4, 3.8), v has a unique decomposition of the form

$$v = v' \cdot n_a \cdot v'', \quad (v', v'' \in U_{(-a)}; \quad n_a \in \mathcal{N}(S), \text{ and } v', v'', n_a \in G_k). \quad (1)$$

Let s be an element of S such that $s^a = -1$. We have

$$v = {}^s(v^{-1}) = {}^s(v''^{-1}) \cdot {}^s(n_a^{-1}) \cdot {}^s(v'^{-1}) = v'' \cdot {}^s(n_a^{-1}) \cdot v',$$

from which, by virtue of the uniqueness of (1), $v' = v''$, ${}^s(n_a^{-1}) = n_a$, and

$$n_a^2 = (n_a, s) \in S_k. \quad (2)$$

Let us set $E_{-a} = \{ {}^s v \mid s \in S \} \cup \{e\}$. We have seen that $U_{(-a)}$ has a vector space structure over k such that conjugation by $s \in S$ is scalar multiplication by s^{-a} . Thus E_{-a} is a k -group k -isomorphic to the additive group. We have, for all $s \in S$,

$$S \cdot E_{-a} \cdot {}^s v = S \cdot E_{-a} \cdot {}^s n_a \cdot {}^s v';$$

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Since ${}^s n_a \in n_a \cdot S = S \cdot n_a$, we obtain, by taking the union over S ,

$$S \cdot E_{-a} \cdot (E_a - \{e\}) = S \cdot E_{-a} \cdot n_a \cdot (E_{-a} - \{e\}). \quad (3)$$

Multiplying side by side this relation and the one deduced from it by replacing each side by its inverse, we obtain, taking (2) into account,

$$S \cdot E_{-a} \cdot E_a \cdot E_{-a} = S \cdot E_{-a} \cdot n_a \cdot E_{-a} \cdot n_a \cdot E_{-a}.$$

Let us set

$$Y_a = S \cdot E_{-a} \cdot \{e, n_a\} \cdot E_{-a}. \quad (4)$$

From (3) multiplied on the right by E_{-a} , it follows that Y_a is also equal to $S \cdot E_{-a} \cdot E_a \cdot E_{-a}$. Therefore

$$S \cdot E_{-a} \cdot n_a \cdot E_{-a} \cdot n_a \cdot E_{-a} = Y_a.$$

It follows that $Y_a \cdot Y_a \subset Y_a$. On the other hand, $Y_a^{-1} = Y_a$. Consequently, Y_a is a group which, being generated by E_a , E_{-a} , and S , is connected and defined over k . From Theorem 5.15 (where the set of positive roots must be replaced by that of negative roots), it follows that $Y_a \cap U_{(-a)} = E_{-a}$. Transforming this relation by n_a , we see that $n_a \cdot E_{-a} \cdot n_a^{-1}$ contains E_a ; for reasons of dimension, it follows that

$$n_a \cdot E_{-a} \cdot n_a^{-1} = E_a. \quad (5)$$

Let us now transform the relation (4) by n_a ; we obtain

$$Y_a = S \cdot E_a \cdot \{e, n_a\} \cdot E_a. \quad (6)$$

Let $a, b \in \Delta$ be any two simple roots, (a, b) the set of all roots of the form $m \cdot a + n \cdot b$ with m and n strictly positive integers, E_{ab} the group generated by E_a and E_b , and $E'_{ab} = (E_a, E_b)$ the group generated by the commutators (x, y) with $x \in E_a$ and $y \in E_b$. We have $E'_{ab} = E'_{ba}$, and by 3.10,

$$E'_{ab} \subset U_{(a,b)}. \quad (7)$$

By virtue of the identity

$${}^v(x, y) = (v \cdot x, y) \cdot (v, y)^{-1},$$

E'_{ab} is normalized by E_a , so the product $E_a \cdot E'_{ab}$ is a group that we will denote by X_{ab} . The latter is normalized by E_b , since E_b normalizes E'_{ab} and $(E_b, E_a) = E'_{ab} \subset X_{ab}$. Consequently,

$$E_{ab} = X_{ab} \cdot E_b = E_a \cdot E'_{ab} \cdot E_b. \quad (8)$$

It then follows from (7) and 3.11 that we have $E_{ab} \cap U_{(a)} = E_a$ and $E_{ab} \cap U_{(b)} = E_b$, which allows us, without risk of contradiction in the notation, to set for all $c \in (a, b) \cup \{a, b\}$,

$$E_c = E_{ab} \cap U_{(c)}.$$

The set $A = \{{}^y x \mid x \in E_b - \{e\}, y \in E_a\}$ is a system of generators of X_{ab} . Indeed, since this set is normalized by S , the same holds for the group it generates, and it then follows from 3.11 that, along with ${}^y x = (y, x) \cdot x$, this group contains (x, y)

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and y^{-1} . Since no linear combination $m \cdot a - n \cdot b$, with m and n strictly positive integers, is a root, the groups E_a and E_{-b} commute as can be seen from 3.10, by using an ordering on $X^*(S)$ such that $a, -b > 0$. Consequently, if we set

$$Y_b^* = (E_b - \{e\}) \cdot S \cdot E_{-b} = Y_b - \{e, n_b\} \cdot S \cdot E_{-b},$$

we have $A = \{{}^x y \mid x \in Y_b^*, y \in E_a\}$. Now, by (2) and the second expression given for the set Y_b^* , the latter is invariant under left multiplication by n_b . It follows that A is normalized by n_b , and therefore the same holds for X_{ab} :

$$n_b \cdot X_{ab} \cdot n_b^{-1} = X_{ab}. \quad (9)$$

From (5), (8), (9) and 3.11, it follows that, if r_a denotes (for any $a \in \Delta$) the fundamental reflection corresponding to a , we have, for all $c \in \{a, b\} \cup \{-a, -b\}$,

$$n_b \cdot E_c \cdot n_b^{-1} = E_{r_b(c)}. \quad (10)$$

Let $m(a, b)$ be the order of the product $r_a \cdot r_b$ in the relative Weyl group ${}_k W$, $n_{ab} = \dots n_a \cdot n_b$ the product of $m(a, b) - 1$ factors alternately equal to n_b and n_a , starting from the right, and $c(a, b)$ the simple root equal to a or b according to whether $m(a, b)$ is even or odd. We then have, by virtue of (10) and Lemma 7.1,

$$n_{ab} \cdot E_a \cdot n_{ab}^{-1} = E_{c(a,b)},$$

and, moreover,

$$n_{ab} \cdot U_{-(a)} \cdot n_{ab}^{-1} = U_{-(c(a,b))}.$$

We now transform the relation (1) by n_{ab} ; we see, taking into account the two preceding relations and Theorem 5.15, that

$$n_{ab} \cdot n_a \cdot n_{ab}^{-1} \in n_{c(a,b)} \cdot S_k,$$

or even, using (2),

$$(n_a \cdot n_b)^{m(a,b)} \in S_k. \quad (11)$$

Relations (2) and (11) and the classical definition of Weyl groups by generators and relations show that, if M denotes the group generated by S and the n_a ($a \in \Delta$), we have

$$M \cap \mathcal{Z}(S) = S, \quad (12)$$

and that the canonical map $M \rightarrow {}_k W$ is surjective.

Let V_a^* be the group generated by the X_{ba} ($b \in \Delta - \{a\}$); since E_a normalizes the X_{ba} , it also normalizes V_a^* , therefore

$$V = E_a \cdot V_a^* = V_a^* \cdot E_a.$$

Let $(a)'$ (resp. $(a)^*$) be the set of positive roots proportional (resp. non-proportional) to a . By virtue of (7), V_a^* is contained in $U_{(a)^*}$, so we have, according to 3.11,

$$V \cap U_{(a)'} = E_a, \quad (13)$$

and

$$V \cap U_{(a)^*} = V_a^*. \quad (14)$$

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On the other hand, it follows from (9) that

$$n_a \cdot V_a^* \cdot n_a^{-1} = V_a^*. \quad (15)$$

We now propose to show that

$$\text{if } m \in M \text{ and if the canonical image } w \text{ of } m \text{ in } {}_k W \text{ transforms } a \text{ into a positive root, then} \quad (16)$$

$$m \cdot E_a \cdot m^{-1} \subset V.$$

The proof will be by induction on the number of positive roots transformed into negative roots by w^{-1} . If this number is zero, $w = e$, and the assertion is obvious. Let $w \neq e$, let $b \in \Delta$ be such that $w^{-1}(b) < 0$, let us set $w' = r_b \cdot w$ and $m' = n_b \cdot m$. By virtue of the induction hypothesis, we have $m' \cdot E_a \cdot m'^{-1} \subset V = E_b \cdot V_b^*$. But $w'(a)$ is not proportional to b (since $w(a) > 0$), hence, by (14), $m' \cdot E_a \cdot m'^{-1} \subset V_b^*$. Transforming both members of this relation by n_b , it follows, taking (15) into account, that $m \cdot E_a \cdot m^{-1} \subset V_b^* \subset V$, which establishes (16).

Making use of (4), (15), (16), and of the fact that Y_a is a group, we can reproduce here the calculations of 5.16, from which it follows that, for all $m \in M$ and all $a \in \Delta$, we have

$$n_a \cdot V \cdot \{n_a, n_a \cdot m\} \cdot V \cdot S = V \cdot \{n_a, n_a \cdot m\} \cdot V \cdot S.$$

It follows immediately (cf. e.g. [33]) that the product $F = V \cdot M \cdot V$ is a group, which, being generated by S , the E_a and the E_{-a} , is connected and defined over k . Let d be any root, let (d) and d' be defined as in the statement, and let m be an element of M_k whose canonical image w in ${}_k W$ transforms d' into an element of Δ that we will denote by a , so that $w((d)) = (a)$ or $(a)'$. It then follows from Theorem 5.15 and from (13) that

$$m \cdot (F \cap U_{(d)}) \cdot m^{-1} = E_a.$$

Consequently, $F \cap U_{(d)} = m^{-1} \cdot E_a \cdot m$ is a subgroup of $U_{(d')}$ isomorphic over k to the additive group, and which we will denote by $E_{d'}$; it is clear that this notation is consistent with those introduced further above. If d' is positive, $E_{d'}$ is contained in V (for example by virtue of (16)), hence we have, by 3.11,

$$V = \prod_{d' \in \Phi^+} E_{d'}. \quad (17)$$

The group Y_a is not solvable, for if it were, its unipotent radical would contain E_a and E_{-a} , hence n_a , hence also every commutator (n_a, s) , with $s \in S$, which is absurd. Any subgroup Q of F containing the product $S \cdot V$ and distinct from it contains at least one element of M whose canonical image w in ${}_k W$ is distinct from e . There then exists at least one simple root a whose transform by w is negative. But then Q contains E_{-a} , hence Y_a , and is not solvable. The subgroup $S \cdot V$ being solvable (since contained in $S \cdot U^+$), it is a Borel subgroup of F , and V is a maximal unipotent subgroup of it. Let m be an element of M whose canonical image in ${}_k W$ transforms positive roots into negative roots. The transform $V^- = {}^m V$ is also

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... a maximal unipotent subgroup of F and we have $R_u(F) \subset V \cap V^- = \{e\}$ (3.22), hence F is reductive. By virtue of Theorem 5.15, $F \cap \mathcal{Z}(S) = \mathcal{Z}_F(S) = S$, consequently S is a maximal torus of F . The $E_{a'}$ ($a' \in \Phi'$) are one-parameter root subgroups of F relative to S , and these are the only ones, by virtue of relation (17) and of its transform by m . The group F is therefore split and Φ' is the root system of F with respect to S .

Finally, let $\bar{F}$ be a split reductive subgroup of G containing $S.V$. The torus S is a maximal torus of $\bar{F}$. By virtue of (17), the $E_{a'}$, with $a' \in \Phi'^+$, are one-parameter root subgroups of $\bar{F}$ relative to S . In particular, the root system of $\bar{F}$ contains Φ'^+ . Since, on the other hand, it is necessarily reduced, and contained in Φ , it can only be Φ' . Let a be a simple root, $v \in E_{a,k}$ and $\bar{E}_{-a}$ the one-parameter root subgroup of $\bar{F}$ relative to $-a$. Applying Theorem 5.15 to the group generated by S , E_a and $\bar{E}_{-a}$, we see that there must exist elements $\bar{n}_a \in \mathcal{N}(S)_k$ and $\bar{v}', \bar{v}'' \in \bar{E}_{-a,k}$ such that $v = \bar{v}' \cdot \bar{n}_a \cdot \bar{v}''$. Since $\bar{E}_{-a}$ is contained in $U_{(-a)}$ (3.19), this decomposition of v must coincide with the decomposition (1). In particular, $\bar{v}' = v'$, hence $\bar{E}_{-a} = E_{-a}$ and $\bar{F} = F$, which completes the proof of the theorem.

7.3. Remarks. — Theorem 7.2 suggests the following questions:

- (i) Are the subgroups F corresponding to various choices of E_a conjugate to one another?
- (ii) When they are conjugate, are two subgroups F conjugate over k ?
- (iii) Is the group F the only split reductive subgroup containing V ?
- (iv) Are the subgroups F of Theorem 7.2 the only maximal split reductive subgroups of G ?

The answer to the four questions is negative, as shown by the following examples.

Let $f = x \cdot y + f_1(z_1, z_2, z_3)$ be a quadratic form in five variables of index 1 (i.e., f_1 is anisotropic) and of defect 1 if the characteristic is 2, and $G = \mathbf{O}^+(f)$. The group G has only two opposite relative roots, $\pm a$, and one sees immediately that (i) (resp. (ii)) has an affirmative answer if and only if two one-dimensional vector subspaces of $U_{(a)}$, defined over k , are conjugate by an element of $\mathcal{Z}(S)$ (resp. $\mathcal{Z}(S)_k$). In the present case, $\mathcal{Z}(S) = S \times \mathbf{O}^+(f_1)$, and the second factor operates on $U_{(a)}$ (which has 3 dimensions) by the usual representation. One therefore obtains a counterexample to (i) by assuming the characteristic is equal to 2 (k is then necessarily non-perfect) and a counterexample to (ii) by assuming the characteristic $\neq 2$, and by taking for f_1 a form whose values do not form a single class of quadratic residues.

Let now $G = \mathbf{SU}_3(f)$ be a special unitary group, corresponding to a Hermitian form f of index 1 (quasi-split group: cf. [35]). This group, of k -rank equal to 1, has four k -roots $\pm d$ and $\pm 2d$. The group $U_{(2d)}$ has dimension one. One obtains a counterexample to (iii) by conjugating F by an element of $U_{(d),k}$ not belonging to $U_{(2d)}$. Suppose now k is of characteristic 0, and let $V' \neq U_{(2d)}$ be a subgroup...

of dimension 1 of $\mathbf{U}_{(d)}$ defined over k . It follows from the Jacobson-Morozov theorem [14, th. 17, p. 100] that V' is contained in a subgroup of G isogenous over k to $\mathbf{SL}_2$, which constitutes a counterexample to (iv), since one easily verifies that V' and $\mathbf{U}_{(2d)}$ are not conjugate (for example, the normalizer of V' is $S.V'$ while that of $\mathbf{U}_{(2d)}$ is $\mathcal{Z}(S) \cdot \mathbf{U}_{(d)}$).

II. — COMPLEMENTS AND APPLICATIONS

§ 8. TRIGONALIZABLE SUBGROUPS (k PERFECT)

8.1. A closed subgroup of $\mathbf{GL}_n$ is said to be *trigonalizable over a field* K if it is defined over K and conjugate over K to a subgroup of the group $\mathbf{T}_n$ of upper triangular matrices. It is then necessarily solvable. Suppose G is connected solvable. Prop. 5 of [23] and the existence of a maximal torus defined over k (2.14, or [26], or 11.4) show that the following properties are equivalent:

- (i) G is k -isomorphic to a matrix group trigonalizable over k .
- (ii) G has a maximal torus split over k , and its unipotent radical G_u is defined over k .
- (iii) Any image of G under a k -morphism $G \rightarrow \mathbf{GL}_m$ is trigonalizable over k .

We shall say that G is *trigonalizable over k* if it satisfies these properties. Any connected solvable group split over k is trigonalizable over k [23, prop. 7]. The converse is not true in general since a unipotent k -group is always trigonalizable over k [23, cor. 1, p. 33] but is not necessarily split over k [23, p. 35]. It is however true over a perfect field [23, cor. 2, p. 34]. By (ii), a torus is split over k if and only if it is trigonalizable over k .

Theorem 0.69 (8.2). *Suppose k is perfect. Then the maximal solvable subgroups split over k (resp. the maximal tori split over k , resp. the maximal connected unipotent k -subgroups) of G are conjugate over k . If H is a maximal solvable subgroup split over k , then $G_k/H_k = (G/H)_k$ is identified with the set of k -rational points of a projective k -variety on which G operates.*

Let H be a solvable subgroup split over k of maximum dimension of G . We choose a linear realization of G as a k -group of automorphisms of a vector space V over k containing a line W defined over k such that $H = \{g \in G \mid g(W) = W\}$, which is always possible [25, p. 222]. Since H is solvable, split over k , its image under the natural morphism into $\mathbf{GL}(V/W)$ is trigonalizable over k (cf. 8.1). There exists therefore a flag F_0 of V , defined over k , stable under H , whose line is W . The group H is then *a fortiori* the stabilizer of F_0 in G . Let B be the stabilizer of F_0 in $\mathbf{GL}(V)$. This is a Borel subgroup of $\mathbf{GL}(V)$, the variety of...

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flags $\mathcal{F}(V)$ is identified with the quotient $\mathbf{GL}(V)/B$, and the canonical projection $\pi : \mathbf{GL}(V) \rightarrow \mathcal{F}(V)$ is surjective for the rational points over k (4.13).

Let us show that

$$G(F_0)_k = \overline{(G(F_0))}_k. \quad (1)$$

Let $P \in \overline{(G(F_0))}_k$ and G_P be the isotropy group of P in G . Since P is a flag defined over k , its stabilizer is trigonalizable over k , hence the identity component of G_P is a solvable group split over k ; consequently, $\dim G_P \leq \dim H$ and $\dim G(P) \geq \dim G(F_0)$. But the closure of $G(F_0)$ is the union of $G(F_0)$ and of orbits of strictly smaller dimensions [1, § 16], hence $P \in G(F_0)_k$.

Now let L be a solvable subgroup of G split over k . It operates on $\mathcal{F}(V)$ leaving the complete variety $\overline{G(F_0)}$ stable. Since the latter is defined over k and has at least one rational point over k , the group L has a fixed point $P \in \overline{(G(F_0))}_k$ (2.7 (i)), which, by (1), is in fact in $G(F_0)_k$. But, H being split over k , the canonical projection $G_k \rightarrow G(F_0)_k$ is surjective (2.7 (ii)). There then exists $g \in G_k$ such that $g \cdot F_0 = P$, hence such that $g^{-1} \cdot L \cdot g \subset H$, which shows that every solvable subgroup of G split over k is conjugate over k to a subgroup of H . The assertion of conjugation of unipotent subgroups then follows from the fact that every connected unipotent k -group is split over k , and that of maximal tori from the fact that, in H , the maximal tori defined over k are conjugate over k (see [26], or also (11.4); one can, moreover, give a simpler proof of this fact in the case of groups split over a perfect field).

Lemma 0.70 (8.3). *Assume G is connected and k is of characteristic zero. Then every unipotent k -subgroup V of G is contained in a minimal parabolic k -subgroup P of G , and we have $P \neq G$ if $V \not\subset R_u(G)$.*

Proof. Let $\pi : G \rightarrow G' = G/R_u(G)$ be the canonical projection. The k -subgroups of G containing $R_u(G)$ are the inverse images of the k -subgroups of G' , and if H is one of them, $G/H \simeq G'/\pi(H)$. The map $H \mapsto \pi(H)$ is therefore a bijection for the minimal parabolic k -subgroups, and we may restrict to the case where G is reductive. Let V be a unipotent k -subgroup $\neq \{e\}$ of G . It contains a unipotent k -subgroup V' of dimension one. According to the Jacobson-Morozov theorem [14, Chap. III, Thm. 17, p. 100], there exists a homomorphism from $\mathfrak{sl}_{2,k}$ into $\mathfrak{g}_k$ whose image contains $\mathfrak{v}'_k$. Since, in characteristic zero, every representation of the Lie algebra $\mathfrak{h}$ of a connected and simply connected semi-simple group H is the differential of a (rational) representation of H , we see that there also exists a k -morphism $f : \mathbf{SL}_2 \rightarrow G$ whose image contains V' . This morphism has finite kernel, so G contains a k -split torus $S \neq \{e\}$ (for example, the image under f of the group of diagonal matrices of $\mathbf{SL}_2$), and consequently a proper parabolic k -subgroup P (4.17). The group V is split over k , hence has a fixed point in $(G/P)_k$ by 2.7, and is then conjugate over k to a subgroup of P according to 4.13 a), whence the lemma. $\square$

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Proposition 0.71 (8.4). *Assume G is connected and k is of characteristic zero. Then the maximal unipotent k -subgroups of G are the unipotent radicals of the minimal parabolic k -subgroups of G .*

As before, we see by considering the projection of G onto $G/R_u(G)$ that we can restrict ourselves to the case where G is reductive.

Let P be a minimal parabolic k -subgroup of G , S a maximal split k -torus of G contained in P . We have the Levi decomposition $P = \mathcal{Z}(S) \cdot R_u(P)$ and $R(P) \supset S \cdot R_u(P)$ (4.16). Let V be a non-trivial unipotent k -subgroup of G . In view of 8.3, it suffices to consider the case where $V \subset P$, and we then have to show that $V \subset R_u(P)$, or, what comes to the same thing, that $V \subset R(P)$. If this were not the case, the reductive group $P/(S \cdot R_u(P)) \cong \mathcal{Z}(S)/S$ would contain a non-trivial unipotent k -subgroup, hence a non-trivial k -split torus (8.3, 4.17), which would contradict the hypothesis made on S , in view of 1.9 b).

Corollary 0.72 (8.5). *Assume G is reductive and k is of characteristic zero. Then the following conditions are equivalent:*

- (i) G is anisotropic over k .
- (ii) G contains no non-trivial unipotent k -subgroup, and $X^*(G^0)_k = \{0\}$.
- (iii) G_k consists of semi-simple elements, and $X^*(G^0)_k = \{0\}$.

The group G is the almost direct product of its derived group, which is semi-simple, by its connected center Z (2.2), so $X^*(G^0)_k$ is identified with a subgroup of finite index of $X^*(Z)_k$ and, in view of 1.7, 1.8, we have

$$X^*(G^0)_k = \{0\} \iff Z_d = \{1\}. \quad (1)$$

In characteristic zero, every unipotent algebraic group is connected, so each of the conditions of 8.5 is equivalent to the same condition for G^0 , and we may assume G is connected. Then, (i) $\Rightarrow$ (ii) by 8.3, 4.17 and (1). The implication (ii) $\Rightarrow$ (iii) results from the existence of a Jordan decomposition for the elements of G_k [1, th. 8.4], and (iii) implies (i) by virtue of (1) and 4.17.

Proposition 0.73 (8.6). *Assume G is connected and k is of characteristic zero. Let P be a minimal parabolic k -subgroup and H a connected k -subgroup containing $R_u(P)$. Then $\mathcal{N}(R_u(H)) = P'$ is a parabolic k -subgroup containing P , and $R_u(P') = R_u(H)$. The group H contains the smallest normal k -subgroup of P' containing $R_u(P)$.*

We reduce as above to the case where G is reductive. Let $\pi : H \rightarrow H' = H/R_u(H)$ be the canonical projection. In view of 8.4, $\pi(R_u(P)) = U'$ is a maximal unipotent k -subgroup of H' . Let V' be the unipotent radical of a minimal parabolic k -subgroup of H' opposite to $\mathcal{N}(U')$ (4.9). By 4.11, U' and V' generate a normal subgroup of H' , so $U = R_u(P) = \pi^{-1}(U')$ and $V = \pi^{-1}(V')$ generate a normal k -subgroup H_1 of H . Since H_1 contains $R_u(H)$, we have $R_u(H) = R_u(H_1)$. Since the group $\mathcal{N}(V)$ is a parabolic k -subgroup (8.4), 4.29 shows that $P' = \mathcal{N}(H_1)$ is a parabolic k -subgroup containing P , whose unipotent radical is equal to that of H_1 . We therefore have

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also $P'' = \mathcal{N}(R_u(H_1)) = \mathcal{N}(R_u(H)) = P'$ and $R_u(P') = R_u(H_1) = R_u(H)$. By construction $H \supset H_1 \supset R_u(P)$ and H_1 is generated by $R_u(P)$ and by a subgroup V which is also conjugate to U in P' by 8.4, so H_1 is the smallest normal subgroup of P' containing $R_u(P)$.

8.7. Proposition. — *Assume G is reductive and k is of characteristic 0. Let H and H' be two conjugate k -subgroups of G . Then if H contains a maximal unipotent k -subgroup (resp. a maximal k -split solvable subgroup) of G , the same holds for H' .*

We may assume H is connected. In both cases considered, there exists, by 8.4, a minimal k -parabolic subgroup P such that $H \supset U = R_u(P)$. Let $Q = \mathcal{N}(R_u(H))$ and $Q' = \mathcal{N}(R_u(H'))$. By 8.6, Q is a k -parabolic subgroup and $R_u(Q) = R_u(H)$. Since H and H' are conjugate over a suitable extension K of k , the groups Q and Q' are conjugate over K , and Q' is also a parabolic subgroup. It is obviously defined over k , hence conjugate over k to

Q (4.13). We can therefore restrict ourselves to the case where $Q = Q'$. Let H_1 be the smallest normal subgroup of Q containing U . By 8.6, it is contained in H . On the other hand, an element $x \in G_K$ such that ${}^xH = H'$ must normalize Q , hence be contained in Q (4.3), hence normalize H_1 , whence $H' = {}^xH \supset H_1 \supset U$.

Suppose now that H contains a maximal k -split solvable subgroup of G . By 8.2, 8.4, we may assume that $H \supset S \cdot R_u(P)$, where S is a maximal k -split torus of G contained in $R(P)$, and it remains to prove that H' contains a maximal k -split torus of G . Using what has already been obtained, and 8.6, we reduce to the case where $Q = \mathcal{N}(H) = \mathcal{N}(H')$ and $R_u(Q) = R_u(H) = R_u(H')$, and where there exists $x \in Q$ such that ${}^xH = H'$. The group Q admits a Levi decomposition $Q = \mathcal{Z}(S') \cdot R_u(Q)$, where $S' = S \cap R(Q)$ is a maximal k -split torus of $R(Q)$, hence is also the largest central k -split torus of $\mathcal{Z}(S')$ (4.15, 4.19). The group $S' \cdot R_u(Q)$ is normal in Q , contained in H , hence also in H' . The group $\mathcal{Z}(S')$ is an almost direct product over k of S' and of a k -group M , which has no non-trivial central k -split torus, and we have $H = (H \cap M) \cdot S' \cdot R_u(Q)$, $H' = (H' \cap M) \cdot S' \cdot R_u(Q)$.

By 4.27, it suffices to show that $H' \cap M$ contains a maximal k -split torus of M . Since maximal k -split tori correspond to each other by separable isogeny (4.25), we see, by considering the images of H and H' in $Q/(S' \cdot R_u(Q))$, that we are reduced to the case where G has no non-trivial central k -split torus and where H has a reductive k -subgroup H_1 , which is normal in G and contains $R_u(P)$. The group G is then an almost direct product of H_1 and of a reductive k -group H_2 . The latter has no non-trivial unipotent k -subgroup (otherwise $R_u(P)$ would not be a maximal unipotent k -subgroup) and has no non-trivial central k -split torus, since G has none; consequently, H_2 is anisotropic over k (8.5), and (4.27) H_1 contains every maximal k -split torus of G .

Since $H' \supset H_1$, this completes the proof.

Remark. — We refer to a paper in preparation by one of us for extensions of 8.3 to 8.7 to more general base fields.

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§ 9. TRIGONALIZABLE SUBGROUPS (LOCALLY COMPACT FIELD OF CHARACTERISTIC ZERO)

9.1. Let K be a locally compact field of characteristic zero. (K is therefore isomorphic to $\mathbf{R}$, $\mathbf{C}$ or to a p -adic field, *i.e.*, to a finite extension of a field $\mathbf{Q}_p$ (p prime).) Let V be an algebraic variety over K . Then V_K is canonically equipped with a Hausdorff locally compact space topology. If $f : V \rightarrow W$ is a K -morphism of V into an algebraic variety W over K , then $f_K : V_K \rightarrow W_K$ is continuous. Finally, if V is irreducible, non-singular, of dimension n , then V_K is a K -analytic manifold of dimension n [39, App. III].

Recall also that if V is a homogeneous space over K of an algebraic K -group H , then V_K is the union of a finite number of orbits of H_K [5, § 6.4], which are open and closed analytic submanifolds of V_K .

Lemma 0.74 (9.2). *Assume k is locally compact of characteristic zero and G is solvable, split over k . Let V be an algebraic k -variety on which G acts k -morphically. Let M be a non-empty compact subset of V_k stable under G_k . Then M contains a fixed point under G_k ¹¹.*

Proof. The proof proceeds by induction on $\dim G$. Suppose first that G is of dimension one, and thus isomorphic to $\mathbf{G}_a$ or $\mathbf{G}_m$. Let $v \in M$ and Y be the closure of $G \cdot v$. Suppose first that $Y \cap M = G \cdot v \cap M$. Then $G \cdot v \cap M$ is closed and therefore compact. But $(G \cdot v)_k$ is the union of a finite number of orbits of G_k , which are open and

¹¹The original statement assumed V to be complete. Independently, J. I. Hano mentioned this lemma to one of us when $k = \mathbf{R}$, which led us to lift the restriction initially made on V .

closed in $(G \cdot v)_k$. Consequently, $G \cdot v \cap M$ is the union of a finite number of orbits of G_k , which are compact. Each is of the form G_k/H_k , where H is a closed k -subgroup of G , and since $G = \mathbf{G}_a$ or $\mathbf{G}_m$, this implies $G = H$; thus v is fixed by G . Suppose now that $Y \cap M \neq G \cdot v \cap M$, and let $y \in M \cap Y$, $y \notin G \cdot v$. Its orbit $G \cdot y$ is of dimension strictly smaller than that of v [1, § 16], hence is of dimension zero, and y is a fixed point.

If $\dim G > 1$, the group G contains a connected normal k -subgroup N of codimension one split over k . The set W of fixed points under N is an algebraic k -subvariety of V , and $W \cap M$ is non-empty by the induction hypothesis. The group G/N acts k -morphically on W [25, prop. 2] and $(G/N)_k$ leaves $W \cap M$ stable. Since G is split over k , G/N is also, and we are reduced to the previous case. $\square$

Proposition 0.75 (9.3). *Assume k is locally compact of characteristic zero, and let H be a closed k -subgroup of G . Then the following conditions are equivalent:*

- (i) H contains a maximal connected subgroup of G^0 which is trigonalizable over k .
- (ii) G_k/H_k is compact.
- (iii) $(G/H)_k$ is compact.

The group $(G^0)_k$ is open, of finite index in G_k . One can therefore restrict to the case where G is connected.

(i) $\Rightarrow$ (ii) Let L be a maximal connected subgroup trigonalizable over k of G contained in H . Then G_k/L_k identifies with the set of k -rational points of a projective k -variety (8.2), and is therefore compact. *A fortiori*, G_k/H_k is compact.

(ii) $\Rightarrow$ (i) Applying 9.2 to the case where $V = G/H$ and $M = G_k/H_k$, we see that every connected solvable subgroup split over k of G has a fixed point in G_k/H_k , and is therefore conjugate over k to a subgroup of H .

$(G/H)_k$ is the union of a finite number of orbits of G_k (9.1). We can therefore find a finite number of k -subgroups H_i ($1 \leq i \leq m$, $H_1 = H$) of G such that $(G/H)_k = \bigcup_i G_k/H_{i,k}$, and each orbit is open and closed in $(G/H)_k$. This proves that (iii) $\Rightarrow$ (ii).

It remains to show that (i) $\Rightarrow$ (iii). The groups H_i are obviously conjugate to $H = H_1$ over a suitable extension K of k (since they are stabilizer groups in the same orbit of G). It therefore follows from 8.7 that for all i , H_i contains a maximal k -split solvable subgroup of G . In view of the implication (i) $\Rightarrow$ (ii) already established, it follows that $G_k/H_{i,k}$ is compact for all i , and thus $(G/H)_k$ is compact.

Remark. — The equivalence of (i) and (ii) was also obtained by I. Satake (*Publ. Math., I.H.E.S.*, Paris, no. 18 (1964), prop. 1, § 1), and for $k = \mathbf{R}$ by R. Hermann (*Proc. Nat. Acad. Sci., U.S.A.*, 51 (1964), 456–461) and J. I. Hano.

Corollary 0.76 (9.4). *The group G_k is compact if and only if G is reductive and anisotropic over k .*

If G_k is compact, then every connected k -subgroup trigonalizable over k is reduced to the identity element, hence G is reductive, and, by 8.5, anisotropic over k . The converse is a consequence of the implication (i) $\Rightarrow$ (ii) of 9.3.

§ 10. CLOSED CONJUGACY CLASSES AND CENTRALIZERS OF TORI

10.0. Notations

Let V be an algebraic variety. The tangent space to V at a simple point v will be denoted by $T(V)_v$. If $f: V \rightarrow W$ is a morphism of V into an algebraic variety, we will denote its differential by df . If $v \in V$ and $f(v) = w$ are simple points, then df induces a linear map from $T(V)_v$ into $T(W)_w$ which will be denoted in general by df_v .

Lemma 0.77 (10.1). *Let H be a closed subgroup of G and s a semi-simple element of G normalizing H . Then the space $\mathfrak{z}(s) \cap \mathfrak{h}$ of fixed points of $\text{Ad } s$ in the Lie algebra $\mathfrak{h}$ of H is the Lie algebra of the group $\mathcal{Z}(s) \cap H$.*

We can obviously assume that $G = \mathbf{GL}_n$. We will first prove the lemma when $H = \mathbf{GL}_n$. For this, we consider a Bruhat decomposition of $\mathbf{GL}_n$, and we resume the notation of 2.3, assuming $s \in T$. Let U^- be the unipotent subgroup generated by the U_a ($a < 0$). Then $(x, t, y) \mapsto x \cdot t \cdot y$ is an open immersion of $U^- \times T \times U$ into G , whose image will be denoted by V . In view of 2.3, the restriction of $\text{Int } s$ to U_a , identified with $\mathbf{G}_a$,

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is the multiplication by $a(s)$, hence the restriction of the differential $\text{Ad } s$ of $\text{Int } s$ to the algebra $\mathfrak{u}_a$ of U_a is also the multiplication by $a(s)$. Consequently $\mathcal{Z}(s) \cap V$ is generated by T and by the groups U_a , where a runs through the roots equal to one on s , and

$$\mathfrak{z}(s) = \mathfrak{t} \oplus \sum_{a(s)=1} \mathfrak{u}_a, \quad (\mathfrak{t} \text{ Lie algebra of } T),$$

thus $\mathfrak{z}(s)$ is the Lie algebra of $\mathcal{Z}(s)$.

Let c_s be the map from G to itself which sends g to the commutator (s, g) . It is clear that $(dc_s)_e = \text{Ad } s - \text{I}$, hence

$$\mathfrak{m} = \text{Im}((dc_s)_e) = \sum_{a(s) \neq 1} \mathfrak{u}_a,$$

$$\mathfrak{g} = \mathfrak{z}(s) \oplus \mathfrak{m}. \quad (1)$$

Let $M = c_s(G)$. This is the translate by s of an orbit of G , acting on itself by inner automorphisms, so M is a non-singular variety. On the other hand, once U_a is identified with $\mathbf{G}_a$, we have $c_s(u) = (a(s) - 1) \cdot u$ ($u \in U_a$), hence $\mathfrak{m}$ is contained in the tangent space at e to M . But $s^{-1} \cdot M$ is the conjugacy class of s^{-1} ; hence its dimension is equal to that of $G/\mathcal{Z}(s)$, that is to say to that of $\mathfrak{m}$, in view of what has been shown above. It follows that $\mathfrak{m} = T(M)_e$.

Let now H be arbitrary. In view of (1) and the fact that ${}^s H = H$, we have

$$\mathfrak{h} = \mathfrak{z}(s) \cap \mathfrak{h} + \mathfrak{m} \cap \mathfrak{h}. \quad (2)$$

Let us also note, in view of 10.3, that since the restriction of $(dc_s)_e$ to $\mathfrak{m}$ is an isomorphism,

$$\mathfrak{m} \cap \mathfrak{h} = (dc_s)_e \mathfrak{h}. \quad (3)$$

The tangent space $\mathfrak{c}$ to $\mathcal{Z}(s) \cap H$ at e is contained in $\mathfrak{z}(s) \cap \mathfrak{h}$ and the tangent space $\mathfrak{m}'$ at e to $M' = c_s(H)$ is contained in $\mathfrak{m} \cap \mathfrak{h}$, since $M' \subset M \cap H$. As c_s is a morphism from H onto M' whose fibers are the left cosets $h \cdot (\mathcal{Z}(s) \cap H)$ ($h \in H$), we must have $\dim \mathfrak{m}' + \dim \mathfrak{c} = \dim \mathfrak{h}$, whence, in view of (2),

$$\mathfrak{c} = \mathfrak{z}(s) \cap \mathfrak{h}, \quad \mathfrak{m}' = \mathfrak{m} \cap \mathfrak{h}. \quad (4)$$

Theorem 0.78 (10.2). *Let g be an element of G , and $\mathcal{C}(g)$ its conjugacy class.*

- (i) *If g is semi-simple, and H a closed subgroup of G normalized by g , then the set $\text{Int}_H(g)$ of conjugates of g by H is closed. In particular, $\mathcal{C}(g)$ is closed.*
- (ii) *If G is semi-simple and connected, the closure $\overline{\mathcal{C}(g)}$ of $\mathcal{C}(g)$ contains the semi-simple part g_s of g ; in particular, $\mathcal{C}(g)$ is not closed when g is not semi-simple.*

We will denote by $C(X, \lambda)$ and $M(X, \lambda)$ the characteristic polynomial and the minimal polynomial of a square matrix X , where λ is an indeterminate.

- (i) $\text{Int}_H(g)$ is the conjugacy class of g in the algebraic group generated by H and the group $\mathcal{A}(g)$ adherent to g (which normalizes H). We can therefore assume $H = G$. We identify G with one of its linear realizations. Let

$$L = \{x \in G \mid M(g, x) = 0, C(\text{Ad } x, \lambda) = C(\text{Ad } g, \lambda)\}.$$

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It is an algebraic subset of G , stable under inner automorphisms. We consider L as a space of transformations of G , acting by inner automorphisms. The orbit of $x \in L$ is the conjugacy class of x in G and its dimension is consequently equal to that of $G/\mathcal{Z}(x)$. Now, if $x \in L$, it annihilates the minimal polynomial of g , hence $M(x, \lambda)$ divides $M(g, \lambda)$ and has only simple factors. Consequently, x is also semi-simple, and 10.1 implies that $\dim \mathcal{Z}(x)$ is equal to the multiplicity of the eigenvalue one of $\text{Ad } x$. But $\text{Ad } x$ and $\text{Ad } g$ have the same characteristic polynomial, hence $\dim \mathcal{Z}(x)$ is also equal to the multiplicity of the eigenvalue one for $\text{Ad } g$, and consequently (10.1) to $\dim \mathcal{Z}(g)$. The orbits of G in L thus all have the same dimension. They are therefore closed [1, §16].

(ii) Let $g = g_s.g_u$ be the multiplicative Jordan decomposition of g . We assume $k = \bar{k}$ and resume the notation of 2.3, assuming that $g_s \in T$, $g_u \in U$. One can find a one-parameter subgroup $\mu : \mathbf{G}_m \rightarrow T$ such that $\langle \mu, a \rangle = m_a$ is > 0 for every simple root, hence for every positive root. Let u_a be the component of g_u in U_a , identified with $\mathbf{G}_a$, and let φ be the morphism of algebraic varieties from $\mathbf{G}_m$ into U which maps x to $\text{Int } \mu(x)(g_u)$. The component in U_a of $\varphi(x)$ is equal to $x^{m_a}.u_a$ ($a \in \Phi^+$). Since the m_a are > 0 , this shows that φ is the restriction to $\mathbf{G}_m$ of a morphism of the affine line which maps the origin to the identity element. The latter therefore belongs to the closure of $\varphi(\mathbf{G}_m)$. Since $\mu(\mathbf{G}_m)$ centralizes g_s , it follows that g_s belongs to the closure of $\text{Int}_G \mu(\mathbf{G}_m)(g)$, hence of $\mathcal{C}(g)$.

10.3. Proposition. — *Let H be a closed subgroup of G , s a semi-simple element of the normaliser of H in G , and $F = \mathcal{Z}(s) \cap H$. Then the map $c_s : h \mapsto (s, h)$ is a separable morphism from H onto $M = c_s(H)$, which induces by passing to the quotient an isomorphism of H/F onto M . If H is connected, and if H, s are defined over k , then F^0 is defined over k .*

(Recall that a morphism $f : V \rightarrow W$ of algebraic k -varieties which is surjective (or generically surjective, i.e., such that $f(V)$ contains a non-empty open subset of W) is *separable* if the field $\bar{k}(V)$ of rational functions on V is a separable extension of the image of $\bar{k}(W)$ under the comorphism f^0 associated with f . For this, it suffices that $df_v : T(V)_v \rightarrow T(W)_{f(v)}$ be surjective when v and $f(v)$ are simple points. Indeed, the tangent space at a generic point being identified with the space of derivations of the field of rational functions (see [39], prop. 15, p. 12 and chap. IV, n° 6), this condition means that every derivation of $f^0(\bar{k}(W))$ over $\bar{k}$ extends to a derivation of $\bar{k}(V)$ over $\bar{k}$, hence that $\bar{k}(V)$ is a separable extension of $f^0(\bar{k}(W))$ according to a well-known criterion (see Samuel-Zariski, *Commutative Algebra*, I, Van Nostrand, chap. II, th. 42).)

It is immediate that $c_s(x) = c_s(y)$ ($x, y \in H$) if and only if $x \in y.F$, hence c_s induces a bijective morphism of H/F onto M . To establish the first assertion, it suffices to show that $(dc_s)_h : T(H)_h \rightarrow T(M)_m$ ($m = c_s(h)$) is surjective for every $h \in H$. Identifying the Lie algebra $\mathfrak{h}$ of H with $T(H)_e$, we have $T(H)_h = h.\mathfrak{h}$, and

$$dc_s(h.X) = s.h.X.s^{-1}.h^{-1} - s.h.s^{-1}.X.h^{-1} \quad (X \in \mathfrak{h}),$$

or also

$$dc_s(h.X) = c_s(h). \text{Ad } s.h. (\text{Ad } s - 1)(X),$$

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whence

$$dc_s(T(H)_h) = c_s(h) \cdot \text{Ad } s \cdot h \cdot (\mathfrak{m}), \quad (1)$$

with

$$\mathfrak{m} = (\text{Ad } s - 1)(\mathfrak{h}) = (dc_s)_e(\mathfrak{h}). \quad (2)$$

But, in view of 10.1, (3) and (4), $\mathfrak{m} = T(\mathbf{M})_e$; hence $(dc_s)_e$ is surjective.

Suppose now H is connected. Let $\Gamma \subset H \times M$ be the graph of the map c_s . We have $\Gamma \cap (H \times \{e\}) = F \times \{e\}$ from the set-theoretic point of view. But the surjectivity of dc_s implies that $H \times \{e\}$ and Γ intersect transversally; according to the multiplicity one criterion [39, VI, § 2, Th. 6], we then have

$$F = \text{pr}_H(F \times \{e\}) = \text{pr}_H(\Gamma \cdot (H \times \{e\})),$$

where the dot denotes here an intersection of cycles. This means that the cycle sum of the irreducible components of F , each with coefficient 1, is rational over k . Since F^0 is stable under $\text{Gal}(k_s/k)$, it is a rational component of F over k . Being irreducible, F^0 is then also defined over k [39, Prop. 1, p. 208].

Corollary 0.79 (10.4). *The map $\varphi : h \mapsto h \cdot s \cdot h^{-1}$ induces an isomorphism of homogeneous spaces of H/F onto $\text{Int}_G H(s)$.*

Indeed, φ is the composition of $c_{s^{-1}}$ and left translation by s .

Corollary 0.80 (10.5). *Suppose G is connected and let S be a k -torus of G . Then $\mathcal{Z}(S)$ is defined over k .*

The group $\mathcal{Z}(S)$ is connected [1, Prop. 18.4] and k -closed. It remains to show that it is defined over k_s . We can assume that $k = k_s$, hence that k is infinite. According to 1.10, there then exists $s \in S_k$ such that $\mathcal{Z}(s) = \mathcal{Z}(S)$, and our assertion follows from 10.3.

Proposition 0.81 (10.6). *Suppose G is connected and let $f : G \rightarrow G'$ be a separable k -morphism of G onto a k -group G' . Then the maximal tori defined over k (resp. the maximal split tori over k) of G' are the images under f of the maximal tori defined over k (resp. the maximal split tori over k) of G .*

It is already known that the image of a maximal torus of G is a maximal torus of G' [1, Th. 22.1]. Let T' be a maximal torus of G' defined over k and let $H = f^{-1}(T')^0$. This group is defined over k . Indeed, according to [22, Prop. 1] and [39, Chap. VIII, Th. 4], the cycle $f^{-1}(T')$, each component of which has coefficient one, is rational over k . Since H is k -closed, it is then also defined over k [39, Chap. VIII, Prop. 1]. The group H contains a maximal torus of G [1, loc. cit.], hence also (2.14) a maximal torus T of G defined over k . Then $T' = f(T)$.

If S is a split torus over k of G , then $f(S)$ is also split over k (1.8). To complete the proof, it suffices to prove that if S is a split torus over k of G and if S' is a maximal split torus over k of G' containing $f(S)$, then S is contained in a split torus S'' over k such that $f(S'') = S'$.

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The centralizer of S' is defined over k (10.5), hence contains a maximal torus T' of G' defined over k (2.14). The group $H = f^{-1}(T')^0$ considered above is defined over k , contains S and is of maximum rank, hence $(\mathcal{Z}(S) \cap H)^0$ is defined over k (10.5) and is of maximum rank. By 2.14, this group consequently contains a maximal torus T of G defined over k containing S . We then have $S \subset T_d$, and $f(T) = T'$, hence also (1.8) $f(T_d) = T'_d = S'$.

§ 11. CARTAN SUBGROUPS OF SOLVABLE GROUPS

This section gives new proofs of existence and conjugacy theorems in solvable groups, due to Rosenlicht [26] and Grothendieck [11]. They use Lemma 11.1, which generalizes and refines Lemma 9.6 of [1].

Lemma 0.82 (11.1). *Let H be a k -group and assume that G is a connected unipotent subgroup of H . Let s be a semisimple element of H_k normalizing G . Let F be the centralizer of s in G , c_s the map $g \mapsto (s, g)$ of G into G and $M = c_s(G)$. Then $M \cap F = \{e\}$; the group F is connected, defined over k ; M is a closed irreducible k -subvariety; the product map $\alpha : M \times F \rightarrow G$ and the restriction of c_s to M are isomorphisms of varieties.*

Let us first note that the restriction of c_s to the center $\mathcal{Z}(G)$ of G is a homomorphism of $\mathcal{Z}(G)$ into itself and that, more generally:

$$c_s(xy) = c_s(x).c_s(y) \quad (x \in G, y \in \mathcal{Z}(G)). \quad (1)$$

Let $v = (s, g) \in F$. In the equality $g.s^{-1}.g^{-1} = s^{-1}.v$, the right-hand side is the product of a semisimple element and a unipotent element commuting with each other, while the left-hand side is semisimple, hence

$$v = e, \quad g \in F \text{ and } M \cap F = \{e\}.$$

M is the translate by s of the set of conjugates by G of the semisimple element s^{-1} . It is therefore a closed irreducible subvariety (10.2), obviously defined over k .

We now want to show that

$$F \text{ is connected and } \alpha \text{ is bijective.} \quad (*)$$

If $\dim G = 1$, or more generally if G is commutative, $(*)$ results from [1, Lemma 9.6]. We therefore proceed by induction on $\dim G$ and assume G is non-commutative. Since $(*)$ is geometric in nature, we can also assume that $k = \bar{k}$. Let $V \neq \{e\}$ be a closed connected subgroup of the center of G , stable under $\text{Int } s$. Let π be the projection of the normalizer N of V in H onto $N' = N/V$. The group $G' = \pi(G)$ is connected unipotent, $s' = \pi(s)$ is semisimple, normalizes G' , and we can apply the induction hypothesis to $c_{s'}$, G' , $M' = c_{s'}(G')$ and $F' = c_{s'}^{-1}(e)$. We obviously have $c_{s'} \circ \pi = \pi \circ c_s$, hence $M' = \pi(M)$ and $\pi(F) \subset F'$.

Let $g' \in F'$ and $g \in \pi^{-1}(g')$. We have $s.g.s^{-1} = g.z$ ($z \in V$) and, taking into account the induction hypothesis, we can write

$$z = s.x.s^{-1}.x^{-1}.y \quad (x \in V, y \in F \cap V),$$

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whence, setting $h = x^{-1} \cdot g = g \cdot x^{-1}$, the equality $s \cdot h \cdot s^{-1} = h \cdot y$, which shows that $y \in M \cap F$, so that $y = e$, and that $h \in \pi^{-1}(g') \cap F$. Consequently, the restriction of π to F is a morphism of F onto F' , with kernel $F \cap V$. Since F' and $F \cap V$ are connected by the induction hypothesis, F is also connected.

From the equality $G' = M' \cdot F'$, we then obtain $G = M \cdot F \cdot V$; but $V = c_s(V) \cdot (F \cap V)$ whence, in view of (1),

$$G = c_s(G) \cdot F \cdot c_s(V) \cdot (F \cap V) = c_s(G) \cdot c_s(V) \cdot F = c_s(G \cdot V) \cdot F = M \cdot F,$$

which shows that α is surjective. It remains to prove that α is injective. Since F is a group, this is equivalent to showing that if $a, b \in M$ and $a = b \cdot f$ ($f \in F$), then $f = e$. We have $\pi(a) = \pi(b) \cdot \pi(f)$, so $\pi(f) = e$, and $f \in V$.

But if $x, y \in G$ are such that $c_s(x) = a$, $c_s(y) = b$, we immediately obtain from the equality $a = b \cdot f$ and the fact that $f \in V$ is central that $f = c_s(y^{-1} \cdot x)$, whence $f \in M \cap F$ and $f = e$, which completes the proof of (*). We have $c_s(x) = c_s(y)$ ($x, y \in G$) if and only if $x \in y \cdot F$, which, in view of (*), shows that the restriction β of c_s to M is a bijective K -morphism of M onto M . We have already seen that α is also bijective. Since the varieties

G, M, F are non-singular, and thus normal, it suffices, to complete the proof of the lemma (by virtue of Zariski's Main Theorem [17, chap. V, §2]), to show that α and β are birational, and thus that their differentials are bijective on the tangent spaces.

Let $\mathfrak{g}, \mathfrak{f}$ be the Lie algebras of G and F respectively. According to 10.3 (1), (2), we have

$$dc_s(T(G)_g) = c_s(g) \cdot \text{Ad } g(\mathfrak{m}) \quad (\mathfrak{m} = (\text{Ad } s - 1)(\mathfrak{g}); g \in G), \quad (2)$$

$$dc_s(T(G)_g) = T(M)_m \quad (m = c_s(g)). \quad (3)$$

Let us show that $T(M)_m \cdot f$ and $m \cdot T(F)_f$ are linearly independent for any $f \in F$. In view of (2), this is equivalent to

$$m \cdot \text{Ad } g(\mathfrak{m}) \cdot f \cap m \cdot \mathfrak{f} \cdot f = \{0\}$$

hence to

$$\text{Ad } g(\mathfrak{m}) \cap \mathfrak{f} = \{0\}. \quad (4)$$

The group $\text{Ad } G$ is a connected unipotent subgroup of $\text{GL}(\mathfrak{g})$, and $\text{Ad } s$ is a semisimple element of $\text{GL}(\mathfrak{g})$ normalizing $\text{Ad } G$. One can find a basis (X_i) ($1 \leq i \leq \dim \mathfrak{g}$) of $\mathfrak{g}$ consisting of eigenvectors of $\text{Ad } s$ with respect to which $\text{Ad } G$ is represented by upper triangular matrices. To see this, we argue by induction on $\dim V$. Let W be the space of fixed points of $\text{Ad } G$ and let W' be a complement of W stable under s . We have $W \neq 0$, so we can assume that $\mathfrak{g}/W$ has a basis (Y_i) with the required properties. We take for (X_i) a basis consisting of eigenvectors of s whose first elements generate W and whose subsequent ones span W' and map to the vectors

Y_i under the canonical projection. Soit $w \in \mathfrak{m} \setminus \{0\}$. We thus have $w = \sum_i m_i \cdot X_i$ with $m_i \in \bar{k}$, $m_i = 0$ if

$X_i \in \mathfrak{f}$. If j is the largest index such that $m_j \neq 0$, then $\text{Ad } g(w) = m_j X_j$ modulo a. . .

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linear combination of $X_1, \dots, X_{j-1}$. Since $m_j \neq 0$, $X_j \notin \mathfrak{f}$, so $\text{Ad } g(w) \notin \mathfrak{f}$, which proves (4). For reasons of dimension, $T(M)_{m \cdot f}$ and $m \cdot T(F)_f$ generate $T(G)_{m \cdot f}$. As they are obviously part of the image of $d\alpha_{(m, f)}$, it follows that $d\alpha_{(m, f)}$ is bijective.

Let $z = c_s(m \cdot f) = c_s(m)$. According to (3), $(dc_s)_{m \cdot f}$ maps $T(G)_{m \cdot f}$ onto $T(M)_z$. Its kernel, which contains $m \cdot T(F)_f$, must therefore be equal to $m \cdot T(F)_f$ for reasons of dimension. But $m \cdot T(F)_f$ has only zero in common with $T(M)_{m \cdot f}$ according to what has already been proven, so the restriction of dc_s to $T(M)_{m \cdot f}$ is injective, and $d\beta(T(M)_m) = T(M)_z$.

Proposition 0.83 (11.2). *Suppose G is connected solvable. Let L be a subgroup of G_k consisting of semi-simple elements. Then $\mathcal{Z}_G(L)$ is connected, defined over k , and L is contained in a maximal torus of G .*

Since the group L is commutative [1, Prop. 10.2], an immediate induction allows us to reduce to the case where L is generated by an element x . This latter belongs to a maximal torus T of G [1, Theorem 12.6], and G is the semi-direct product of T by its unipotent radical G_u [1, Theorem 12.2]. Consequently, $\mathcal{Z}_G(x) = T \cdot (\mathcal{Z}_G(x) \cap G_u)$. Since $\mathcal{Z}_G(x) \cap G_u$ is connected (11.1), the group $\mathcal{Z}_G(x)$ is also connected. It is then defined over k according to 10.3.

The following lemma is an immediate consequence of some results of Rosenlicht.

Lemma 0.84 (11.3). *Suppose G is connected unipotent commutative, and such that $G^p = \{e\}$ if the characteristic p of k is non-zero. Let T be a k -torus which operates k -morphically on G , and has only the identity element of G as a fixed point. Then G is split over k .*

According to [24, Prop. 1.2], G is $\bar{k}$ -isomorphic to a product of groups $\mathbf{G}_a$. Since the group T is split over k_s (1.5), the lemma on p. 109 of [26] (in which we have $G_0 = \{e\}$ in view of our hypothesis) shows that G is k_s -isomorphic to a product of groups $\mathbf{G}_a$. Thus, G is split over k_s , and therefore over k [26, Theorem 3].

Theorem 0.85 (11.4). *Let G be connected solvable. Then G possesses a Cartan subgroup (resp. a maximal torus) defined over k . Two such subgroups are conjugate by $(\mathcal{C}^\infty G)_k$.*

A Cartan subgroup C possesses a unique maximal torus T of which it is the centralizer [1, § 20]. If C (resp. T) is defined over k , the same is true of T (resp. C) according to [23, Prop. 9] (resp. (10.3)). To prove 11.4, we can therefore restrict ourselves to considering Cartan subgroups.

(i) We will first give a proof, different from that of [11], of the existence of a k -Cartan subgroup when k is infinite. (If k is finite, we refer to [23, Note p. 45].) We proceed by induction on $\dim G$.

Assume first that $\mathcal{C}^\infty G$ possesses a proper connected k -subgroup $N \neq \{e\}$ normal in G , and let

$\pi: G \rightarrow G' = G/N$ be the canonical projection. Let C' be a k -Cartan subgroup of G' . Since

$\mathcal{C}^\infty G' = \pi(\mathcal{C}^\infty G) \neq \{e\}$, the group G' is not nilpotent, so $C' \neq G'$ and $H = \pi^{-1}(C')$ is a proper connected k -subgroup of G . It contains a Cartan subgroup of G [1, Theor. 22.2], hence any subgroup

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of Cartan of H is a Cartan subgroup of G ; by the induction hypothesis, there exists at least one which is defined over k .

If $\mathcal{C}^\infty G$ does not contain any subgroup N satisfying the conditions above, then $\mathcal{C}^\infty G$ is commutative, and $(\mathcal{C}^\infty G)^p = \{e\}$ when the characteristic p of k is non-zero. Let us show the existence of a semisimple element

$$a \in G_k \text{ such that } \mathcal{Z}(a) \cap \mathcal{C}^\infty G = \{e\}.$$

Let $\pi: G \rightarrow G' = G/\mathcal{C}^\infty G$ be the canonical projection. The quotient G' is a nilpotent k -group, its unique maximal torus T' is defined over k [23, Prop. 9], and $H = \pi^{-1}(T')$ is a connected k -group. It operates by inner automorphisms on $\mathcal{C}^\infty G$; since $\mathcal{C}^\infty G$ is commutative, it operates trivially on itself, thus, by passing to the quotient, G' operates k -morphically on $\mathcal{C}^\infty G$ [25, Prop. 2]. The intersection of $\mathcal{C}^\infty G$ with the centralizer of a maximal torus of H is reduced to $\{e\}$ [1, Theorem 13.4], hence the only fixed point of T' in $\mathcal{C}^\infty G$ is the identity element. Consequently, $\mathcal{C}^\infty G$ is split over k (11.3), and the fibration of G over G' has a regular section s defined over k [26, Theorem 1].

Let V be the set of elements of T' whose set of fixed points in $\mathcal{C}^\infty G$ is different from the identity element. It is the union of a finite number of proper closed subgroups, and one can find $x \in T'_k - V_k$ (1.10). Let $y = s(x)$. If k is perfect, then the semisimple part y_s of y is rational over k , and $\pi(y_s) = x$, hence y_s is the desired element. If $p \neq 0$, then there exists a power $q = p^m$ of p such that $z = y^q$ is semisimple, and obviously rational over k . We then have $\pi(z) = \pi(y)^q = x^q$. But $t \mapsto t^q$ is a bijection of T' onto itself leaving stable every subgroup; we therefore still have $x^q \in T'_k - V_k$, whence $\mathcal{Z}(z) \cap \mathcal{C}^\infty G = \{e\}$, which establishes our assertion.

The group $\mathcal{Z}(a)$ is connected, defined over k (10.3) and contains a Cartan subgroup C of G . We have $G = C \cdot \mathcal{C}^\infty G$ and $\mathcal{Z}(a) \cap \mathcal{C}^\infty G = \{e\}$, hence $C = \mathcal{Z}(a)$ is defined over k .

(ii) Let T be a maximal torus of G defined over k and g a semisimple element of G_k . We want to show the existence of $x \in (\mathcal{C}^\infty G)_k$ such that $x.g.x^{-1} \in T$.

The projection $\pi: G \rightarrow G' = G/\mathcal{C}^\infty G$ induces an isomorphism of T onto the maximal k -torus T' of G' . Indeed, $\pi: T \rightarrow T'$ is bijective, $\pi(T)$ is a maximal torus of G' [1, Theor. 22.2], $\pi: G \rightarrow G'$ is separable, and the kernel of $d\pi: \mathfrak{g} \rightarrow \mathfrak{g}'$ is the Lie algebra of $\mathcal{C}^\infty G$; but G is, as a variety, the product of T by G_u [1, Theor. 12.2], hence $d\pi$ induces an isomorphism of $\mathfrak{t}$ onto $\mathfrak{t}'$, whence our assertion. There exists therefore a unique element $t \in T_k$ such that $\pi(t) = \pi(g)$. We have $t^{-1} = g^{-1}.u$, with $u \in (\mathcal{C}^\infty G)_k$. The conjugacy theorem over k of maximal tori shows the existence of $v \in \mathcal{C}^\infty G$ such that $v.g^{-1}.v^{-1} = t^{-1}$, whence

$$u = (g, v) \in c_g(\mathcal{C}^\infty G),$$

and 11.1 implies the existence of $x \in (\mathcal{C}^\infty G)_k$ such that $u = (g, x)$. We then have

$$t^{-1} = g^{-1}.g.x.g^{-1}.x^{-1} = x.g^{-1}.x^{-1},$$

whence (ii).

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(iii) Let C, C' be two k -Cartan subgroups of G , and T, T' their respective maximal tori. Suppose first that k is infinite. In view of 1.10, we can find $a' \in T'_k$ such that $\mathcal{Z}(a') = \mathcal{Z}(T') = C'$. In view of (ii), there exists

$x \in (\mathcal{C}^\infty G)_k$ such that $x \cdot a' \cdot x^{-1} = a \in T$. Obviously, $x \cdot \mathcal{Z}(a') \cdot x^{-1} = \mathcal{Z}(a)$. Thus, $\mathcal{Z}(a)$ is a Cartan subgroup containing C , hence equal to C , whence $x \cdot C' \cdot x^{-1} = C$.

Suppose now that k is finite, or more generally perfect, and let

$$V = \{x \in \mathcal{C}^\infty G \mid x \cdot C \cdot x^{-1} = C'\}.$$

This is an algebraic set [1, 2.5] defined over k since C and C' are (and k is perfect), non-empty in view of the conjugation theorem of Cartan subgroups over $\bar{k}$, and which is a homogeneous space for the group

$H = \mathcal{N}_G(T) \cap \mathcal{C}^\infty G$. But $\mathcal{N}_G(T) = \mathcal{Z}_G(T) = C$ [1, Prop. 10.2, Theor. 12.2], and 11.1 shows that H is connected. Being unipotent, H is then split over k [23, Cor. 2, p. 34], hence V possesses a rational point over k [22, Theor. 10].

11.5. Corollary. — *Let T be a maximal torus defined over k of G , S a k -torus of G and L a subgroup of G_k consisting of semisimple elements. Then S (resp. L) is conjugate to a subgroup of T by an element of $(\mathcal{C}^\infty G)_k$, and the group $\mathcal{A}(L)$ adherent to L [1, §3] is defined over k .*

The groups $\mathcal{Z}(S)$ and $\mathcal{Z}(L)$ are connected and defined over k according to 10.5 and 11.2. Let T' be a maximal torus of $\mathcal{Z}(S)$ (resp. $\mathcal{Z}(L)$) defined over k , which exists by 11.4. It contains S (resp. L) and is a maximal torus of G (11.2). The first assertion therefore results from 11.4 applied to T and T' . For the second, we can consequently assume $L \subset T$. Then $\mathcal{A}(L)$ is a subgroup of T which is k -closed, hence defined over k (1.6).

Remark. — Theorem 11.4 for maximal tori is due to Rosenlicht [26, Theor. 4]. The existence of k -Cartan subgroups defined over k in any connected algebraic k -group was proved by Grothendieck [11, Exp. XII, Theor. 1.1]. In [11, Exp. XIV, §6] it is also shown that the homogeneous space G/C (G solvable connected, C a Cartan subgroup defined over k) is k -isomorphic to an affine space; this results from the fact that $\mathcal{C}^\infty G$ is split over k [26, Theor. 4, Cor. 2] and from the fact that a homogeneous k -space of a unipotent group split over k which possesses a rational point over k is k -isomorphic to an affine space [11, Exp. XIV, lemma 6.5]. Let us also

remark that if a maximal torus T of G defined over k possesses a point s rational over k such that $\mathcal{Z}(s) = \mathcal{Z}(T)$, which is always the case when k is infinite (1.10), then the variety $M = c_s(\mathcal{C}^\infty G) = \{s \cdot x \cdot s^{-1} \cdot x^{-1}, (x \in \mathcal{C}^\infty G)\}$ is k -isomorphic to G/C . Indeed, $\mathcal{C}^\infty G$ is k -isomorphic, as a variety, to the product $(\mathcal{Z}(s) \cap \mathcal{C}^\infty G) \times M = (C \cap \mathcal{C}^\infty G) \times M$, according to 11.1, hence the projection of G onto G/C is an isomorphism of M onto G/C .

We end this section with an application to not necessarily solvable groups.

11.6. Proposition. — *Suppose that $R_u(G^0)$ is split over k . Then the maximal k -split tori of G are conjugate by elements of G_k^0 .*

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(If k is perfect, the assumption made on $R_u(G^0)$ is automatically satisfied [23, Cor. 1, 2 to Prop. 5], thus 11.6 generalizes the assertion of conjugacy of split tori made in 8.2.)

For the proof, we may assume G is connected. Let π be the canonical projection of G onto $G' = G/R_u(G)$, and let S, S' be two maximal k -split tori of G . In view of 10.6, $\pi(S)$ and $\pi(S')$ are maximal k -split tori of G' , and thus are conjugate over k (4.21). Since $R_u(G)$ is split over k , the map $\pi : G_k \rightarrow G'_k$ is surjective; there exists therefore $g \in G_k$ such that ${}^g S \subset \pi^{-1}(\pi(S')) = S' \cdot R_u(G)$, and we are reduced to 11.4.

§ 12. RATIONALITY QUESTIONS FOR LINEAR REPRESENTATIONS

In this section, the characteristic of k is zero, G is connected, semisimple, and $\tilde{G}$ denotes the universal covering of G . We fix a maximal torus T of G and a Borel subgroup B of G containing T , and we denote by Δ the corresponding set of simple roots. We write Γ for $\text{Gal}(\bar{k}/k)$.

We first recall some classical properties of linear representations of G (see [10, 14, 29, 31]).

12.1. Let $\rho : G \rightarrow \text{GL}(V)$ be an irreducible representation of G in the vector space V . There exists in V one and only one line D_ρ stable under B . The orbit C_ρ of D_ρ under G will be called the *cone of the representation* ρ . The stabilizer groups of the lines of C_ρ form a conjugacy class of parabolic subgroups of G , denoted by $\mathcal{P}_\rho$, the *class of parabolic subgroups* of ρ .

Let V' be the projective space of lines of V , and $\text{Aut } V'$ the group of automorphisms of V' . The representation ρ defines a projective representation ρ' of G , i.e., a morphism $\rho' : G \rightarrow \text{Aut } V'$, which is irreducible, that is to

say, it does not leave invariant any proper projective subspace of V' . To C_ρ is associated a closed subvariety C'_ρ of V' , which is identified with the homogeneous space G/P , $P \in \mathcal{P}_\rho$, and is the unique closed orbit of G in V' .

An element $P' \in \mathcal{P}_\rho$ has only one fixed point in V' , and this fixed point belongs to C'_ρ .

12.2. The representation of T in D_ρ induced by ρ defines a character l_ρ of T , the *dominant weight* of ρ , which characterizes ρ up to equivalence. If $(,)$ denotes an admissible scalar product on $X^*(T) \otimes \mathbf{R}$, we have

$$2(l_\rho, a) = (a, a) \cdot m_a \quad (m_a \in \mathbf{N}, a \in \Delta), \quad (1)$$

and conversely every element of $X^*(T)$ satisfying the conditions (1) is the dominant weight of an irreducible representation.

Suppose $G = \tilde{G}$. There then exists for any set of integers $m_a \geq 0$ ($a \in \Delta$) a dominant weight satisfying (1). The dominant weights l_a defined by

$$2(l_a, a) = (a, a), \quad (l_a, b) = 0 \quad (a \neq b) \quad (a, b \in \Delta), \quad (2)$$

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are the *fundamental dominant weights*, and the dominant weights of G are the linear combinations with integer coefficients ≥ 0 of the fundamental weights. Even if G is not simply connected, we will express every dominant weight l_ρ as a linear combination of the l_a , which is justified by considering either l_ρ as a dominant weight of $\tilde{G}$, or the l_a as elements of $X^*(T) \otimes \mathbf{Q}$. The coefficient of l_a will sometimes be denoted by $c_a(l_\rho)$. We therefore have

$$l_\rho = \sum_{a \in \Delta} c_a(l_\rho) \cdot l_a, \quad c_a(l_\rho) = 2(l_\rho, a) \cdot (a, a)^{-1} \in \mathbf{N}. \quad (3)$$

Recall also that $\mathcal{P}_\rho = \mathcal{P}_\theta$, where θ is the set of $a \in \Delta$ orthogonal to l_ρ .

12.3. We will denote by $\mathcal{R}(G)$ or $\mathcal{R}$ (resp. $\mathcal{R}'(G)$ or $\mathcal{R}'$) the set of equivalence classes of irreducible linear (resp. projective) representations of G . The notions introduced previously which depend only on the equivalence class of ρ or of ρ' will also be indexed by the class of ρ or ρ' , and $d(\xi)$ ($\xi \in \mathcal{R}$) (resp. $d(\xi')$, $\xi' \in \mathcal{R}'$) will denote the dimension of the vector (resp. projective) space of any representation contained in ξ (resp. ξ').

It is clear that $\mathcal{R}'(\tilde{G}) = \mathcal{R}'(G)$; we have a natural map from $\mathcal{R}(G)$ to $\mathcal{R}'(G)$, which associates to every linear representation the associated projective representation, and which is bijective when G is simply connected.

An element $\xi \in \mathcal{R}$ is *rational over k* if it contains a k -morphism $G \rightarrow \mathbf{GL}_d$ ($d = d(\xi)$). Since two vector spaces of the same dimension defined over k are k -isomorphic, it amounts to the same to require the existence in ξ of a k -morphism $G \rightarrow \mathbf{GL}(V)$, where V is a vector space defined over k , of dimension $d(\xi)$. Such a morphism is determined up to equivalence over k . Indeed, if $\rho : G \rightarrow \mathbf{GL}(V)$ and $\rho' : G \rightarrow \mathbf{GL}(V')$ are two k -morphisms belonging to ξ , there exists a $\bar{k}$ -isomorphism $A : V \rightarrow V'$ such that

$$A \cdot \rho(x) = \rho'(x) \cdot A, \quad (x \in G_{\bar{k}})$$

hence, for all $\gamma \in \Gamma$ and $x \in G_{\bar{k}}$,

$$\gamma A \cdot \rho(\gamma x) = \rho'(\gamma x) \cdot \gamma A,$$

and Schur's lemma implies the existence of $a_\gamma \in \bar{k}^*$ such that $A = a_\gamma \cdot \gamma A$. It is immediate that $\gamma \mapsto a_\gamma$ is a 1-cocycle of Γ with values in $\bar{k}^*$, hence a coboundary [30, prop. 2, p. 158], which means that there exists $b \in \bar{k}^*$ such that $a_\gamma = b^{-1} \cdot \gamma b$ for all $\gamma \in \Gamma$. We then have $\gamma(b \cdot A) = b \cdot A$, hence B is a k -isomorphism of V onto V' which realizes an equivalence of ρ with ρ' .

It is clear that if ξ is rational over k , the class of parabolic subgroups $\mathcal{P}_\xi$ associated with ξ is stable under Γ , that is to say, is defined over k (5.24), but the converse does not hold, as can already be seen by considering a unitary form of $\mathbf{SL}_2$.

An element $\xi \in \mathcal{R}$ is *strongly rational over k* if it is rational over k and if $\mathcal{P}_\xi$ contains a k -group.

An element $\eta \in \mathcal{R}'$ is *rational over k* if there exists a k -form V' of the projective space

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$\mathbf{P}_d$ ($d = d(\eta)$) and a k -morphism $\sigma : G \rightarrow \text{Aut } V'$ belonging to η . Since $\mathbf{P}_d$ has in general non-trivial k -forms, the “Severi-Brauer varieties” [30, chap. X, 6], this condition is weaker than the existence of a k -morphism $G \rightarrow \mathbf{PGL}_{d+1} \cong \text{Aut } \mathbf{P}_d$ belonging to η . If η comes from $\xi \in \mathcal{R}$, this latter condition is equivalent to the fact that ξ is rational over k . Indeed, if $\pi : \mathbf{GL}_d \rightarrow \mathbf{PGL}_d$ is the natural projection and if $\rho : G \rightarrow \mathbf{GL}_d$ is such that

$\rho' = \pi \circ \rho$ is a k -morphism, then ρ is defined over k , because it is the unique morphism from G to $\mathbf{GL}_d$ such that $\rho' = \pi \circ \rho$. The converse is obvious.

In the following, we will be interested in the following three conditions concerning an element $\xi \in \mathcal{R}(G)$ and its image $\xi' \in \mathcal{R}'(G)$:

- a) ξ' is rational over k ,
- b) ξ is rational over k ,
- c) ξ is strongly rational over k .

It is clear that c) $\Rightarrow$ b) $\Rightarrow$ a). If G is split over k , then $\tilde{G}$ is also, and every element of $\mathcal{R}(G)$ (resp. $\mathcal{R}'(G)$) is strongly rational (resp. rational) over k . This follows from the fact that the irreducible representations of g can be constructed rationally over the base field [14, chap. VII]. One can also see this by noting that all linear systems of divisors on G/B that arise in the construction of irreducible representations of G (see [29, 31] in characteristic zero, [10] in the general case) are rational over k . It follows in particular that every element of $\mathcal{R}(G)$ is strongly rational over $\bar{k}$.

12.4. The group Γ operates in a natural way on the linear or projective representations, preserving equivalence, and thus also on $\mathcal{R}$ and $\mathcal{R}'$. If $\rho : G \rightarrow \mathbf{GL}_d$ is a $\bar{k}$ -morphism, we have

$$\gamma(\rho)(g) = \gamma(\rho(\gamma^{-1}(g))) \quad (\gamma \in \Gamma, g \in G_{\bar{k}}), \quad (1)$$

in other words,

$$\gamma(\rho) = \gamma \circ \rho \circ \gamma^{-1}. \quad (2)$$

Lemma 0.86 (12.5). *We preserve the preceding notations. Let $\xi \in \mathcal{R}(G)$ and $\gamma \in \Gamma$. Then the dominant weight of $\gamma(\xi)$ is ${}^\gamma\Delta(l_\xi)$, in the notations of 6.2.*

Proof. Let $\rho : G \rightarrow \mathbf{GL}(V)$ be an element of ξ , D the line in V stable under B , u an element of $G_{\bar{k}}$ such that $u \cdot {}^\gamma B \cdot u^{-1} = B$ and $E = {}^\gamma\rho(u) \cdot {}^\gamma D \subset V$. Let also $b \in B$, $x \in D - \{0\}$ and $y = {}^\gamma\rho(u)({}^\gamma x)$. The dominant weight $l_\rho \in X^*(T)$ has a unique extension to B ; we will also denote it by l_ρ . We then have

$${}^\gamma\rho(u \cdot {}^\gamma b \cdot u^{-1}) \cdot y = {}^\gamma\rho(u) \cdot \gamma(\rho(b)) \cdot {}^\gamma\rho(u^{-1}) \cdot y = {}^\gamma\rho(u) \cdot \gamma(\rho(b) \cdot x) = \gamma(l_\rho(b)) \cdot y,$$

thus E is stable under B , acting in ${}^\gamma V$ through the intermediary of ${}^\gamma\rho$, and the character l' of B in E is given by

$$\gamma(l_\rho(b)) = l'(u \cdot {}^\gamma b \cdot u^{-1}),$$

which (6.2 (2)) means precisely that $l' = {}^\gamma\Delta(l_\rho)$. □

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12.6. Proposition. — *Let $\xi \in \mathcal{R}(G)$. Then the following conditions are equivalent:*

- (i) ${}^\Delta\gamma(l_\xi) = l_\xi$ for all $\gamma \in \Gamma$.
- (ii) ξ is stable under Γ .
- (iii) The class of projective representations ξ' is rational over k .
- (iv) There exists a central division algebra C_k over k and a k -morphism

$$G \rightarrow \mathbf{GL}_m(C) \quad (d(\xi) = m \cdot c, \quad c^2 = [C : k]).$$

The equivalence of (i) and (ii) follows from 12.2 and 12.5.

(ii) $\Rightarrow$ (iii), (iv). Let $\rho : G \rightarrow \mathbf{GL}_n$ ($n = d(\xi)$) be an element of ξ defined over $\bar{k}$. Given hypothesis (ii), there exists for every $\gamma \in \Gamma$ a matrix $A_\gamma \in \mathbf{GL}_{n, \bar{k}}$ such that

$${}^\gamma\rho(g) = A_\gamma^{-1} \cdot \rho(g) \cdot A_\gamma \quad (g \in G_{\bar{k}}).$$

By Schur's lemma, A_γ is determined up to multiplication by an element of $\bar{k}^*$ near, hence the image $A'_\gamma = \pi(A_\gamma)$ of A_γ in $\mathbf{PGL}_n$ under the canonical projection is uniquely determined. If $\delta \in \Gamma$, we have

$${}^{\delta \cdot \gamma}\rho(g) = \delta({}^\gamma\rho(\delta^{-1}(g))) = \delta(A_\gamma^{-1} \cdot \rho(\delta^{-1}(g)) \cdot A_\gamma) = \delta(A_\gamma^{-1}) \cdot {}^\delta\rho(g) \cdot \delta(A_\gamma),$$

from which we deduce that $A_\delta \cdot \delta(A_\gamma)$ is the product of $A_{\delta\gamma}$ by a scalar, hence that $A' : \gamma \mapsto A'_\gamma$ is a 1-cocycle of Γ , with values in $\mathbf{PGL}_n$, in the sense of Galois cohomology [5, 30]. As $\mathbf{PGL}_n \cong \text{Aut } \mathbf{P}_{n-1} \cong \text{Aut } \mathbf{M}_n$, this cocycle defines a k -form V' of $\mathbf{P}_{n-1}$ and a k -form M' of $\mathbf{M}_n$ (see [5, §2.6] or [30, Chap. X, §6]). There exist

then $\bar{k}$ -isomorphisms $\varphi : \mathbf{P}_{n-1} \rightarrow V'$ and $\psi : \mathbf{M}_n \rightarrow M'$ tels que ${}^\gamma\varphi = \varphi \cdot A'_\gamma$, ${}^\gamma\psi = \psi \cdot A'_\gamma$ ($\gamma \in \Gamma$). Let then $\sigma = \varphi \circ \rho' \circ \varphi^{-1}$. This is a $\bar{k}$ -morphism from G into $\text{Aut } V'$. For every $\gamma \in \Gamma$, we have

$${}^\gamma\sigma = {}^\gamma\varphi \cdot {}^\gamma\rho' \cdot {}^\gamma\varphi^{-1} = \varphi \cdot A'_\gamma \cdot A'^{-1}_\gamma \cdot \rho' \cdot A'_\gamma \cdot A'^{-1}_\gamma \cdot \varphi^{-1} = \sigma,$$

hence σ is defined over k , which shows that (ii) $\Rightarrow$ (iii). We see similarly that $\psi \circ \rho$ is a k -morphism from G into the group M'^* of invertible elements of M' . But M'_k is a central simple algebra of degree n^2 over k , hence is of the form $\mathbf{M}_m(C_k)$ where C_k is a central division algebra over k , whence (ii) $\Rightarrow$ (iv).

Conversely, let V' be a k -form of $\mathbf{P}_{n-1}$ (resp. M' a k -form of $\mathbf{M}_n$), $\sigma : G \rightarrow \text{Aut } V'$ (resp. $\sigma : G \rightarrow M'^*$) a k -morphism of ξ and $\varphi : \mathbf{P}_{n-1} \rightarrow V'$ (resp. $\psi : \mathbf{M}_n \rightarrow M'$) a $\bar{k}$ -isomorphism. Then $A' : \gamma \mapsto A'_\gamma = \varphi^{-1} \circ {}^\gamma\varphi$ is a 1-cocycle of Γ with values in $\mathbf{PGL}_n$. If we set $\rho = \varphi^{-1} \circ \sigma \circ \varphi$, the preceding calculation shows that ${}^\gamma\rho = A'^{-1}_\gamma \cdot \rho \cdot A'_\gamma$, whence the implications (iv) $\Rightarrow$ (ii) and (iii) $\Rightarrow$ (ii).

12.7. To every class ξ of representations satisfying the conditions of 12.6, there corresponds an element β_ξ of the Brauer group $H^2(k^*)$: the image under the coboundary map $\delta : H^1(\Gamma, \mathbf{PGL}_n) \rightarrow H^2(k^*)$ of the cohomology class of the cocycle A' considered above. Recall that δ is part of the non-commutative cohomology exact sequence associated with the exact sequence

$$1 \rightarrow \mathbf{G}_m \rightarrow \mathbf{GL}_n \rightarrow \mathbf{PGL}_n \rightarrow 1,$$

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and that, since $H^1(\Gamma, \mathbf{GL}_n) = 0$, we in fact have an exact sequence

$$0 \rightarrow H^1(\Gamma, \mathbf{PGL}_n) \xrightarrow{\delta} H^2(k^*).$$

Of course, the image β_ξ of the cohomology class of the cocycle A' of 12.6 under δ corresponds to the division algebra C_k (see [30, Chap. X] for more details).

12.8. Proposition. — *Let $\xi \in \mathcal{R}(G)$ and ξ' be its image in $\mathcal{R}'(G)$. The following two conditions are equivalent:*

- (i) ξ is rational over k .
- (ii) ξ' is rational over k , and the element β_ξ of the Brauer group associated to ξ' in 12.7 is zero.

If ξ is rational over k , then ξ' is also rational, and the cocycle A' of 12.6 is trivial, hence (i) $\Rightarrow$ (ii). Conversely, if $\beta_\xi = 0$, then, by 12.7, A' is a coboundary, so there exists $B \in \mathbf{PGL}_{n,\bar{k}}$ such that $A'_\gamma = B^{-1} \cdot {}^\gamma B$ ($\gamma \in \Gamma$). If we set $\sigma'(g) = B \cdot \rho'(g) \cdot B^{-1}$, we have

$${}^\gamma\sigma'(g) = {}^\gamma B \cdot {}^\gamma\rho'(g) \cdot {}^\gamma B^{-1} = {}^\gamma B \cdot A'^{-1}_\gamma \cdot \rho'(g) \cdot A'_\gamma \cdot {}^\gamma B^{-1} = B \cdot \rho'(g) \cdot B^{-1},$$

hence ${}^\gamma\sigma' = \sigma'$, and σ' is defined over k , which shows that ξ' is represented by a k -morphism of G into $\mathbf{PGL}_n$. Thus ξ is rational over k (12.3).

12.9. In the following, we shall assume that T contains a maximal k -split torus S and that B is contained in a minimal k -parabolic subgroup P . We thus have $P = P_\theta$ ($\theta = \{a \in \Delta \mid a|_S = 0\}$).

12.10. Proposition. — *Let $\xi \in \mathcal{R}(G)$. Then the following two conditions are equivalent:*

- (i) l_ξ is stable under Γ , and the coefficient $c_a(l_\xi)$ is zero if $a|_S = 0$.
- (ii) ξ is strongly rational over k .

Suppose (ii) is satisfied. Then ξ is stable under Γ by 12.6. On the other hand, $\mathcal{P}_\xi$ contains an element defined over k , hence an element $P' \supset P$. We then have $P' = P_{\theta'}$ with $\theta \subset \theta' \subset \Delta$ (5.14), and l_ξ is orthogonal to θ' (12.2) hence to θ , whence the second part of (i).

Suppose now that ξ satisfies (i). By 12.6, there exists a k -form V' of $\mathbf{P}_{d-1}$ ($d = d(\xi)$) and a k -morphism $\rho' : G \rightarrow \text{Aut } V'$ belonging to ξ' . Furthermore, $\mathcal{P}_\xi \succ \mathcal{P}_\theta$, so $\mathcal{P}_\xi$ has an element defined over k (4.7).

Let P' be a k -subgroup of $\mathcal{P}_\xi$. According to what was recalled in 12.1, P' has a unique fixed point Q in V' , which is then necessarily rational over k . A result of F. Châtelet (*Annales E.N.S.*, 61 (1944), 249–300) then implies that V' is k -isomorphic to $\mathbf{P}_{d-1}$; consequently, ξ' contains a k -morphism of G into $\mathbf{PGL}_d$ and ξ is rational over k (12.3).

12.11. Corollary. — ξ is strongly rational over k if and only if l_ξ is the restriction to T of an element of $X^*(P)_k$.

Suppose ξ is strongly rational over k . Let $\rho : G \rightarrow \mathbf{GL}_d$ be a k -morphism contained in ξ and let D be the line invariant under B . The class $\mathcal{P}_\xi$ contains an element defined over k . The unique element P' of $\mathcal{P}_\xi$ containing P is therefore defined over k (4.7), and the same is true for the line D , which is the unique line stable under P' . The dominant weight can be viewed

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as the character of P' in D ; it is consequently the restriction to T of an element of $X^*(P')_k$, thus *a fortiori* of an element of $X^*(P)_k$.

Suppose now that l_ξ is in the image of $X^*(P)_k \rightarrow X^*(T)_k$. It is fixed by Γ according to 6.2 (3). On the other hand, l_ξ is trivial on the derived group of $\mathcal{Z}(S)$, hence orthogonal to the simple roots which are equal to 1 on S .

We thus find again condition (i) of 12.10.

12.12. Let π be an irreducible representation of G . The k -weights of π are the restrictions to S of the weights of π . If G is split over k , there is therefore no distinction to be made between weights and k -weights. The dominant k -weight (restriction of the dominant weight) will be denoted by m_π or m_ξ , ξ being the class of π .

In the following, we will assume, in addition to the conditions of 12.9, that T is defined over k , and that we have chosen on $X^*(T)$ and $X^*(S)$ compatible orders, whose positive sets of roots correspond to P and B . We denote by j the restriction $X^*(T) \rightarrow X^*(S)$ and by $(,)$ admissible and compatible scalar products on

$$X^*(T) \otimes \mathbf{R} \text{ and } X^*(S) \otimes \mathbf{R} \quad (6.10).$$

For all $b \in {}_k\Delta$, let $l_{(b)}$ be the sum of the l_a ($a \in \rho^{-1}(b)$). It follows from 12.10 that the dominant weight of a strongly rational representation over k is a linear combination with integer coefficients ≥ 0 of the $l_{(b)}$ (and conversely if G is simply connected). If m_b denotes the restriction of $l_{(b)}$ to S , we have, according to 6.11:

$$2(m_b, b) = c_b \cdot (b, b), \quad (m_b, c) = 0 \quad (b, c \in {}_k\Delta; b \neq c), \quad (1)$$

with c_b rational > 0 , (in fact $c_b = (a, a)/(b, b)$, where $a \in \Delta$ is any element of $\rho^{-1}(b)$), whence the following proposition:

Proposition 0.87 (12.13). *Let $\xi \in \mathcal{R}(G)$ be strongly rational over k . Then m_ξ is a linear combination with integer coefficients ≥ 0 of the weights m_b ($b \in {}_k\Delta$). There exists an integer $d_b \geq 1$ such that $z \cdot d_b \cdot m_b$ is the dominant k -weight of a strongly rational representation over k for all $z \in \mathbf{N}$.*

Let us call a strongly rational representation over k *fundamental* if its dominant k -weight is the smallest possible multiple of some m_b . There exist therefore $r = r_k(G)$ fundamental strongly rational representations over k , and their dominant weights form a basis of $X^*(S) \otimes \mathbf{Q}$. Furthermore, any linear combination with integer coefficients ≥ 0 of these latter is the dominant k -weight of a strongly rational representation over k , but the converse is false in general. It is however true if G is simply connected, and in this case the fundamental dominant k -weights are the m_b ($b \in {}_k\Delta$).

12.14. We will end this section with some properties of the k -weights of an irreducible representation π (not necessarily rational over k). We know that any weight of π is of the form $l = l_\pi - \sum c_a(l) \cdot a$ ($a \in \Delta$, $c_a(l) \in \mathbf{N}$), where l_π is the dominant weight of π . Since $\rho(\Delta) \subset {}_k\Delta \cup \{0\}$, it follows that a k -weight q of π can be written

$$q = m_\pi - \sum_{b \in {}_k\Delta} d_b(q) \cdot b, \quad (d_b(q) \in \mathbf{N}), \quad (1)$$

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where m_π is the dominant k -weight of π . We will denote by $\theta(l)$ (resp. $\theta(q)$) the set of $a \in {}_k\Delta$ (resp. $b \in {}_k\Delta$) such that $d_a(l) \neq 0$ (resp. $d_b(q) \neq 0$), and by $|l|$ (resp. $|q|$) the sum of the $d_a(l)$ (resp. $d_b(q)$).

The set of k -weights of π is obviously invariant under the Weyl group ${}_kW(G)$ relative to k (5.1).

Lemma 0.88 (12.15). *Let q be a k -weight of π , different from the dominant k -weight. Then there exists $c \in \theta(q)$ such that $q + c$ is a k -weight of π .*

Proof. Let V be the space of the representation π . It is the direct sum of the eigenspaces V_r of S corresponding to the different k -weights r of π , spaces which are stable under $\mathcal{Z}(S)$. By an elementary and well-known argument, we have

$$\rho(X)(V_r) \subset V_{r+c} \quad (c \in {}_k\Phi, X \in \mathfrak{g}_c), \quad (1)$$

where we have set

$$\mathfrak{g}_c = \{X \in \mathfrak{g} \mid \text{Ad } s(X) = s^c \cdot X \ (s \in S)\} \quad (c \in {}_k\Phi).$$

The unipotent radical U of B is generated by the root groups U_a ($a \in {}_k\Delta$) (cf. 2.3), as follows for example from 2.5 and the remark to 2.5 (this is moreover true in any characteristic). If $q + c$ were not a k -weight for any $c \in {}_k\Delta$, V_q would be stable under U , by (1), hence by B (since $B \subset \mathcal{Z}(S) \cdot U$), and would consequently contain a line fixed by B , which is absurd, by 12.1. There exists then $c \in {}_k\Delta$ such that $q + c$ is a k -weight of π . But, if $c \notin \theta(q)$, then $q + c$ is not a k -weight by 12.14, whence the lemma. $\square$

The following proposition is due to Satake when $k = \mathbf{R}, \mathbf{C}$ [27, lemmas 5, 7].

Proposition 0.89 (12.16). *We preserve the preceding notation. A subset θ of ${}_k\Delta$ is of the form $\theta(q)$, where q is a k -weight of the irreducible representation π , if and only if $\{m_\pi\} \cup \theta$ is connected.*

Proof. First, let us prove that $\theta(q) \cup \{m_\pi\}$ is connected. We proceed by induction on the number of elements of $\theta(q)$, the assertion being vacuous if $\theta(q)$ is empty. By repeated application of 12.15, we see that there exists a k -weight l of π and an element $c \in \theta(q)$ such that $\theta(q) = \theta(l) \cup \{c\}$, $c \notin \theta(l)$, and that $l' = l - c$ is a k -weight of π . By the induction hypothesis, $\{m_\pi\} \cup \theta(l)$ is connected. It remains then to show that c is not orthogonal to $\{m_\pi\} \cup \theta(l)$. If it were, then $(l', c) = -(c, c)$, so the image

$$r_c(l') = l' - 2(l', c) \cdot (c, c)^{-1} \cdot c = l' + 2c = l + c$$

of l' under the reflection r_c would be a k -weight, which is absurd by 12.14 and the fact that $c \notin \theta(l)$.

Conversely, let $\theta \subset {}_k\Delta$ be such that $\{m_\pi\} \cup \theta$ is connected. We can number the elements b_i of θ in such a way that $(m_\pi, b_1) > 0$, and that for all i ($2 \leq i \leq m = \text{card}(\theta)$) there exists a j ($1 \leq j < i$) such that $(b_j, b_i) \neq 0$ (hence < 0). One then immediately verifies, by induction on i , that

$$q_i = r_{b_i} \cdots r_{b_1}(m_\pi)$$

is a k -weight such that $\theta(q_i) = \{b_1, \dots, b_i\}$ ($1 \leq i \leq m$) and that $(q_i, b_{i+1}) > 0$ if $i < m$. $\square$

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Proposition 0.90 (12.17). *We preserve the previous notation. Let $w \in {}_kW(G)$ and $q = w(m_\pi)$. Then there exist $w_1, w_2 \in {}_kW(G)$ such that $w_2(m_\pi) = m_\pi$, that w_1 is a product of reflections r_b ($b \in \theta(q)$), and that $w = w_1 \cdot w_2$.*

Proof. We proceed by induction on $|q|$, the proposition being obvious if $|q| = 0$.

It is known that the closure of a Weyl chamber is a fundamental domain for ${}_kW(G)$ [6]. Consequently, any element of the orbit of m_π under ${}_kW(G)$, different from m_π , has a scalar product < 0 with at least one element of ${}_k\Delta$. Thus there exists $c \in {}_k\Delta$ such that $(q, c) < 0$. Let then

$$q' = r_c(q) = q - 2(q, c) \cdot (c, c)^{-1} \cdot c.$$

This is a k -weight. In view of 12.14 and the inequality $(q, c) < 0$, we must have $c \in \theta(q)$, hence $\theta(q') \subset \theta(q)$, and $|q'| < |q|$. It then suffices to apply the induction hypothesis to q' and to notice that we also have $q = r_c(q')$. $\square$

§ 13. COMPACTLY GENERATED p -ADIC GROUPS

13.1. A topological group is compactly generated if it is equal to the subgroup generated by a suitable compact subset. It is known that any real or complex Lie group, having only a finite number of connected components, is compactly generated. This is not always the case for the group of rational points of an algebraic group H over a p -adic field K . For example, if H is the additive group $\mathbf{G}_a$, then H_K is non-compact, but is the union of an increasing sequence of open compact subgroups. The goal of this section is to give a necessary and sufficient condition for H_K to be compactly generated. The following remark will be useful:

- (i) Let L be a locally compact topological group, M a closed compactly generated subgroup. Then if L/M is compact, L is compactly generated.

This follows from the fact that every compact subset of L/M is the image of a compact subset of L under the canonical projection.

Lemma 0.91 (13.2). *Let G be unipotent and k of characteristic zero. Then $(\mathcal{D}G)_k = \mathcal{D}(G_k)$.*

Proof. The lemma is obvious if G is commutative. Otherwise, let V be a non-trivial central k -subgroup of G contained in $\mathcal{D}G$, for example the last non-trivial subgroup of the lower central series. If x, y are elements of the Lie algebra $\mathfrak{g}$ of G whose bracket belongs to the Lie algebra $\mathfrak{v}$ of V , then the Campbell-Hausdorff formula immediately implies that $(\exp x, \exp y) = \exp[x, y]$. Consequently,

$$V_k \subset \mathcal{D}(G_k). \quad (1)$$

Let $\pi : G \rightarrow G' = G/V$ be the canonical projection. Proceeding by induction on the dimension, we can assume that $\mathcal{D}(G'_k) = (\mathcal{D}G')_k$. As $G'_k = G_k/V_k$ (2.7), this gives $(\mathcal{D}G)_k \subset \mathcal{D}(G_k) \cdot V_k$, and the lemma then follows from (1). $\square$

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13.3. Lemma. — Assume k has characteristic zero. Let U be a normal unipotent k -subgroup of G , $\pi : G \rightarrow G' = G/\mathcal{D}U$ the canonical projection, and L a subset of G_k . If $\pi(L)$ generates G'_k , then L generates G_k .

Proof. Proof by induction on $\dim G$. If U is commutative, there is nothing to prove. Otherwise, let V be the last non-trivial subgroup of the descending central series of U . It is defined over k , and normal in G . We obviously have $G' = (G/V)/\mathcal{D}(U/V)$.

Let L^* be the subgroup generated by L . The induction hypothesis, and the equality $G_k/V_k = (G/V)_k$ (2.7) show that $G_k = L^* \cdot V_k$, hence also that $U_k = (L^* \cap U_k) \cdot V_k$.

Now, if

$$x = a \cdot v, \quad y = b \cdot w \quad (a, b \in L^* \cap U_k; v, w \in V_k),$$

we have, since v, w are in the center of U , $(x, y) = (a, b)$, whence $\mathcal{D}(U_k) \subset L^* \cap U_k$ and also, by 13.2, $V_k \subset L^*$, so $G_k = L^*$. $\square$

13.4. Theorem. — Assume that k is a p -adic field of characteristic zero. Let U be the unipotent radical of G , $V = U/\mathcal{D}U$, and ρ the representation of G in the Lie algebra $\mathfrak{v}$ of V defined from the adjoint representation of G . Then the following two conditions are equivalent:

- (i) G_k is compactly generated.
- (ii) $\mathfrak{v}$ contains no non-zero subspace $\mathfrak{w} \neq 0$ defined over k , stable under G , such that the image of G_k in $\mathrm{GL}(\mathfrak{w})$ under ρ is compact.

In particular, every reductive k -group is compactly generated¹².

Since the group $(G^0)_k$ is of finite index in G_k [5, 6.4], we may assume G is connected. We will first prove that (ii) $\Rightarrow$ (i), by distinguishing several cases.

- a) $G = \mathbf{G}_m$. Then $G_k = k^*$ is generated by the group M of units of k , which is compact, and by t , where t is a uniformizer of k (an element whose image in k^*/M , which is infinite cyclic, generates this group).
- b) $G = T$ is a torus. Then T contains a split torus T_d such that $T' = T/T_d$ is anisotropic over k (1.8). The group T'_k is compact (9.4) and we have $T_k/T_{d,k} = T'_k$ (2.7). Our assertion then follows from a), and 13.1 (i).
- c) $G = T \cdot U$ is the semi-direct product of a k -torus T and a normal unipotent k -subgroup U of dimension one.

Hypothesis (ii) implies that T acts non-trivially by inner automorphisms on U . Therefore, there exists $\chi \in X^*(T)_k$, $\chi \neq 0$, and a k -isomorphism $\theta : \mathbf{G}_a \rightarrow U$, such that

$$\theta(\chi(t) \cdot x) = t \cdot \theta(x) \cdot t^{-1} \quad (x \in \mathbf{G}_a, t \in T).$$

One can then find a k -morphism $\lambda : \mathbf{G}_m \rightarrow T$ such that $\chi \circ \lambda$ is of the form $s \mapsto s^m$ where m is an integer > 0 (cf. § 1). Consequently, $\chi(T_k) \subset (k^*)^m$. If L is a compact generating system of T_k , which exists by b), then G_k is generated by L and by the $\theta(x^i \cdot y)$, where x is a uniformizer of k , $i \in \mathbf{N}$, $i < m$, and y runs through the integers of k , whence c).

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- d) G is reductive. Let P be a minimal parabolic k -subgroup of G . Write it in the form $P = M.S.U$ where U is the unipotent radical of P , S a maximal k -split torus of G , and M the k -anisotropic part of $\mathcal{Z}(S)$ (cf. 4.28).

¹²This theorem answers a question of M. Kneser, and is used in [15]. We are also indebted to him for several suggestions for the transition from the reductive case to the general case.

The group U admits a composition series over k whose successive quotients are vector spaces on which S acts by non-trivial homotheties (3.18, remark). It then follows immediately from c) that $(S.U)_k = S_k.U_k$ is compactly generated. The group $\mathcal{Z}(S)/S$ is isogenous to M , hence anisotropic over k , and $(\mathcal{Z}(S)/S)_k$ is compact (9.4). Since S is split, we have $(\mathcal{Z}(S)/S)_k = \mathcal{Z}(S)_k/S_k$, so in view of b) and 13.1 (i), $\mathcal{Z}(S)_k$ is compactly generated. Then $P_k = \mathcal{Z}(S)_k.U_k = \mathcal{Z}(S)_k.(S.U)_k$ is also compactly generated, and the same holds for G_k , which is the union of a finite number of double cosets modulo P_k (5.15).

e) G is not reductive. It is the semi-direct product of its unipotent radical U and of a maximal reductive k -subgroup H (cf. 0.8). Let $\mathfrak{q}$ be the smallest vector subspace of $\mathfrak{u}$ containing every line defined over k which is stable under a k -split torus acting non-trivially on it. This subspace is defined over k , stable under H_k , hence also under H , since H_k is Zariski dense in H [23]. Since the rational representations of H are completely reducible, one can find a complement $\mathfrak{r}$ of $\mathfrak{q}$ in $\mathfrak{u}$, defined over k and stable under H . Every k -split torus of H acts trivially on $\mathfrak{r}$, so the image of the homomorphism $\sigma : H \rightarrow \mathrm{GL}(\mathfrak{r})$ defined by ρ is anisotropic over k , and $\sigma(H_k)$ is compact (9.4). In view of hypothesis (ii), we therefore have $\mathfrak{r} = \{0\}$. We then immediately deduce from the definition of $\mathfrak{q}$ and from c) the existence of a compact subset M of $(G/\mathcal{Z}U)_k$ such that V_k is contained in the subgroup M^* generated by M . In view of d), it follows that $(G/\mathcal{Z}U)_k = H_k.V_k$ is compactly generated. Since $(G/\mathcal{Z}U)_k = G_k/(\mathcal{Z}U)_k$, we see that there exists a compact subset $L \subset G_k$ whose image in $(G/\mathcal{Z}U)_k$ generates this group; L then generates G_k according to 13.3, which completes the proof of the implication (ii) $\Rightarrow$ (i).

(i) $\Rightarrow$ (ii). The group $(G/\mathcal{Z}U)_k = G_k/(\mathcal{Z}U)_k$ is also compactly generated; we can therefore assume U is abelian, $U = V$. Let $\mathfrak{q}$ be a k -subspace of $\mathfrak{v}$ stable under G , and let $\mathfrak{r}$ be a k -complement of $\mathfrak{q}$ in $\mathfrak{v}$, stable under the Levi subgroup H of G , and hence also under G . Let $Q = \exp \mathfrak{q}$ and $R = \exp \mathfrak{r}$ be the corresponding subgroups. V is therefore the direct product of Q and R , and G the semi-direct product of H, Q, R . Let L be a compact set generating G_k , which we assume to be of the form $L = A.B.C$ ($A \subset H_k, B \subset Q_k, C \subset R_k$), which is obviously permissible. Suppose that the image of G_k in $\mathrm{GL}(\mathfrak{q})$ is compact. Then, up to replacing B by $\bigcup_{g \in G_k} {}^g B$, which is also compact, we can assume that B is invariant under the inner automorphisms of G_k . For every integer $n \geq 1$, we then have

$$H_k.B^n.R_k.A.B.C \subset H_k.B^{n+1}.R_k,$$

which shows that G_k is the union of the subsets $H_k.B^n.R_k$ ($n = 1, 2, \dots$), so Q_k is generated by the compact set B . Since Q_k is the additive group of a vector space over k , this implies that $Q = \{e\}$, whence (ii).

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§ 14. REAL GROUPS

14.1. As is customary, $\pi_0(X)$ denotes the set of path-connected components of the topological space X . This is a group if X is a topological group. It is known that if X is an algebraic $\mathbf{R}$ -variety, then $\pi_0(X_{\mathbf{R}})$ is finite [40]. (Here and in the following, $X_{\mathbf{R}}$ is of course considered as being equipped with the usual topology, defined from that of $\mathbf{R}$, cf. 9.1.) The topological dimension of $X_{\mathbf{R}}$ is at most equal to $\dim X$, and if X is a non-singular variety, each connected component of $X_{\mathbf{R}}$ is an analytic manifold of dimension $\dim X$ [40]. When X is a group, we will also denote by $(X_{\mathbf{R}})^0$ the identity component of $X_{\mathbf{R}}$.

14.2. Proposition. — *Assume G is connected, defined over $\mathbf{R}$, and let P be a parabolic $\mathbf{R}$ -subgroup of G . Then $(G/P)_{\mathbf{R}}$ is a connected compact manifold.*

$(G/P)_{\mathbf{C}}$ is a non-singular complex projective variety, so $(G/P)_{\mathbf{R}}$ is a real (9.1), projective, and therefore compact variety.

Let $G = H \cdot U$ be a Levi decomposition of G . The group P contains U and $G/P = H/(H \cap P)$. The group $H \cap P$ is therefore a parabolic $\mathbf{R}$ -subgroup of H , and we may assume that G is reductive.

Let V be the unipotent radical of a parabolic $\mathbf{R}$ -subgroup opposite to P . We have seen (4.10) that the projection of G onto G/P identifies V with an open subset of G/P , and thus (9.1) $V_{\mathbf{R}}$ with an open subset of $(G/P)_{\mathbf{R}}$. The complement Y of V in G/P is an algebraic $\mathbf{R}$ -subset of dimension strictly smaller than $m = \dim G/P$. The topological dimension of $Y_{\mathbf{R}}$ is therefore $< m$ (14.1), and $V_{\mathbf{R}}$ is dense in $(G/P)_{\mathbf{R}}$. Since $V_{\mathbf{R}}$ is connected, the proposition is proved.

14.3. In [8, Chap. VI, § 5, no. 2, prop. 2] it is shown that “every compact linear group is algebraic.” In the terminology followed here, this means that a compact subgroup H of $\mathbf{GL}(n, \mathbf{R})$ is the set of real points of an algebraic subgroup of $\mathbf{GL}(n, \mathbf{C})$ defined over $\mathbf{R}$, which can be chosen to be the Zariski closure $\mathcal{A}(H)$ of H . This implies in particular that if $G \subset \mathbf{GL}(n, \mathbf{C})$ is connected, defined over $\mathbf{R}$, and if $G_{\mathbf{R}}$ is compact, then $G_{\mathbf{R}}$ is connected. Indeed, $G_{\mathbf{R}}^0 = \mathcal{A}((G_{\mathbf{R}})^0)_{\mathbf{R}}$, so if $G_{\mathbf{R}} \neq (G_{\mathbf{R}})^0$, the group $\mathcal{A}((G_{\mathbf{R}})^0)$ is a proper subgroup of the same dimension as G , and G is therefore not connected.

14.4. Theorem (Matsumoto [18]). — Assume G is connected, defined over $\mathbf{R}$, and let S be a maximal $\mathbf{R}$ -split torus of G . Then $G_{\mathbf{R}} = (G_{\mathbf{R}})^0 \cdot S_{\mathbf{R}}$.

We have to show that the natural homomorphism $\pi_0(S_{\mathbf{R}}) \rightarrow \pi_0(G_{\mathbf{R}})$ is surjective. Let H be a maximal reductive $\mathbf{R}$ -subgroup of G containing S , and U the unipotent radical of G . The group $U_{\mathbf{R}}$ is connected and $G_{\mathbf{R}}$ is homeomorphic to the product $H_{\mathbf{R}} \times U_{\mathbf{R}}$. The homomorphism $\pi_0(H_{\mathbf{R}}) \rightarrow \pi_0(G_{\mathbf{R}})$ is therefore bijective, and it suffices to carry out the proof when G is reductive.

Let P be a minimal parabolic $\mathbf{R}$ -subgroup of H containing S . It is the product...

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semidirect product of its unipotent radical by $\mathcal{Z}(S)$, hence $\pi_0(\mathcal{Z}(S)_{\mathbf{R}}) \rightarrow \pi_0(P_{\mathbf{R}})$ is bijective. But $(G/P)_{\mathbf{R}}$ is equal to $G_{\mathbf{R}}/P_{\mathbf{R}}$ (4.13) and is connected (14.2). The exact homotopy sequence

$$\pi_0(P_{\mathbf{R}}) \xrightarrow{\iota} \pi_0(G_{\mathbf{R}}) \rightarrow \pi_0(G_{\mathbf{R}}/P_{\mathbf{R}}) \quad (1)$$

therefore shows that ι is surjective. It remains to prove that $\pi_0(S_{\mathbf{R}}) \rightarrow \pi_0(\mathcal{Z}(S)_{\mathbf{R}})$ is surjective.

The group $M' = \mathcal{Z}(S)/S$ is anisotropic over $\mathbf{R}$, connected, hence $M'_{\mathbf{R}}$ is compact (9.4) and connected (14.3). Since S is split over $\mathbf{R}$, we have $M'_{\mathbf{R}} = \mathcal{Z}(S)_{\mathbf{R}}/S_{\mathbf{R}}$ and our assertion follows from the homotopy sequence (1), applied to the fibration of $\mathcal{Z}(S)$ by S .

14.5. Corollary. — We have $\pi_0(G_{\mathbf{R}}) \cong (\mathbf{Z}/2\mathbf{Z})^d$, where $d \leq r_{\mathbf{R}}(G)$.

Indeed, $S_{\mathbf{R}}$ is a product of $r = r_{\mathbf{R}}(G)$ factors $\mathbf{R}^*$, hence $\pi_0(S_{\mathbf{R}}) \cong (\mathbf{Z}/2\mathbf{Z})^r$, and $\pi_0(G_{\mathbf{R}})$ is a quotient of $\pi_0(S_{\mathbf{R}})$ according to the theorem.

14.6. Corollary. — Suppose G is reductive. Then every element of the relative Weyl group ${}_{\mathbf{R}}W(G)$ (cf. 5.1) admits a representative in $(G_{\mathbf{R}})^0$.

Indeed, if $x \in {}_{\mathbf{R}}W(G)$, there exists, by 14.4, $y \in S_{\mathbf{R}}$ such that $y \cdot x \in (G_{\mathbf{R}})^0$, and $y \cdot x$ obviously represents the same element of ${}_{\mathbf{R}}W(G)$ as x .

14.7. We conclude with a few remarks to connect the theory of real semi-simple Lie groups and algebras with certain notions introduced in this work. We assume G is semi-simple, defined over $\mathbf{R}$. The main point is to see that the group $A.N$ of the Iwasawa decomposition is the identity component of the set of real points of a maximal trigonalizable $\mathbf{R}$ -group, and that ${}_{\mathbf{R}}W(G)$ and ${}_{\mathbf{R}}\Phi$ are identified with the Weyl group and the root system of the symmetric pair $(G_{\mathbf{R}}, K)$, where K is a maximal compact subgroup of $G_{\mathbf{R}}$.

We first recall the Cartan and Iwasawa decompositions. (For more details and proofs, see for example [13], as well as [4, §1] for various complements, concerning in particular non-connected real groups.)

Let K be a maximal compact subgroup, $\mathfrak{k}$ its Lie algebra, $\mathfrak{p}$ the orthogonal complement of $\mathfrak{k}$ in the Lie algebra $\mathfrak{g}_{\mathbf{R}}$ of $G_{\mathbf{R}}$ with respect to the Killing form, $\mathfrak{a}$ a maximal subalgebra of $\mathfrak{p}$, necessarily commutative, and

$$P = \exp \mathfrak{p}, A = \exp \mathfrak{a}.$$

It is known that $(k, p) \mapsto k \cdot \exp p$ is a homeomorphism of $K \times \mathfrak{p}$ onto $G_{\mathbf{R}}$ (Cartan decomposition) and that $s : k \cdot p \mapsto k \cdot p^{-1}$ ($k \in K, p \in P$) is an involutive automorphism of $G_{\mathbf{R}}$ (Cartan involution), of which K is the set of fixed points. The endomorphisms $\text{ad } a$ ($a \in \mathfrak{a}$) are simultaneously diagonalizable over $\mathbf{R}$, and $\mathfrak{g}_{\mathbf{R}}$ is the direct sum of the centralizer $\mathfrak{z}(\mathfrak{a})$ of $\mathfrak{a}$ and of eigenspaces $\mathfrak{g}_a$, where a runs through a finite set Ψ of non-zero linear forms on $\mathfrak{a}$, the *roots* of $\mathfrak{g}_{\mathbf{R}}$ with respect to $\mathfrak{a}$. We choose an order on $\mathfrak{a}^*$ and let $\mathfrak{n} = \sum_{a>0} \mathfrak{g}_a$. Then $\mathfrak{n}$ is a Lie algebra consisting of elements x such that $\text{ad } x$ is nilpotent, hence nilpotent, which is normalized by $\mathfrak{a}$. Let

$A = \exp \mathfrak{a}$ and $N = \exp \mathfrak{n}$ be the analytic subgroups of $G_{\mathbf{R}}$ generated by $\mathfrak{a}$ and $\mathfrak{n}$. They are closed, and $(k, a, n) \mapsto k \cdot a \cdot n$ is a homeomorphism of $K \times A \times N$ onto $G_{\mathbf{R}}$ (Iwasawa decomposition). Furthermore, N is unipotent, A is diagonalizable over $\mathbf{R}$, and $A.N$ is trigonalizable over $\mathbf{R}$. Let $L = \mathcal{A}(A.N)$

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the Zariski closure of $A.N$ in G . Then L is a maximal $\mathbf{R}$ -subgroup trigonalizable over $\mathbf{R}$ of G , and $A.N = (L_{\mathbf{R}})^0$. Indeed, L is trigonalizable over $\mathbf{R}$, since $A.N$ is. On the other hand, $G_{\mathbf{R}}/A.N$ is compact (it is homeomorphic to K), so $G_{\mathbf{R}}/L_{\mathbf{R}}$ is also, and 9.3 implies that L contains a maximal $\mathbf{R}$ -subgroup trigonalizable over $\mathbf{R}$ of G .

This also implies that $A = (S_{\mathbf{R}})^0$ where $S = \mathcal{A}(A)$ is a maximal $\mathbf{R}$ -split torus, hence $\mathcal{N}(S) = \mathcal{N}(A)$. The Cartan involution s leaves A stable, hence also $\mathcal{N}(S)_{\mathbf{R}}$. We have $\mathcal{N}(S)_{\mathbf{R}} = (K \cap \mathcal{N}(S)_{\mathbf{R}}) \cdot \exp \mathfrak{m}$, where $\mathfrak{m}$ is the intersection of $\mathfrak{p}$ with the Lie algebra of $\mathcal{N}(S)_{\mathbf{R}}$ (cf. [4, 1.10]). But the relation $[\mathfrak{p}, \mathfrak{p}] \subset \mathfrak{k}$ then shows that $[\mathfrak{m}, \mathfrak{a}] = 0$, whence $\mathfrak{m} = \mathfrak{a}$ and

$$\mathcal{N}(S)_{\mathbf{R}} = (K \cap \mathcal{N}(S)_{\mathbf{R}}) \cdot A,$$

whence also, since $A.N$ is connected,

$$(G_{\mathbf{R}})^0 \cap \mathcal{N}(S) = (K^0 \cap \mathcal{N}(S)) \cdot A.$$

Taking 14.6 into account, we see that every element of ${}_{\mathbf{R}}W(G)$ has a representative in K^0 , whence also

$${}_{\mathbf{R}}W(G) = (K^0 \cap \mathcal{N}(A)) / (K^0 \cap \mathcal{Z}(A)).$$

The group ${}_{\mathbf{R}}W(G)$ therefore coincides with the group introduced in the theory of symmetric spaces by E. Cartan; moreover, since $\mathfrak{a}$ is the real part of the Lie algebra of S , the roots of $\mathfrak{g}_{\mathbf{R}}$ with respect to $\mathfrak{a}$ are identified with the $\mathbf{R}$ -roots of G (or more precisely with their differentials). We thus recover the original definitions of these notions, formulated without reference to algebraic groups.

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